

**Faculty of Computers
& Artificial Intelligence**



Benha University

Esaltare Dekör

Graduation Project

Computer Science Department

Team Names:

- 1) Ziad Essam Abdelaziz AbdelMoula**
- 2) Saad AbdelRazek Saad AbdelRazek**
- 3) Karim Hazem Hamed Mohamed**
- 4) Al-Shaimaa Gamal Ahmed Hassan**
- 5) Baraa AbdelMoez Ibrahim Al-Sayed Al-Deeb**

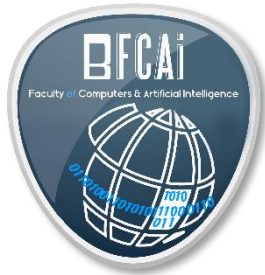
Under the supervision of:

Dr. Shadia Sarhan

Eng.Nada Adel

Team Leader: Ziad Essam Abdelaziz AbdelMoula

June 2025



**Faculty of Computers
& Artificial Intelligence**



Benha University



Graduation Project

Computer Science Department

Team Names:

- 1) Ziad Essam Abdelaziz AbdelMoula**
- 2) Saad AbdelRazek Saad AbdelRazek**
- 3) Karim Hazem Hamed Mohamed**
- 4) Al-Shaimaa Gamal Ahmed Hassan**
- 5) Baraa AbdelMoez Ibrahim Al-Sayed Al-Deeb**

Under the supervision of:

Dr. Shadia Sarhan

Eng.Nada Adel

Team Leader: Ziad Essam Abdelaziz AbdelMoula

June 2025



ABSTRACTION

This project focuses on creating a modern e-commerce platform tailored for the home decor industry, offering an enhanced and interactive shopping experience. Inspired by top brands like Home Zmart and IKEA, the platform includes a user-friendly interface where customers can browse products, view high-quality images, compare items, save favorites, and manage wish lists and carts easily.

A key feature is the integration of advanced 3D Design and visualization tools, allowing users to preview decor items in their own spaces, making the shopping process more informed and personalized. Real-time order tracking and delivery updates add transparency and build customer trust.

Vendors benefit from a subscription-based model, gaining access to intuitive tools for managing product listings, updating inventory, adjusting pricing, and tracking performance. An admin dashboard provides full control over users, vendors, and orders, alongside analytics for monitoring platform efficiency.

Additionally, a built-in chatbot offers real-time assistance and recommendations, supporting both users and vendors with inquiries, updates, and troubleshooting. Secure payment methods ensure financial data is protected, while multiple checkout options offer flexibility.

Overall, the platform delivers seamless, efficient, and secure experience, combining innovative features with robust management tools to serve the needs of customers, vendors, and administrators alike.

DECLARATION

We hereby certify that this material, which we now submit for assessment on the program of study leading to the award of Bachelor of Computers and Artificial Intelligence in Computer Science is entirely our own work, that we have exercised reasonable care to ensure that the work is original, and does not to the best of our knowledge breach any law of copyright, and has not been taken from the work of others save and to the extent that such work has been cited and acknowledged within the text of our work.

Signed: _____

Date: June 17, 2025

ACKNOWLEDGMENT

We would like to express our utmost gratitude to the following individuals who have contributed to the successful completion of our graduation project:

First and foremost, we would like to thank our advisor **Dr. Shadia Sarhan** for her unwavering support, guidance, and expertise throughout this project. Her valuable insights and constructive feedback have been instrumental in shaping our project and helping us overcome various challenges along the way.

We would also like to extend our deepest gratitude to **Eng.Nada Adel** for her continuous support and valuable guidance throughout this project. Her constructive feedback and dedicated assistance have played a significant role in shaping our ideas and overcoming the challenges we faced.

We extend our heartfelt gratitude to our College for its continuous support throughout our academic journey. These four years have been filled with challenges and achievements, and thanks to the dedication of the college administration and faculty members, we have gained the knowledge and experience that will accompany us in our professional future.

We also wish to express our deep appreciation for the incredible teamwork that united us throughout this journey. Dedication, collaboration, and perseverance were key factors in reaching our goals, and we are proud of what we have accomplished together.

Table of Contents

Chapter 1: Introduction	10
1.1 Project Overview	10
1.1.1 Project Description:	10
1.1.2 Goals	10
1.2 Objectives	11
1.2.1 Primary Objectives.....	11
1.2.2 Secondary Objectives.....	12
1.3 Scope	13
1.4 Audience and Stakeholders.....	14
1.5 Problem Statement	17
1.6 Significance of the Project	18
1.7 Methodology Overview	20
1.8 Limitations	23
1.9 Assumptions.....	24
 Chapter 2: Planning and Selection	 26
2.1 Problem Statement	26
2.2 Project Planning	30
2.3 Project Milestones	32
2.4 Resource Allocation.....	35
2.5 Risk Management	38
2.6 Timeline and Schedule	40
2.7 Work Break Down Structure	42
2.8 Gantt Chart and Network Diagram	43
2.9 Communication Plan	45
2.10 Business Plan	48
 Chapter 3: Analysis To The System	 55
3.1 Gather our information (Esaltare Dekor).....	55
3.1.1 Interviewing	59
3.1.2 Questionnaire	61
3.1.3 classification To The Requirements	64
3.2 Esaltare Dekor Requirements Structuring	68
3.2.1 Process Modeling	68
3.2.1.1 DFD and Context Diagram	68

3.2.1.2 Use Case Diagrams and Analysis	70
3.2.2 Data Modeling	86
3.2.2.1 ERD Diagram (Conceptual Layer) (1st Layer Of Database).....	86
Chapter 4: EDAI Model	87
4.1 Chatbot overview	87
4.2 Prompt Engineering	88
4.3 Retrieval Augmented Generation (RAG)	90
4.4 Agent Based System Architecture	92
4.5 Workflow (All Working Together).....	95
4.6 Key Functionalities	97
4.7 System Architecture	98
4.8 Personality Analysis	100
Chapter 5: Design The System.....	101
5.1 Class Diagram.....	101
5.2 Sequence Diagram	102
5.3 Activity Diagrams.....	107
5.4 State Diagram	111
5.5 Database Design	113
5.5.1 Database schema design.	113
5.5.2 Relational Model or Logical Data Model. (2nd Layer Of Database)	115
5.6 System Architecture and Structure	116
5.7 User Interface and Experience Design.....	117
5.7.1 Design of The User Interface.....	117
5.7.2 Client User Interface	121
5.7.2.1 Home Page.....	121
5.7.2.2 Category Page	122
5.7.2.3 Product Page	123
5.7.2.4 Add To Favorite Page	124
5.7.2.5 Add To Card Page.....	125
5.7.2.6 Search Product Page	126
5.7.2.7 Compare Product Page.....	127
5.7.2.8 User Profile Page	128
5.7.2.9 Track Order Page	129

5.7.2.10 3D Design Page.....	130
5.7.2.11 Chatbot Page	131
5.7.2.12 Vendor Package Page	132
5.7.2.13 Vendor Profile Page.....	133
5.7.2.14 Add Product Page	134
5.7.2.15 Login Page	135
5.7.2.16 Signup Page	136
5.7.2.17 Forget Password Page.....	137
5.7.3 Admin User Interface	140
5.7.3.1 Dashboard Page	140
5.7.3.2 Categories Page.....	141
5.7.3.3 Add Category Page	142
5.7.3.4 Edit Category Page	143
5.7.3.5 Subcategory Page.....	144
5.7.3.6 Add Subcategory Page.....	145
5.7.3.7 Edit Subcategory Page	146
5.7.3.8 Products Page.....	147
5.7.3.9 Add Product Page	148
5.7.3.10 Edit Product Page.....	149
5.7.3.11 Brand Page	150
5.7.3.12 Add Brand Page	151
5.7.3.13 Edit Brand Page	152
5.7.3.14 Posts Page	153
5.7.3.15 Add Post Page.....	154
5.7.3.16 Edit Post Page	155
5.7.3.17 Announcement Page	156
5.7.3.18 Add Announcement Page	157
5.7.3.19 Edit Announcement Page.....	158
5.7.3.20 Coupon Page	159
5.7.3.21 Add Coupon Page	160
5.7.3.22 Edit Coupon Page	161
5.7.3.23 Customer List Page.....	162
5.7.3.24 Customer Details Page.....	163
5.7.3.25 Vendor List Page.....	164
5.7.3.26 Vendor Details Page	165
5.7.3.27 Order Page	166
5.7.3.28 Order Details Page	167

5.7.3.29 Review Page.....	168
5.7.3.30 Signup Page	169
5.7.3.31 Sign in Page	170
5.7.3.32 Forget Password Page	171
5.7.3.33 Reset Password Page.....	172
5.7.3.34 Inbox Page	174
5.7.3.35 Setting Page	175

Chapter 6: Implementation.....176

6.1 Implementation Journy	176
6.2 API Documentation	178
6.3 Frontend Development.....	180
6.4 Backend Development.....	182
6.5 Tools Used	183
6.6 Deployment For The Project	119
6.7 Results After Implementation	185

Chapter 7: Conclusion.....192

7.1 References.....	195
---------------------	-----

List Of Images

Figure 1-1 ED Platform	12
Figure 1-2 ED Scope.....	14
Figure 1-3 ED Ecosystem	16
Figure 1-4 ED Significant.....	19
Figure 1-5 ED Assumptions Success	25
Figure 2-1 ED WHY?	29
Figure 2-2 ED Planning	31
Figure 2-3 ED Milestones.....	34
Figure 2-4 ED Resource Allocation.....	37
Figure 2-5 ED Communication Plane.....	47
Figure 4-1 EDAI Pipeline	99
Figure 5-1 ED System Architecture	116

Chapter 1: Introduction

1.1 Project Overview

1.1.1 Project Description:

This project involves the development of an innovative e-commerce platform specifically designed for home decor, like industry leaders such as Home zmart and IKEA. The platform will be built to enhance the online shopping experience by providing users with a wide array of features that make browsing, selecting, and purchasing home decor items intuitive and enjoyable.

Users will benefit from detailed product information, the ability to compare products, save favorites, and add items to their shopping cart. Furthermore, the platform will integrate advanced 3D design tools that allow users to visualize how products look in their spaces, real-time updates on order locations and estimated delivery times.

The platform will also support brand vendors by offering a subscription-based model for accessing various features and services, including product submissions and admin approval processes. Vendors will have robust tools for listing and managing their products efficiently.

Administrators will have comprehensive control over platform operations through a user-friendly dashboard that includes user and vendor management, order processing, and analytics reporting. Additionally, the platform will feature a chatbot assistant to assist users and vendors, provide personalized product recommendations, and offer real-time support.

1.1.2 Goals:

- To provide users with a seamless and engaging shopping experience and to offer vendors efficient tools for listing and managing products.
- To ensure effective platform management through comprehensive admin tools.
- To enhance customer satisfaction and engagement through personalized support and real-time updates and payment options, safeguarding user financial information.
- To integrate advanced 3D design tools that allow users to visualize how home decor items will look in their spaces.
- To provide a subscription-based model for vendors, ensuring they have access to various features and services.
- To offer detailed product comparisons to help users make informed purchasing decisions.
- To provide comprehensive order management, from initial purchase to delivery, ensuring timely and accurate fulfillment.
- To leverage data analytics for personalized recommendations, enhancing the user experience by tailoring suggestions to individual preferences.
- To facilitate real-time user support through an integrated chatbot assistant, enhancing the overall customer experience.

1.2 Objectives

To develop a comprehensive **Esaltare Dekör** e-commerce platform specialized in home decor that provides a seamless shopping experience for users, a robust product management system for vendors, and an efficient administration tool for platform management

1.2.1 Primary objectives

User Experience (UX) Enhancement:

- **Seamless Navigation:** Develop an intuitive and user-friendly interface that allows users to effortlessly browse through various home decor categories.
- **Detailed Product Information:** Provide comprehensive information for each product, including descriptions, specifications, high-quality images, and user reviews to help customers make informed decisions.
- **Favorites and Shopping Cart:** Enable users to save their preferred products to a favorites list and add items to a shopping cart, making it easy to organize and manage their purchases.
- **Product Comparison:** Allow users to compare multiple products side by side in terms of features, prices, and reviews to facilitate better decision-making.
- **3D Design Capability:** Integrate tools that allow users to visualize how products will look in their spaces using interactive 3D designs.
- **Order Tracking:** Offer real-time tracking of orders so users can stay informed about the status and location of their shipments.
- **Secure Payment:** Provide a variety of secure payment options to ensure a safe and convenient checkout process for users.

Vendor Brand Management:

- **Subscription Packages:** Require vendors to subscribe to a package plan to access the platform's features and services. Offer various subscription tiers with different levels of access and benefits.
- **Product Submission:** Develop a portal where brand vendors can submit their products for listing on the platform.
- **Admin Approval:** Implement a review system where administrators can evaluate and approve or reject product submissions from vendors to ensure quality and relevance.

Administrative Control:

- **Dashboard:** Create an admin dashboard that allows administrators to monitor and manage all platform activities, including user and vendor interactions, orders, and overall performance.
- **User and Vendor Management:** Enable administrators to manage user accounts and vendor profiles, including handling approvals, updates, and removals.
- **Order Processing:** Streamline the order processing workflow from purchase to delivery to ensure timely and accurate fulfillment.
- **Analytics and Reporting:** Provide tools for tracking key metrics such as sales, user behavior, and inventory levels to help administrators make data-driven decisions.

1.2.2 Secondary objectives

Chatbot Assistant Integration:

- **User Support:** Develop a chatbot assistant designed to interact with users in a conversational manner, helping them with their queries and providing real-time assistance. The chatbot will be equipped to answer frequently asked questions, guide users through the navigation of the platform, and provide information about products, orders, and services.
- **Availability:** Offer round-the-clock support through the chatbot, ensuring users can receive assistance at any time, regardless of their location or time zone. This will improve customer satisfaction and provide a reliable support channel.

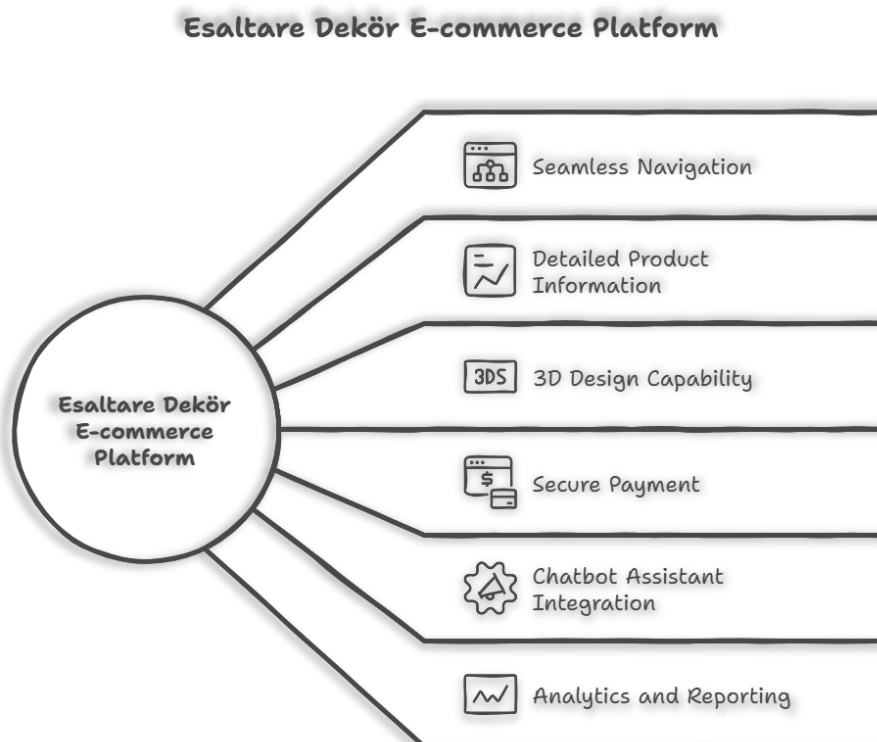


Figure 1-1 ED Platform

By achieving these objectives, we aim to create a user-centric platform that simplifies the process of home decoration, empowers users to express their creativity, and fosters a vibrant community of home decor enthusiasts.

1.3 Scope

Shopping Products:

- **In Scope:** Offering a variety of home decoration products for users to browse, filter, explore, and purchase, including furniture, lighting, wall decor, and accessories.
- **Out of Scope:** Non-decorative products or unrelated categories.

Add to Cart and Favorites:

- **In Scope:** Implementing cart functionality for users to add products they're interested in. Favorites allow users to save items for future reference.
- **Out of Scope:** Complex cart management features beyond basic add-to-cart and favorites functionality.

Payment Methods:

- **In Scope:** Providing multiple payment options including credit/debit cards, digital wallets (like PayPal or stripe), and potentially cryptocurrency. Ensuring secure transactions is crucial.
- **Out of Scope:** Developing custom payment gateways or handling financial integrations beyond standard secure payment methods.

Chatbot Assistant:

- **In Scope:** Integrating an AI-powered chatbot that uses large language models (LLMs) and natural language processing (NLP) to assist users with inquiries, make product recommendations, and facilitate adding items to the cart and favorites.
- **Out of Scope:** Handling sensitive personal data or financial transactions directly through the chatbot and engaging in extensive conversational depth beyond shopping assistance and basic inquiries.

Vendor Brand Furniture Listings (Admin Access):

- **In Scope:** Allowing users with admin privileges to list their own furniture for sale.
- **Out of Scope:** Managing user-generated content moderation or extensive admin features beyond the basic listing functionality.

Drag-and-Drop Furniture Section (Simulating Canva):

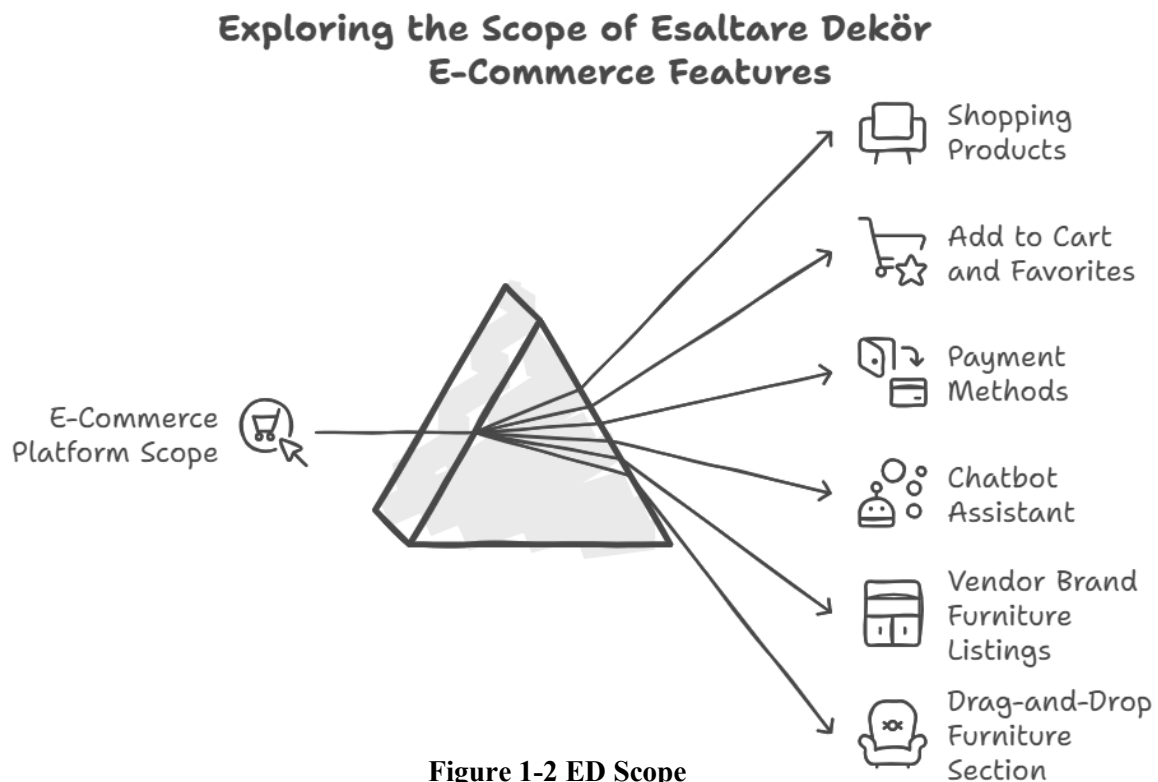
- **In Scope:** Creating a user-friendly interface for designing spaces using 3D modeling tools.
- **Out of Scope:** Developing complex 3D rendering features or professional-grade interior design tools beyond basic 3D modeling capabilities.

Vendor Management and Subscription Packages:

- **In Scope:** Allowing vendors to subscribe to various package plans, submit their products for listing, and providing tools for managing listings. Implementing an admin approval process for quality control.
- **Out of Scope:** Managing vendor relationships beyond the provided tools and admin approval process.

Admin Dashboard and Control:

- **In Scope:** Developing an admin dashboard to monitor and manage platform activities, including user and vendor management, order processing, and generating analytics reports.
- **Out of Scope:** Custom development of advanced business intelligence or enterprise resource planning tools beyond basic analytics and reporting.



1.4 Audience and Stakeholders

Audience:

1. Consumers:

- **Homeowners and Renters:** Individuals looking to enhance their living spaces with stylish and functional home decor products.
- **Interior Designers:** Professionals seek a diverse range of home decor items to incorporate into their design projects for clients.
- **DIY Enthusiasts:** Users who enjoy personalizing their homes and are looking for unique decor pieces and tools to create their desired look.
- **Eco-conscious Consumers:** Prefer sustainable, environmentally friendly furniture options and need transparency about materials used and sustainable practices of the company

2. Brand Vendors:

- **Home Decor Manufacturers:** Companies that produce home decor items and want to list their products on a dedicated platform.
- **Local Artisans:** Small businesses and individual artisans offering handcrafted and unique home decor items looking for a wider audience.
- **Furniture Retailers:** Retailers specializing in furniture and home decor who want to expand their reach through an online platform.

Stakeholders:

1. Project Owners/Investors:

- **Company Executives and Investors:** Individuals and entities funding the project, who are interested in the platform's financial performance and growth.
- **Business Development Team:** Responsible for identifying new opportunities, partnerships, and strategies to expand the platform's market presence.

2. Project Management Team:

- **Project Managers:** Overseeing the development and implementation of the platform, ensuring that milestones are met, and objectives are achieved.
- **Development Team:** Software developers, UX/UI designers, and IT specialists building and maintaining the platform.

3. Operations and Support Team:

- **Customer Support Representatives:** Aiding users and vendors, handling inquiries, and resolving issues.
- **Vendor Relations Managers:** Managing relationships with vendors, onboarding new vendors, and ensuring product quality and compliance.
- **Marketing and Sales Team:** Promoting the platform, managing advertising campaigns, and driving user and vendor engagement.

4. Users:

- **End Users:** The primary users of the platform who will browse, purchase, and interact with the platform's features.
- **Admin Users:** Administrators who will manage the platform, oversee user and vendor activities, and ensure smooth operations.

5. Technology Partners:

- **Payment Gateway Providers:** Ensuring secure and reliable payment processing for transactions on the platform.
- **Logistics and Delivery Partners:** Coordinating the delivery of products to users, integrating GPS tracking for real-time order updates.
- **AI and Analytics Providers:** Supplying AI and data analytics tools to enhance the platform's capabilities, such as the chatbot assistant and personalized recommendations.

Importance of Understanding Audience and Stakeholders

1. **Targeted Marketing:** Tailoring marketing messages to different audience segments increases engagement and sales.
2. **Product Development:** Gathering insights from the audience helps in designing products that fulfill their needs and preferences.
3. **Collaboration Opportunities:** Building relationships with designers and contractors can lead to joint projects and increased exposure.
4. **Customer Satisfaction:** Addressing the needs of both audiences and stakeholders enhances overall satisfaction and drives repeat business.
5. **Brand Reputation:** Positive experiences from customers and partners can boost your brand's reputation in the market.
6. **Market Adaptation:** Understanding trends and feedback allows the company to adapt quickly to changing market demands.

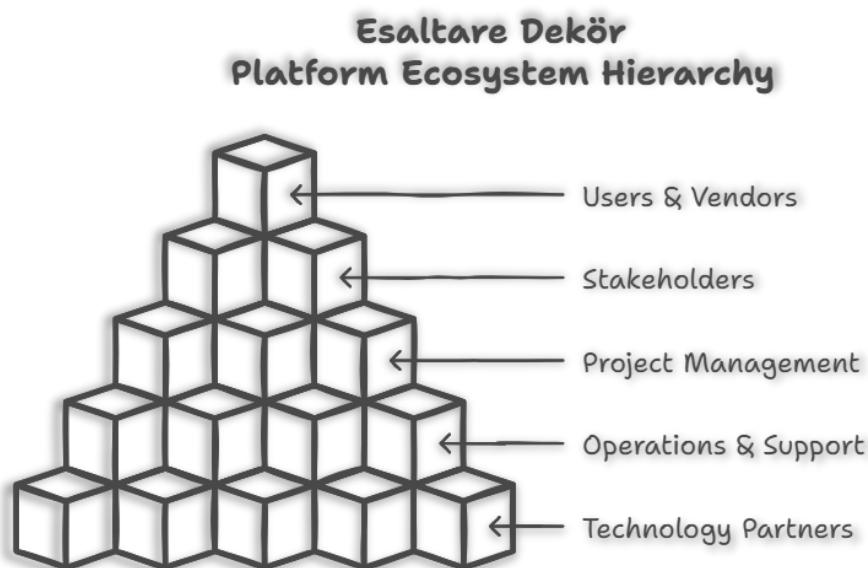


Figure 1-3 ED Ecosystem

1.5 Problem Statement

In today's busy world, people often struggle to find the time and expertise needed to decorate their spaces to match their personal style and practical needs. Traditional decor shopping lacks personalization, and users face challenges in visualizing how products will fit into their specific spaces. Moreover, finding inspiration and managing decor purchases across multiple platforms can be overwhelming.

Many consumers are faced with a fragmented shopping experience, where they must navigate between various websites and stores to find the right decor items. This process is time-consuming and often frustrating, as it does not cater to individual preferences or provide a cohesive solution. Additionally, the inability to visualize how decor items will look in their own homes leads to uncertainty and hesitation in making purchases.

Brand vendors also face challenges in reaching a broader audience and effectively managing their product listings. They require a platform that not only provides visibility but also offers tools to manage and showcase their products in an appealing way.

Administrators of e-commerce platforms need efficient tools to oversee platform operations, manage user and vendor activities, and ensure the smooth functioning of the platform. The lack of comprehensive administrative controls can lead to operational inefficiencies and poor user experiences.

This project aims to address these issues by developing an innovative e-commerce platform specifically designed for home decor, offering features like detailed product information, advanced 3D design tools and a robust admin dashboard. By integrating AI-powered chatbot assistance, personalized recommendations, and a seamless shopping experience, this platform will cater to the needs of users, vendors, and administrators alike, bridging the gap between traditional decor shopping and modern technological capabilities.

1.6 Significance of the Project

The development of this innovative **Esaltare Dekör** e-commerce platform holds significant value for several key stakeholders, addressing critical pain points and enhancing overall user experience. Here are the key points highlighting its importance:

1. Enhancing User Experience:

- **Personalization:** By integrating advanced 3D design tools and AI-powered chatbot assistance, the platform offers a personalized shopping experience, allowing users to visualize products in their own spaces and receive tailored recommendations based on their preferences.
- **Convenience:** The platform simplifies the shopping process by providing a wide range of home-decorated products in one place, reducing the time and effort needed to find and purchase the perfect items. The addition of features like favorites, shopping cart, and order tracking further enhances convenience.
- **Visual Representation:** The ability to use 3D modeling tools helps users overcome the common challenge of visualizing how products fit into their specific spaces, leading to more confident purchasing decisions.

2. Supporting Brand Vendors:

- **Increased Visibility:** Brand vendors benefit from a dedicated platform that showcases their products to a targeted audience interested in home decor, increasing their reach and potential sales.
- **Efficient Management:** The platform provides vendors with robust tools to manage their product listings, track performance, and ensure quality control through the admin approval process.
- **Subscription Models:** By offering various subscription packages, the platform enables vendors to access different levels of features and services, catering to their specific needs and budgets.

3. Empowering Administrators:

- **Comprehensive Control:** The admin dashboard allows administrators to efficiently monitor and manage all aspects of the platform, from user interactions and vendor activities to order processing and analytics reporting.
- **Data-Driven Decisions:** With access to detailed analytics and reporting tools, administrators can make informed decisions to optimize platform performance and user satisfaction.

4. Market Differentiation:

- **Innovative Features:** The integration of AI-powered tools, such as the chatbot assistant and personalized recommendations, along with advanced 3D modeling capabilities, sets the platform apart from traditional e-commerce sites, offering a unique value proposition to users.

- **Seamless Integration:** The platform's ability to combine multiple functionalities, such as shopping, design visualization, and order tracking, into a single, cohesive user experience makes it a market leader in the home decor industry.

5. Economic Impact:

- **Boosting Local Businesses:** By providing a platform for local artisans and small businesses to list their unique home decor items, the project supports local economies and encourages entrepreneurship.
- **Job Creation:** The development and maintenance of the platform will create job opportunities in technology, customer support, marketing, and logistics.

6. Customer Engagement and Loyalty:

- **Loyalty Programs:** Implementing loyalty programs and targeted promotions helps build long-term relationships with customers, encouraging repeat purchases and enhancing brand loyalty.
- **User Feedback:** Continuous engagement through the chatbot assistant and other interactive features allows for the collection of valuable user feedback, which can be used to further improve the platform.

Potential Impact:

Empowering Creativity: By leveraging AI, this e-commerce home decoration project empowers people to unleash their creativity. It encourages experimentation, allowing users to envision their dream spaces and turn them into reality.

Business Opportunities: For e-commerce platforms, integrating AI-driven decor tools opens new revenue streams. Providing personalized recommendations and seamless shopping experiences can boost sales and customer satisfaction.

Sustainable Choices: AI can also promote sustainable design by suggesting eco-friendly materials and energy-efficient solutions. As environmental consciousness grows, this impact becomes increasingly relevant.

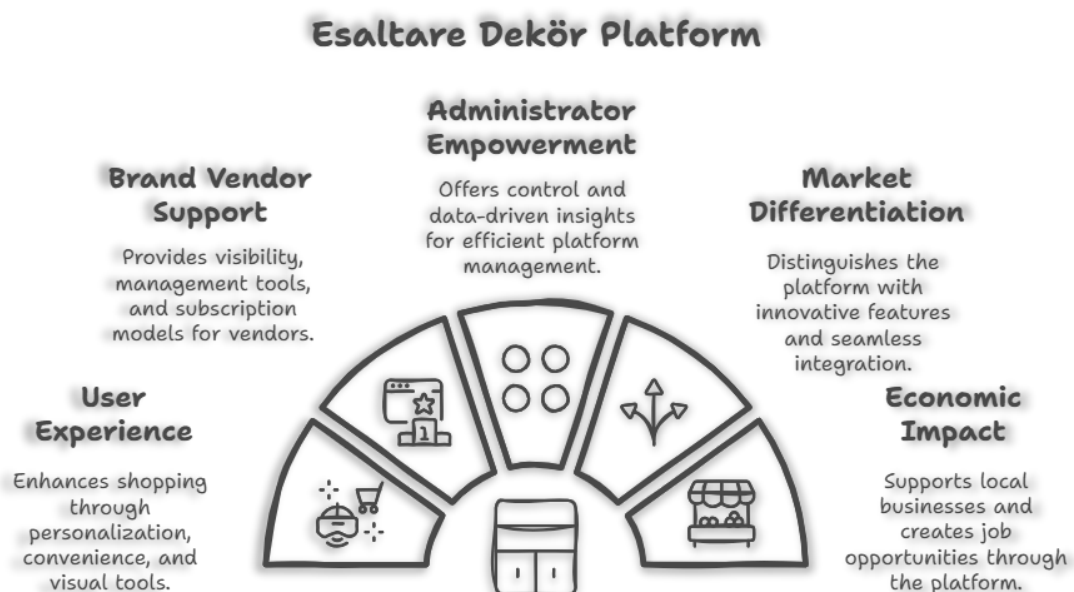


Figure 1-4 ED Significant

1.7 Methodology Overview

Methodology Overview

In this **Esaltare Dekör** e-commerce platform, we will implement a comprehensive and systematic approach to ensure effective sales and customer satisfaction. Below are the detailed methods we will use:

1. Market Research

- **Objective:** Understand current trends, customer needs, and the competitive landscape.
- **Actions:**
 - Analyze industry reports, surveys, and studies to identify popular styles and preferences.
 - Conduct focus groups and interviews with potential customers to gather qualitative insights.

2. Development of Survey Model

- **Objective:** Collect specific data from customers regarding their decoration preferences.
- **Actions:**
 - Design a detailed survey that includes questions about preferred styles, colors, materials, and budget.
 - Use online survey tools to distribute the survey and encourage participation through incentives.

3. Data Analysis

- **Objective:** Identify patterns and preferences from customer responses.
- **Actions:**
 - Use statistical analysis software to analyze survey data for trends and common preferences.
 - Customer segment based on their responses to tailor product offerings and recommendations.

4. Product Selection

- **Objective:** Curate a diverse range of products that align with customer preferences.
- **Actions:**
 - Collaborate with suppliers to source high-quality furniture and decor items and Create a catalog that shows various styles, ensuring a balance between modern, traditional, and eclectic designs.

5. Personalized Consultation

- **Objective:** Offer tailored decoration solutions for individual customers.
- **Actions:**
 - Analyze survey results to provide personal recommendations on furniture and decor.
 - Schedule one-on-one consultation sessions (virtual or in-person) to discuss design ideas and preferences.

6. Customer Service

- **Objective:** Maintain high levels of customer support throughout the purchasing process.
- **Actions:**
 - Establish a dedicated customer service team to handle inquiries and support requests promptly.
 - Create a comprehensive FAQ section on the website to address common questions.

7. Post-Sale Follow-Up

- **Objective:** Ensure customer satisfaction and gather feedback for improvement.
- **Actions:**
 - Reach out to customers after delivery to confirm satisfaction with their purchases.
 - Use follow-up surveys to gather feedback on both the products and the shopping experience.

8. Evaluation and Improvement

- **Objective:** Continuously enhance project performance based on data-driven insights.
- **Actions:**
 - Monitor sales data, customer feedback, and market trends to assess the effectiveness of strategies.
 - Regularly review and update the product selection and marketing approaches based on findings.

10. Technology Integration

- **Objective:** Leverage technology to enhance user experience and streamline operations.
- **Actions:**
 - Integrate AI-powered chatbots for customer assistance and personalized recommendations.
 - Implement 3D design tools to help customers visualize products in their own spaces.
 - Develop a mobile-friendly platform to ensure accessibility on all devices.

11. Marketing Strategy

- **Objective:** Increase brand awareness and drive traffic to the platform.
- **Actions:**
 - Use social media, email marketing, and search engine optimization (SEO) to reach a wider audience.
 - Partner with influencers and home decor bloggers to promote the platform.
 - Offer promotions and discounts to attract new customers and retain existing ones.

12. Vendor Management

- **Objective:** Build strong relationships with suppliers and ensure a diverse product offering.
- **Actions:**
 - Create a vendor portal for easy product listing and management.
 - Implement a subscription model with different tiers of access and benefits.
 - Conduct regular quality checks to maintain high standards.

1.8 Limitations

This project faces several constraints that may impact its development and functionality:

1. Time Constraints:

- **Developing and Fine-Tuning AI Models:** Building sophisticated AI models for personality analysis and space-specific recommendations is time-intensive. Limited time may restrict the depth of testing and optimization achievable within the project period, potentially affecting the accuracy and reliability of the AI functionalities.

2. Resource Constraints:

- **Budget Limitations:** Budget constraints can limit access to advanced tools, third-party APIs, and premium data sources essential for developing accurate AI recommendations and effective GPS tracking for products. This may influence the scope of features that can be implemented.
- **Access to Expertise:** High-quality development requires skilled professionals in AI, UX design, and data security. Budget limitations may impact the ability to hire or consult with the necessary experts.

3. Technological Challenges:

- **AI Algorithm Implementation:** Accurate interpretation of user personality and space requirements demands significant processing power and extensive data. Achieving efficiency and scalability in these algorithms may be challenging.

4. Data Privacy Concerns:

- **Sensitive User Information:** Collecting and managing sensitive user information, such as personality traits and location data, necessitating stringent data privacy and security measures. Ensuring compliance with global privacy regulations like GDPR and CCPA adds complexity to the development process.
- **User Trust:** Maintaining user trust through transparent data handling practices and robust security measures is crucial but challenging.

5. User Interface and Experience:

- **Balancing Sophistication and Usability:** Integrating advanced AI-backed features with a seamless and intuitive user interface is challenging. Ensuring a positive user experience while maintaining the sophistication of the system requires careful design and thorough user testing.
- **Design Constraints:** Time and resource limitations can restrict the extent of user interface customization and refinement, potentially impacting overall user satisfaction.

1.9 Assumptions

Technology Stack:

1. We assume a solid understanding of web development technologies like HTML, CSS, JavaScript, Bootstrap, React.js, and a backend framework like Laravel.
2. We assume familiarity with database systems like MySQL.
3. We assume knowledge of cloud platforms like AWS or Azure for deployment and scalability.
4. We assume a basic understanding of AI and machine learning concepts, and the ability to integrate AI services or libraries.

User Behavior and Expectations:

1. We assume users will be tech-savvy enough to navigate the platform and use its features effectively.
2. We assume users will be willing to share personal information, such as address and payment details, to complete transactions.
3. We assume users will be open to using AI-powered design tools and chatbots to enhance their shopping experience.

Data Privacy and Security:

1. We assume that robust security measures will be implemented to protect user data, including encryption, secure authentication, and regular security audits.
2. We assume compliance with relevant data privacy regulations such as GDPR and CCPA.

Product Sourcing and Inventory:

1. We assume a reliable and consistent supply of home decor products from various vendors.
2. We assume efficient inventory management and order fulfillment processes.

AI Model Performance:

1. We assume the AI model will be accurate and effective in generating design suggestions and product recommendations.
2. We assume the AI model will be continuously improved and updated based on user feedback and data analysis.

Chatbot Effectiveness:

1. We assume the chatbot will be able to handle a wide range of user queries and provide accurate and helpful responses.
2. We assume the chatbot will be able to learn and improve its responses over time.

User Experience:

1. We assume a user-friendly and intuitive interface that is easy to navigate.
2. We assume a responsive design that adapts to different screen sizes and devices.

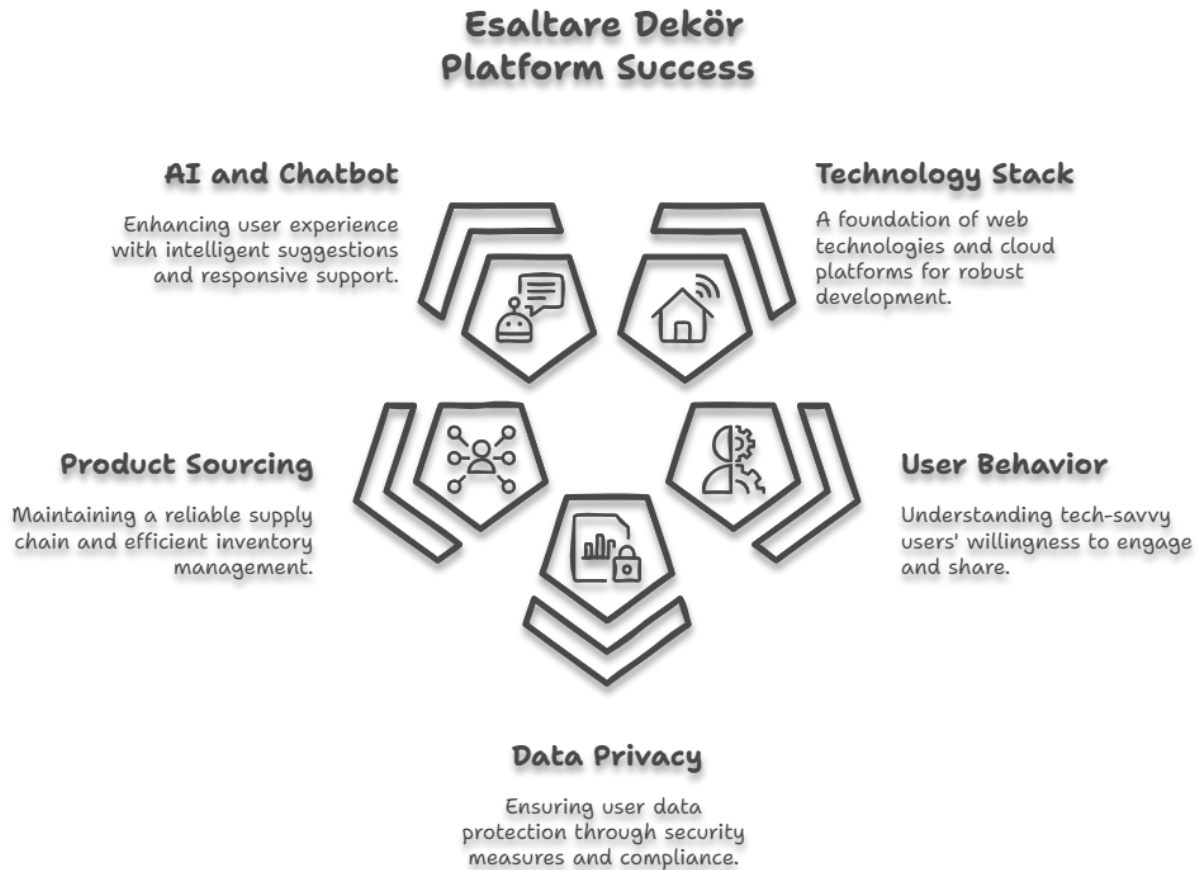


Figure 1-5 ED Assumptions Success

As we conclude our comprehensive introduction to the project, it's clear that a well-structured and innovative approach is essential to address the challenges and seize the opportunities in the home decor e-commerce landscape. However, the journey is just beginning. The next critical step involves meticulous planning and careful selection of technologies, resources, and strategies to transform this ambitious vision into reality.

In the next chapter we will delve into the heart of our project's foundation. This chapter will explore the strategic decisions and planning processes that are crucial for successful implementation. From choosing the right technology stack to defining project milestones, every element will be scrutinized to ensure alignment with our goals. Prepare to dive into a detailed roadmap that will guide us through each phase of development, setting the stage for a groundbreaking and user-centric platform.

Chapter 2: Planning and Selection

2.1 Problem Statement

In the rapidly evolving landscape of digital commerce, the demand for online shopping experiences has skyrocketed. The furniture and home decor market, traditionally reliant on physical showrooms, is now faced with the necessity to adapt to the digital realm. However, the transition to online platforms presents a myriad of challenges for vendors who seek to sell their products effectively. These challenges include high subscription costs, cumbersome product approval processes, and limited customization options. Concurrently, consumers are seeking comprehensive and engaging shopping experiences that offer convenience, personalization, and seamless transactions. The current gap in the market for a platform that addresses these needs underscores the necessity for a solution that can bridge this divide.

Context and Relevance

The furniture and home decor industry has been historically dominated by brick-and-mortar stores, where consumers can physically interact with products before making purchase decisions. With the advent of e-commerce, consumer behavior has shifted significantly towards online shopping, driven by the convenience of browsing and purchasing from the comfort of one's home. This shift is further accelerated by the increasing penetration of internet access and the proliferation of smart devices. Despite this trend, many vendors in the furniture and home decor sector face significant obstacles in establishing a robust online presence.

Furthermore, the integration of artificial intelligence (AI) into digital platforms represents a transformative shift in how businesses interact with consumers. AI technologies, such as chatbot assistants, enhance the efficiency and effectiveness of customer service, providing real-time support and personalized interactions. The inclusion of AI in e-commerce platforms can significantly elevate the user's experience, offering tailored recommendations and instant assistance, thus addressing many of the pain currently faced by both vendors and consumers.

Specific Challenges for Vendors

1.High Costs: One of the primary barriers for vendors is the high cost associated with subscription fees for popular e-commerce platforms. These platforms often charge substantial fees for basic access, with additional costs for premium features that are necessary to create a compelling online storefront. Small and medium-sized vendors find these costs prohibitive, limiting their ability to compete effectively in the digital marketplace. The financial burden of these fees can deter vendors from fully leveraging online platforms, resulting in missed opportunities for market expansion and revenue growth.

2.Cumbersome Approval Processes: The product approval process on many e-commerce platforms can be cumbersome and time-consuming. Vendors often face delays in getting their products listed, as they must navigate through complex and opaque approval procedures. These delays hinder vendors' ability to quickly introduce new products to the market, especially those that are trendy or seasonal. The lack of a streamlined approval process not only impacts on the vendors' agility but also their competitiveness, as they are unable to promptly respond to market demands and consumer preferences.

3.Limited Customization and Personalization Options: Customization is a critical aspect of brand differentiation in the crowded e-commerce space. However, many existing platforms offer limited tools for vendors to personalize their storefronts. This lack of customization options restricts vendors' ability to showcase their products in unique and engaging ways, making it difficult to stand out from competitors. Vendors need robust tools that allow them to create visually appealing and interactive online experiences that reflect their brand identity and attract consumers.

User Needs and Expectations

1. Comprehensive Shopping Experience: Modern consumers seek a comprehensive and seamless shopping experience that allows them to browse, compare, and purchase products with ease. The absence of a dedicated platform for furniture and home decor that meets these expectations creates a significant gap in the market. Consumers want a one-stop shop where they can find a wide variety of products, access detailed information, and make informed purchase decisions without having to navigate multiple websites.

2. Personalization and Customization: Personalization is a key driver of consumer satisfaction in e-commerce. Features such as 3D design tools that enable consumers to visualize furniture in their homes can significantly enhance the shopping experience. Consumers also appreciate personalized recommendations based on their preferences and browsing history. However, such advanced features are often lacking or inadequately implemented on many existing platforms, limiting the overall user experience.

3. Seamless Transactions: A smooth and hassle-free transaction process is crucial for converting browsing into purchases. Consumers expect intuitive interfaces, secure payment options, and efficient order management systems. Any shortcomings in these areas, such as complicated checkout processes or unreliable payment gateways, can lead to abandoned carts and negative perceptions of the platform. Ensuring a seamless transaction process is essential for building consumer trust and loyalty.

4. Responsive Customer Support: Customer support is a vital component of the e-commerce experience. Consumers expect quick and helpful responses to their queries and concerns. The integration of AI-powered chatbots and dedicated support teams can enhance customer support by providing instant assistance and resolving issues promptly. Effective customer support not only improves user satisfaction but also encourages repeat business.

5. 3D Design Tools: 3D design tools enable consumers to visualize how furniture will look and fit in their homes. By inputting room dimensions and other parameters, users can create virtual room layouts and experiment with different furniture arrangements. This feature helps consumers make informed purchasing decisions and enhances their engagement with the platform

The Role of Artificial Intelligence

The integration of artificial intelligence into the proposed e-commerce platform aims to address both the vendors' and consumers' needs by enhancing the overall functionality and user experience.

1. AI-Powered Chatbot Assistant: An AI-powered chatbot assistant can provide real-time support to both vendors and consumers, significantly enhancing the customer service experience. For vendors, the chatbot can help with the platform's features, provide guidance on product listings, and expedite the approval process by answering queries related to compliance and standards. For consumers, the chatbot can handle common inquiries, offer personalized product recommendations based on browsing history and preferences, assist with order tracking, and provide support throughout the purchase process. This real-time interaction ensures that both vendors and consumers receive timely and relevant assistance, improving overall satisfaction and efficiency.

2. Personalized Recommendations: AI algorithms can analyze consumer behavior and preferences to offer personalized product recommendations. By leveraging data on past purchases, browsing history, and user interactions, the platform can present consumers with products that align with their tastes and needs. This level of personalization not only enhances the shopping experience but also increases the likelihood of purchase, benefiting vendors by driving sales.

3. Enhanced Functionalities: AI can improve search functionalities by understanding natural language queries and providing more accurate and relevant results. This allows consumers to find products more easily, reducing the time and effort required to locate specific items. Advanced search capabilities, including visual and voice search, can further streamline the shopping experience, making it more intuitive and user-friendly.

Objectives and Goals

To address these challenges and meet the needs of both vendors and consumers, the proposed e-commerce platform aims to:

1. Provide Affordable Subscription Packages: Design tiered subscription packages that cater to vendors of all sizes, ensuring affordability without compromising on essential features. This approach will help vendors manage costs while accessing the tools they need to succeed online. By offering flexible pricing options, the platform can attract a diverse range of vendors, from small businesses to established brands.

2. Streamline Product Approval Processes: Implement efficient and transparent product approval processes that minimize delays and allow vendors to quickly list new products. This will enable vendors to stay agile and responsive to market trends. The platform should leverage automation and clear guidelines to expedite approvals, reducing the administrative burden on vendors and ensuring timely product listings.

3. Offer Robust Customization Tools: Develop advanced customization tools that allow vendors to create unique and engaging storefronts. Features such as customizable templates, multimedia support, and virtual showrooms will help vendors showcase their products

effectively. These tools should be user-friendly and flexible, allowing vendors to tailor their online presence to their specific needs and brand identity.

4. Enhance User Experience with Innovative Features: Integrate innovative features such as 3D design tools, personalized recommendations, and advanced search functionalities. These features will enhance the user experience, making it easier for consumers to find, visualize, and purchase products. The platform should prioritize user-centric design, ensuring that the interface is intuitive and accessible to a wide range of users.

5. Ensure Secure and Seamless Transactions: Implement secure and user-friendly payment gateways, along with features like saved payment methods and order tracking. These enhancements will ensure a smooth and trustworthy transaction process for users. The platform should also incorporate robust security measures to protect user data and transactions, fostering a safe and reliable shopping environment.

6. Provide Excellent Customer Support: Integrate AI-powered chatbots and a dedicated support team to offer responsive and helpful customer support. This will ensure that users' queries and issues are promptly addressed, enhancing overall satisfaction. The platform should provide multiple support channels, including live chat, email, and phone support, to cater to diverse user preferences.

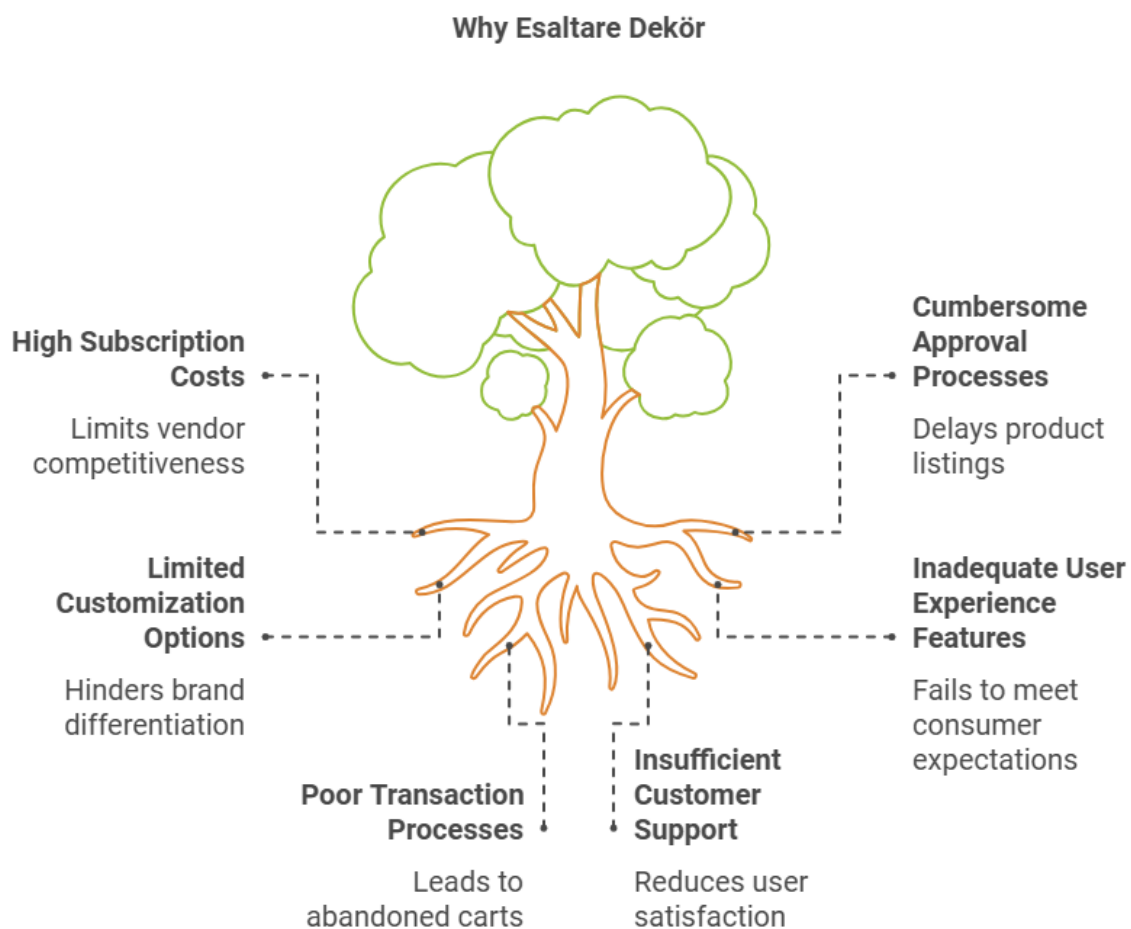


Figure 2-1 ED WHY?

2.2 Project Planning

The planning phase is integral to the success of any project. It serves as the blueprint that guides the entire process from initiation to completion. In this section, we will define the core components necessary for a well-structured and efficient e-commerce project plan. This includes identifying the project's scope, objectives, and requirements, as well as selecting the appropriate methodologies to ensure flexibility and control throughout the project lifecycle. To ensure successful delivery within the defined timelines and budget, we will explore various project management methodologies such as Agile, Waterfall, Scrum, and Kanban. Each methodology brings its own strengths, providing the necessary structure and adaptability based on the nature of the project. Furthermore, careful consideration will be given to risk identification, resource allocation, and the development of a comprehensive communication plan. By addressing these elements, we aim to create a strong foundation for the project, mitigating potential challenges and maximizing efficiency to meet the desired outcomes.

"In order to achieve the desired outcomes, careful planning is essential to anticipate challenges and define clear goals."

So, we will be starting by our planning of our project Esaltare Dekor planning let's get started:

1. Understanding the Basics:

In managing our e-commerce project, we're leveraging a blend of project management skills, tools, and techniques to ensure everything runs smoothly. Our primary goals are to complete the project successfully, on time, and within budget. By doing so, we aim to boost efficiency and elevate customer satisfaction.

2. Choosing Methodologies:

Selecting the right project management methodology is crucial. Here are the options we're considering:

- **Agile:** We're breaking the project into smaller phases, known as iterations, to focus on different elements. This approach gives us the flexibility to adapt as we progress.
- **Waterfall:** For clear structure, we might use the Waterfall method, where we sequentially complete phases like requirements, design, development, testing, and deployment.
- **Scrum:** Scrum appeals to us with its short development cycles (sprints) and regular feedback sessions, allowing us to stay agile.
- **Kanban:** By visualizing our work on a Kanban board, we can continuously improve and maintain a steady workflow.

3. Project Planning:

In our planning phase, we are:

- Defining the project's scope, objectives, and requirements to ensure everyone is aligned.
- Crafting a product roadmap to guide our development process.
- Identifying potential risks and challenges early on to mitigate them effectively.
- Allocating resources, including time, budget, and team members, to ensure we stay on track.
- Developing a robust communication plan to facilitate clear and consistent communication among stakeholders.

Additional Elements:

- **Stakeholder Analysis:** We're identifying and understanding the needs and influence of all stakeholders involved in the project. This will help us manage expectations and secure their buy-in throughout the project lifecycle.
- **Risk Management Plan:** We are formulating a comprehensive plan to identify, assess, and mitigate potential risks. This includes creating contingency plans for various scenarios.
- **Quality Assurance:** Establishing quality standards and procedures is vital. We're committed to regular testing, reviews, and feedback loops to ensure our deliverables meet or exceed expectations.
- **Timeline and Milestones:** By creating a detailed timeline with key milestones, we can track our progress and maintain accountability.
- **Documentation:** Maintaining thorough documentation of all project aspects, from requirements to design processes, ensures clarity and provides a valuable reference for future projects.
- **Training and Support:** We're planning training sessions and support resources for our team and stakeholders to ensure everyone is well-equipped to contribute to the project's success.
-

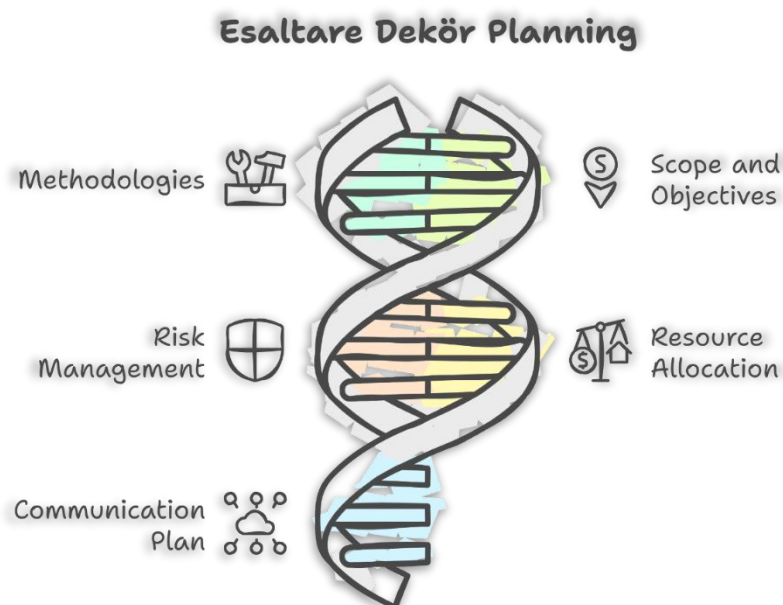


Figure 2-2 ED Planning

2.3 Project Milestones

Milestone 1: Project Planning

- **Objective:** Establish a foundational framework and define the scope, objectives, and deliverables.
- **Key Tasks:**
 - Articulate the project's objectives and scope, defining its functional (e-commerce platform, vendor subscriptions, admin approval system, AI chatbot, etc.) and non-functional requirements.
 - Develop a comprehensive project timeline, identifying milestones and deliverable due dates.
 - Determine the technical stack, frameworks, and tools (such as React for frontend, Node.js for backend, AI technologies for chatbot) that align with project requirements.
 - Formulate a team structure, outlining roles, responsibilities, and accountability for each member.
- **Expected Outcomes:**
 - A finalized project plan, including defined objectives and scope.
 - Timeline with milestones for efficient tracking.
 - Resource allocation plan detailing team roles and responsibilities.

Milestone 2: Analysis

- **Objective:** Perform an in-depth analysis to refine the project's technical and functional specifications.
- **Key Tasks:**
 - Conduct a granular analysis of all features, focusing on functionality (e.g., vendor product listings, user shopping cart), user interaction, and technical feasibility.
 - Define a robust system architecture, covering backend, frontend, AI integration for chatbot, and GPS tracking for shipments.
 - Identify potential risks (e.g., data security, system scalability) and formulate mitigation strategies.
- **Expected Outcomes:**
 - Detailed analysis documentation, specifying requirements and features.
 - Architectural blueprint that aligns with project requirements.
 - Risk assessment report to ensure proactive issue management.

Milestone 3: UI/UX Design

- **Objective:** Design an intuitive, user-centered interface with a seamless user experience.
- **Key Tasks:**
 - Develop UI designs for each core feature, emphasizing usability and aesthetic appeal.
 - Design UX flows to ensure intuitive navigation and optimized user journeys.
 - Refine prototypes based on stakeholder feedback to align with user expectations.

- **Expected Outcomes:**

- Detailed wireframes for primary screens (e.g., vendor dashboards, admin approval interface, user shopping experience).
- High-fidelity prototypes with documented UX flows.
- Approved design assets ready for frontend implementation.

Milestone 4: Frontend Development

- **Objective:** Implement responsive and interactive frontend components for a cohesive user experience.
- **Key Tasks:**
 - Construct the app's frontend using HTML, CSS, JavaScript, and React.
 - Ensure design responsiveness across all devices and platforms.
 - Integrate frontend interfaces with backend APIs for dynamic data retrieval and updates.
- **Expected Outcomes:**
 - Interactive, responsive frontend aligned with UI/UX design specifications.
 - Seamless integration with backend services.

Milestone 5: Backend Development

- **Objective:** Build a secure and scalable backend to manage data, services, and user transactions.
- **Key Tasks:**
 - Design and configure a database schema to manage user, vendor, and product data efficiently.
 - Develop robust API endpoints to support data flow and functionality for each app feature.
 - Implement backend security measures to safeguard user data and ensure data integrity.
- **Expected Outcomes:**
 - Fully operational database schema and APIs.
 - Scalable and secure backend environment.

Milestone 6: AI Model & Chatbot Development

- **Objective:** Implement AI-driven personalization features and integrate a conversational chatbot to enhance user engagement and recommendations.
- **Key Tasks:**
 - Develop and train AI models to analyze user preferences and provide personalized decor recommendations.
 - Create recommendation algorithms based on apartment layout and decor preferences.
 - Develop and integrate a conversational chatbot to guide users and provide recommendations based on AI model outputs.
 - Test AI models and chatbot for accuracy, performance, and seamless user interaction.
 - Implement AI-driven features within the chatbot to provide personalized responses and support.

- **Expected Outcomes:**
 - Deployed AI models for personalized recommendations.
 - Fully functional chatbot integrated with AI capabilities for enhanced user support.
 - AI-powered recommendation system and chatbot functionalities seamlessly integrated within the application.

Milestone 7: System Integration & Testing

- **Objective:** Ensure system cohesion and optimal performance through comprehensive integration and testing.
- **Key Tasks:**
 - Integrate frontend, backend, and AI components into a unified and operational system.
 - Conduct end-to-end testing, focusing on system performance, stability, and user experience.
 - Perform user acceptance testing (UAT) to verify that all functionalities meet project requirements.
- **Expected Outcomes:**
 - A fully integrated system with all components working seamlessly.
 - Comprehensive test reports and identified resolutions for issues.

Milestone 8: Launch Preparation & Marketing

- **Objective:** Prepare for market entry and promote the app to the target audience.
- **Key Tasks:**
 - Finalize deployment preparations and initiate the release process across chosen platforms.
 - Develop a marketing strategy to generate user interest and drive initial adoption.
 - Set up a customer support system to handle post-launch queries and technical support.
- **Expected Outcomes:**
 - App deployed and accessible to users.
 - Marketing plan in place with an active user acquisition strategy.
 - Customer support protocol established.



Figure 2-3 ED Milestones

2.4 Resource Allocation

To effectively allocate resources for our **Esaltare Dekör** platform project, which includes selling furniture, tracking shipments, AI-driven chatbots, and creating a comprehensive admin system, we can divide the resources into several key categories:

1. Human Resources

Staff:

- **Product Managers:** Oversee the overall project, coordinate between teams, and ensure alignment with business goals.
- **Interior Designers:** Curate furniture and decor collections, provide design consultations.
- **Sales Team:** Engage in selling furniture, interact with clients, and manage vendor relations.
- **Logistics Team:** Manage the shipment tracking and delivery process.
- **Technical Support Team:** Assist in managing the online platform and handling technical issues.
- **AI Specialists:** Develop and maintain AI models and chatbots for personalized recommendations and customer support.

Training:

- Provide training sessions for staff on the latest design trends, effective technology use, and customer service skills.
- Conduct workshops for technical teams on new tools, AI integration, and security protocols.

2. Financial Resources

Budget Planning:

- Establish the overall budget for the project, including costs for platform development, furniture procurement, salaries, marketing, and technological systems.
- Allocate funds for AI model development, chatbot integration, and marketing campaigns.

Cost Analysis:

- Conduct periodic cost analyses to ensure budget adherence.
- Review supplier offers to secure the best prices for furniture and technology solutions.
- Monitor ongoing expenses and adjust the budget as necessary to accommodate unexpected costs.

3. Material Resources

Furniture Inventory:

- Identify the furniture to be sold, ensure its availability in stock, and maintain relationships with reliable vendors.
- Implement an inventory management system to minimize excess stock and avoid product shortages.

Tools and Materials:

- Provide necessary tools for designers, such as design software (e.g., SketchUp, AutoCAD) and high-performance computers.
- Equip the technical team with development tools, servers, and testing environments.

4. Technological Resources

Platform Development:

- Develop a robust and scalable e-commerce platform using modern frameworks and technologies (e.g., React, Node.js, AI tools).
- Ensure the platform supports vendor management, product listings, user accounts, and transaction processing.

AI Integration:

- Develop and integrate AI-driven features such as personalized recommendations, chatbots for customer support, and predictive analytics.
- Regularly update and maintain AI models to ensure accuracy and relevance.

Shipment Tracking System:

- Implement an effective shipment tracking system that allows clients to easily monitor their shipments.
- Ensure the system is regularly updated to avoid delivery issues and enhance user experience.

Website and Application:

- Develop a user-friendly website and mobile application showcasing the furniture and services offered.
- Ensure the platform is mobile-responsive and optimized for various devices.

5. Time Management

Project Scheduling:

- Establish a clear timeline for each phase of the project, from initial design to final launch.
- Set deadlines for key milestones and regularly monitor progress to ensure timely completion.

Task Prioritization:

- Identify the most critical tasks that need to be completed first, such as platform development, AI integration, and vendor onboarding.
- Schedule regular meetings with the team to review progress, address challenges, and ensure deadlines are met.

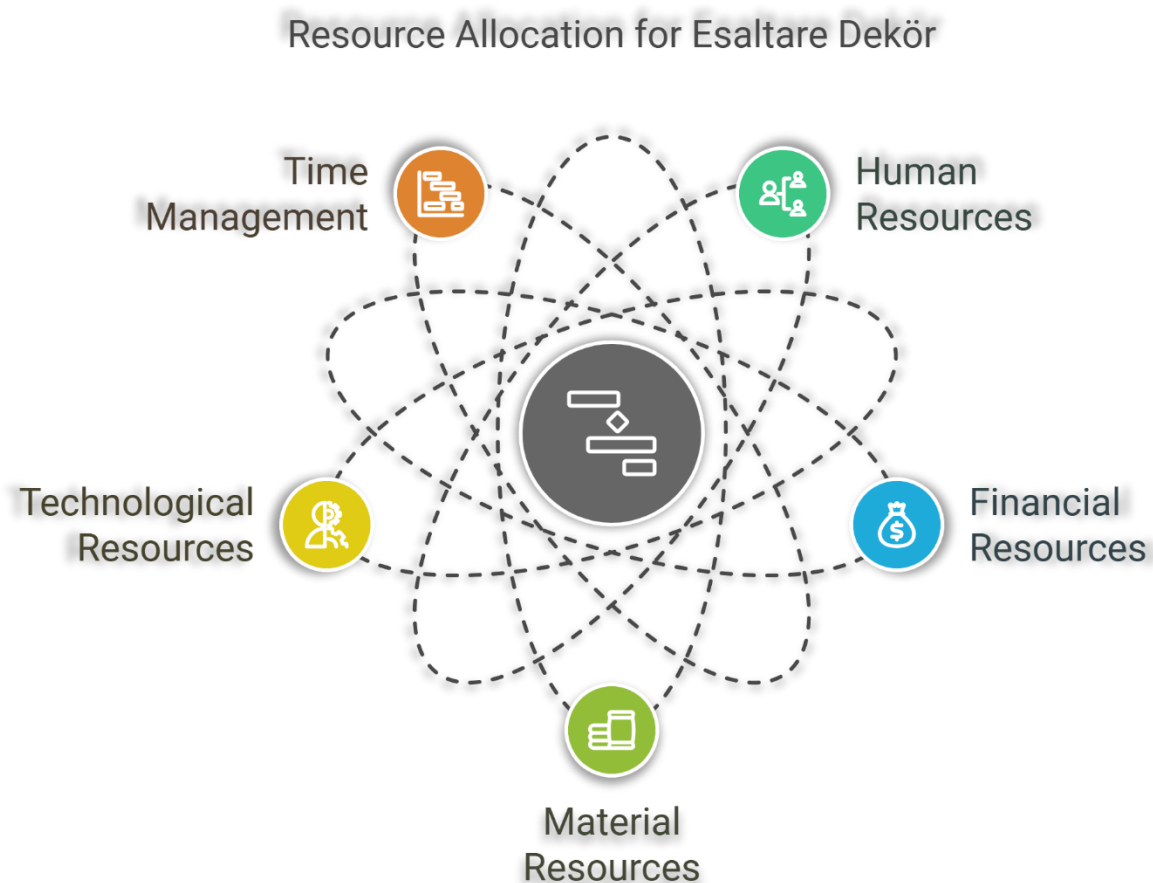


Figure 2-4 ED Resource Allocation

2.5 Risk Management

In the context of our **Esaltare Dekör** platform project, risk management involves identifying, assessing, and mitigating potential risks that could impact the project's success. Here are some key aspects tailored to your project:

Risk Identification

At the outset of the project, project managers and their teams proactively identify potential risks. These risks can be anything that might affect the project's timeline, performance, or budget. Specific to e-commerce projects, these risks include:

- **Operational Risks:** Potential issues in managing day-to-day operations, such as delays in product approvals, logistical challenges, and platform downtime.
- **Financial Risks:** Pricing challenges, unexpected costs, currency fluctuations, and tax compliance that could affect the project's budget.
- **Technological Risks:** Risks related to technology, such as system failures, integration challenges, and data breaches.
- **Legal and Regulatory Risks:** Compliance with e-commerce legal frameworks, data protection laws, and other regulations.
- **Reputational Risks:** Negative reviews, chargebacks, and changing consumer expectations affecting the platform's reputation.

Tools like risk registers, brainstorming sessions, and historical data analysis help identify risks.

Specific Risks:

1. **Consumer Behavior and Reputation Management:**
 - Negative reviews, chargebacks, and changing consumer expectations can affect your e-commerce business.
 - Develop strategies to handle these issues proactively, such as robust customer support and reputation management practices.
2. **Financial Management:**
 - Pricing challenges, unexpected costs, currency fluctuations, and tax compliance are financial risks.
 - Plan ahead and allocate resources wisely to manage these risks effectively.
3. **Supply Chain and Vendor Management:**
 - Build strong relationships with vendors and have contingency plans for supply chain disruptions.
 - Ensure vendors meet quality and delivery standards consistently.
4. **Technology and Data Security:**
 - Protect against credit card fraud, phishing attacks, and ensure secure payment gateways.
 - Regular software updates, employee training, and robust cybersecurity measures are crucial.
5. **Legal and Regulatory Compliance:**
 - Understand e-commerce legal frameworks and comply with data protection laws.
 - Develop clear privacy policies and educate customers in security practices.
6. **Privacy Policies and Data Security:**

- Develop clear privacy policies and educate customers on security practices.
- Implement measures to safeguard customer data and ensure compliance with relevant laws.
- 7. **Payment Processing and Financial Transactions:**
 - Offer clear payment choices and mitigate payment processing risks.
 - Ensure secure and reliable payment gateways.
- 8. **Crisis Management and Recovery Planning:**
 - Prepare for crises and have business continuity strategies in place.
 - Develop a recovery plan to ensure quick resumption of operations after a disruption.

Risk Assessment

Once identified, these risks are assessed. Project managers evaluate their likelihood of occurring and their potential impact. Some risks are minor blips on the radar, while others could be major hurdles.

Risk Response

After assessing the risks, the project team develops strategies to respond to them. These strategies can fall into several categories:

- **Avoidance:** If possible, avoid the risk altogether. For instance, if there's a risk related to using a certain technology, consider an alternative.
- **Mitigation:** Implement measures to reduce the impact of the risk, such as improving security protocols to mitigate data breaches.
- **Transfer:** Shift the risk to someone else. Insurance is a classic example of risk transfer.
- **Acceptance:** Sometimes, you just must accept a certain level of risk. But even then, you'll have a plan for dealing with it.

By proactively managing these risks, you can ensure the smooth execution of our **Esaltare Dekör** platform project and achieve your desired goals.

2.6 Timeline and Schedule

Project Timeline:

Over the first four weeks, we will gather comprehensive project requirements and translate them into detailed user stories, along with developing initial design mockups. From weeks five to eight, we will focus on frontend development using React Js, building interactive and responsive user interface components. During weeks nine to twelve, our attention will turn to backend development, where we will set up the Laravel framework, develop robust API endpoints, and integrate the database schema. In weeks thirteen to fifteen, we will integrate AI into the platform, developing and training AI models for personalized recommendations and an AI-driven chatbot for enhanced customer support. Weeks sixteen and seventeen will be dedicated to rigorous testing to identify and fix bugs, optimize the platform, and ensure a seamless user experience. Finally, in week eighteen, we will prepare comprehensive documentation, including technical and user guides, and develop a presentation showcasing the project's features and achievements. We'll also apply final touches and prepare for the launch. This structured approach ensures all key activities are completed efficiently, setting us up for a successful project execution.

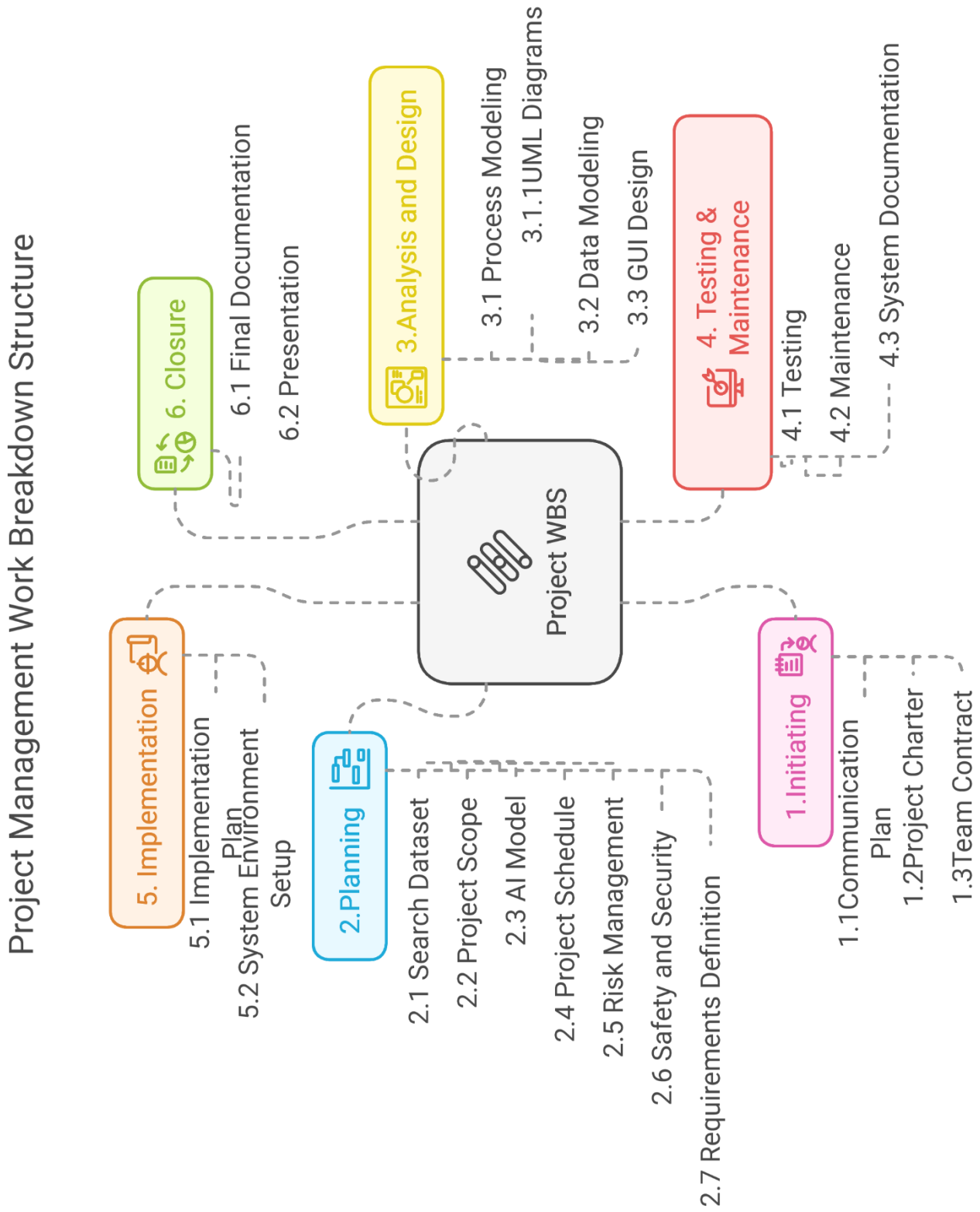
- **Week 1-4: Requirements Gathering, User Stories, and Initial Design**
 - Collect and document project requirements.
 - Create detailed user stories to capture end-user needs.
 - Develop initial design mockups for key features.
- **Week 5-8: Frontend Development (React Js)**
 - Set up the project environment and architecture.
 - Build the user interface components using React Js.
 - Ensure responsive design for various devices.
- **Week 9-12: Backend Development (Laravel Setup, API Endpoints)**
 - Configure the Laravel framework for backend development.
 - Develop API endpoints to support frontend interactions.
 - Implement database schema and integrate it with the backend.
- **Week 13-15: AI Integration (AI Model)**
 - Develop and train AI models for personalized recommendations.
 - Integrate AI features within the platform, including the chatbot.
 - Test and optimize AI models for accuracy and performance.
- **Week 16-17: Testing, Bug Fixes, and Optimization**
 - Conduct thorough testing of the platform, including unit, integration, and end-to-end tests.
 - Identify and fix bugs to ensure stability and performance.
 - Optimize the platform for a seamless user experience.
- **Week 18: Documentation, Presentation, and Final Touches**
 - Prepare comprehensive project documentation, including technical and user guides.
 - Develop a presentation to showcase the project's features and achievements.
 - Apply final touches and prepare for launch.

Project Schedule

Name	Activity	Effort (days)	Start date	End date
A	Initiating Communication plan and Business case	9	3/10/2024	12/10/2024
B	Project Charter	7	13/10/2024	20/10/2024
C	Team Contract	6	21/10/2024	27/10/2024
D	Search dataset	12	28/10/2024	10/11/2024
E	Determine Project Scope	13	11/11/2024	24/11/2024
F	The AI Model	45	25/11/2024	8/1/2025
G	WBS	2	9/1/2025	11/1/2025
H	Project Schedule	7	12/1/2025	19/1/2025
I	List of Prioritized Risks	6	20/1/2025	26/1/2025
J	How to achieve safety and security	5	27/1/2025	1/2/2025
K	Defining Requirement	6	2/2/2025	8/2/2025
L	Process Modeling	12	9/2/2025	21/2/2025
M	UML Diagrams	12	22/2/2025	5/3/2025
N	Data Modeling ERD	2	6/3/2025	8/3/2025
O	GUI Design	11	9/3/2025	20/3/2025
P	Implementation plan and Establish system environment	5	21/3/2025	26/3/2025
Q	Implementation	21	27/3/2025	16/4/2025
R	Testing and Maintenance	8	17/4/2025	25/4/2025
S	System Documentation (Q)	6	26/4/2025	2/5/2025
T	Presentation (R)	3	3/5/2025	6/5/2025

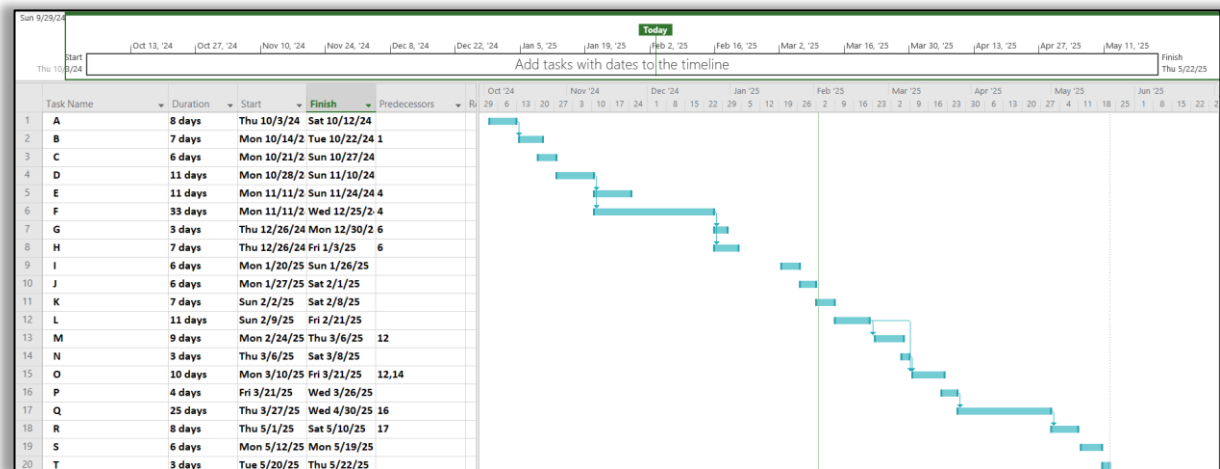
The project schedule spans from October 3, 2024, to April 25, 2025. Key activities include initiating the communication plan and business case, drafting the project charter, and forming the team contract in the initial weeks. Following this, we focus on dataset searches, defining project scope, and developing work breakdown structures. Subsequent tasks involve scheduling, risk prioritization, and establishing safety protocols. Core development phases, including process modeling, UML diagrams, and GUI design, occur from late December to early February. Implementation and testing take place from mid-February to mid-April, concluding with system documentation and a final presentation by late April.

2.7 Work Break Down Structure

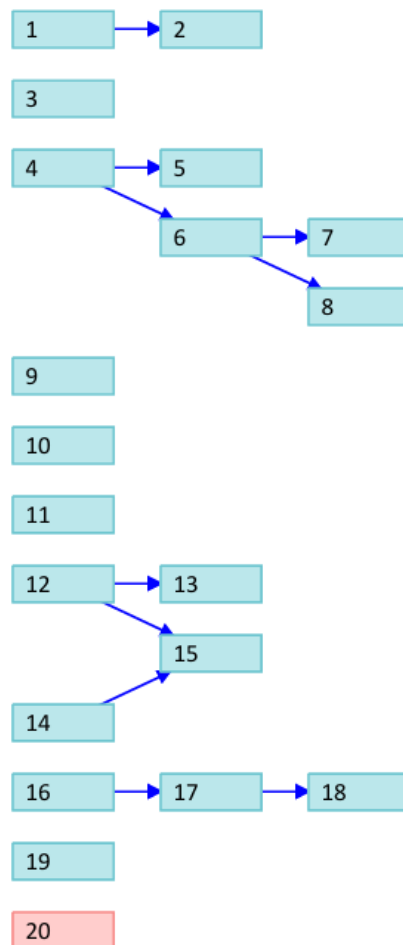


2.8 Gannt Chart and Network Diagram

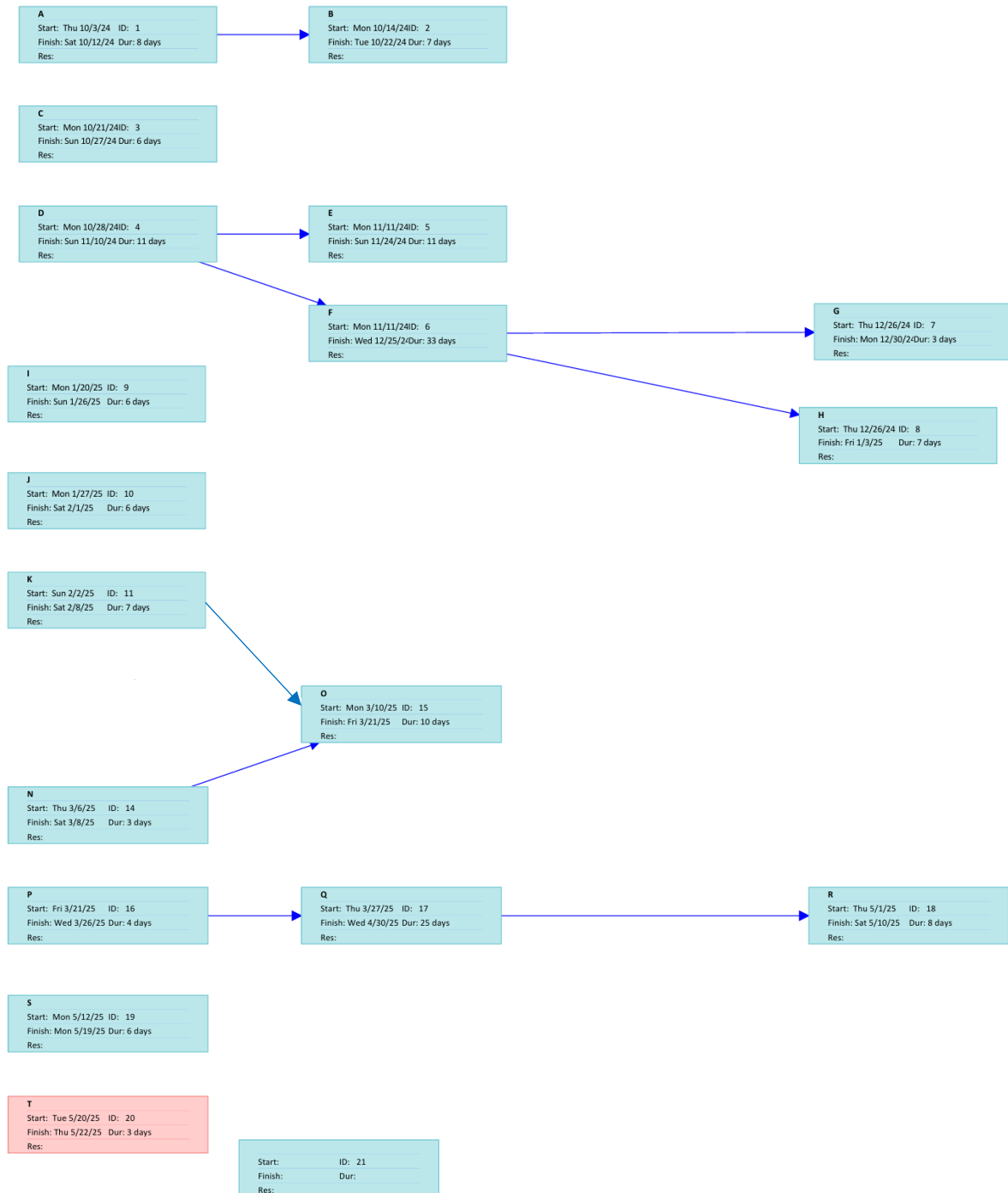
2.8.1 Gannt Chart



2.8.2.1 Network Diagram Summary



2.8.2.2 Network Diagram



2.9 Communication Plan

Communication Plan

The communication plan aims to ensure that all stakeholders are kept informed about the project's progress and updates. Here's a detailed plan for your e-commerce platform project focused on selling furniture, tracking shipments, and integrating AI-driven chatbots:

1. Objectives of the Communication Plan:

- Regularly update stakeholders on project progress.
- Facilitate effective communication among all parties involved.
- Address any issues or inquiries promptly.
- Enhance collaboration and engagement with all stakeholders.

2. Identifying Stakeholders:

Internal Stakeholders:

- Product Managers
- Interior Designers
- Sales Team
- Logistics Team
- Technical Support Team
- AI Specialists
- Senior Management

External Stakeholders:

- Vendors
- Clients
- Suppliers
- Contractors
- Investors

3. Communication Methods:

Meetings:

- **Weekly Team Meetings:** Discuss project progress, challenges, and upcoming tasks (Duration: 1 hour).
- **Monthly Stakeholder Meetings:** Provide a comprehensive project update and gather feedback (Duration: 1.5 hours).

Reports:

- **Weekly Progress Reports:** Summary of completed tasks, current status, and next steps. Sent via email.
- **Monthly Status Reports:** Detailed reports highlighting achievements, budget status, and any issues. Sent to all stakeholders.

Emails:

- Immediate updates regarding important milestones or changes in the project plan.
- Notifications about any critical issues requiring stakeholder input.

Collaboration Tools:

- **Project Management Software (like Trello or Asana):** For task tracking, deadlines, and progress.
- **Instant Messaging Platforms (like Slack or Microsoft Teams):** For quick information exchange among team members.

Visual Updates:

- **Progress Dashboards:** Visual representation of project status, accessible to all stakeholders via the project management platform.
- **Design Model Presentations:** Showcasing proposed designs to clients through online galleries.

4. Communication Frequency:

- **Daily Updates:** Through instant messaging for any immediate concerns or updates.
- **Weekly Meetings:** To discuss project progress and coordinate efforts.
- **Monthly Reports:** To provide comprehensive updates to all stakeholders.

5. Feedback Mechanism:

- Encourage stakeholders to provide feedback during meetings and through reports.
- Use surveys or feedback forms after each major phase to gather client and stakeholder opinions.

6. Documentation:

- Maintain a central repository for all communications and documentation related to the project, including:
 - Meeting minutes
 - Reports
 - Emails
 - Feedback forms

7. Responsibilities:

- **Project Manager:** Responsible for implementing the communication plan, preparing reports, and ensuring ongoing updates.
- **Team Leaders:** Responsible for gathering information and communicating with their teams.
- **Team Members:** Responsible for providing updates and information regularly.

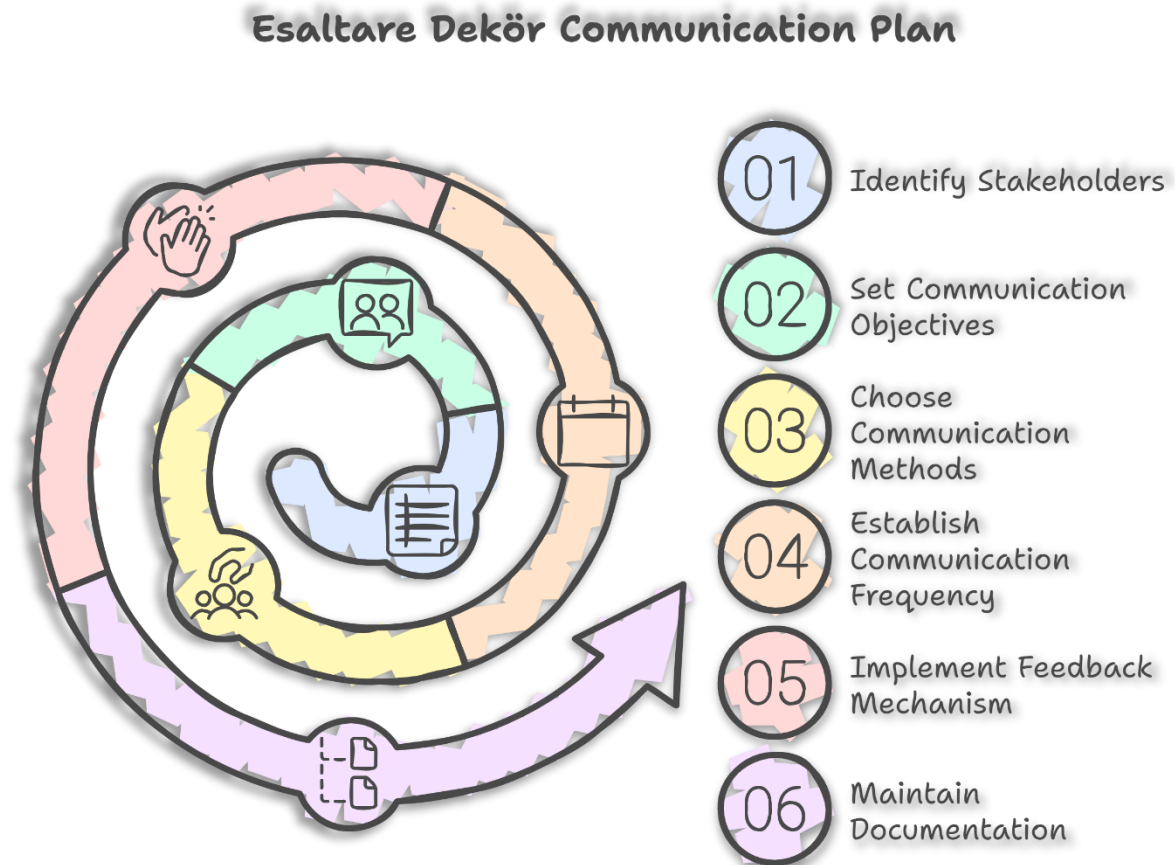


Figure 2-5 ED Communication Plane

2.10 Business Plan For Esaltare Dekör

WHAT + HOW + WHO	WHAT do we do?	We're creating a revolutionary home decoration website that leverages the power of AI to provide personalized recommendations for our users. Our platform will showcase a diverse range of home decor products, from furniture and lighting to textiles and accessories. By utilizing AI algorithms, we'll analyze user preferences, room dimensions, and existing decor to offer tailored suggestions that transform their living spaces.
	HOW do we do it?	• AI-powered recommendations: Our platform will employ sophisticated AI algorithms to analyse user data and provide personalized product recommendations based on their individual tastes and needs. • Curated product selection: We'll carefully curate a collection of high-quality home decor products from trusted suppliers, ensuring variety, style, and affordability. • User-friendly interface: Our website will feature an intuitive and visually appealing interface, making it easy for users to browse products, explore different styles, and save their favourites. • Virtual design tools: We'll offer virtual design tools that allow users to visualize how different products would look in their own homes, helping them make informed decisions.
	WHO do we serve?	• Homeowners looking to update or decorate their living spaces. • Interior designers seeking inspiration and product recommendations. • Renters wanting to personalize their rental properties. • Individuals interested in creating stylish and functional home environments.
	WHY DEFINE CUSTOMER PROBLEM	• Overwhelm by choice: The vast array of home decor products available can be overwhelming and confusing. • Lack of design expertise: Many people don't have the skills or knowledge to create a cohesive and stylish home environment. • Time constraints: Busy lifestyles can make it difficult to find the time to shop for and decorate a home. • Budget limitations: Home decor products can be expensive, making it difficult to achieve a desired look without breaking the bank.

DEFINE SOLUTION PROVIDED	• Providing personalized recommendations: Our AI-powered platform analyses user preferences and offers tailored product suggestions, simplifying the decision-making process. • Offering expert guidance: We provide resources and tools to help users create stylish and functional spaces, even if they lack design experience. • Saving time: Our platform allows users to browse products and explore different design options conveniently online, saving them time and effort. • Offering affordable options: We curate a selection of high-quality home decor products at competitive prices to cater to various budgets.
REVENUE PRICING + BILLING STRATEGIES	Pricing Model: • Subscription-based: Offer monthly or annual subscription plans to provide continuous access to our platform and personalized recommendations. • Freemium model: Provide a basic level of service for free, with premium features available for a subscription fee. • Per-product pricing: Charge a fee for each individual product purchased through the platform. • Combination model: Combine subscription fees with additional charges for premium services or custom design consultations. Pricing Tiers: • Basic: Offer essential features and a limited number of recommendations. • Premium: Provide advanced features, such as virtual design tools, personalized styling consultations, and exclusive access to curated collections. • Enterprise: Cater to businesses and professionals with specialized needs, offering bulk discounts and custom solutions.

MARKETING	INCOME STREAMS	Billing and Payment: • Secure payment gateways: Integrate with reputable payment processors to ensure secure transactions. • Flexible payment options: Offer various payment methods, including credit cards, debit cards, PayPal, and digital wallets. • Subscription management: Implement a user-friendly subscription management system that allows customers to easily upgrade, downgrade, or cancel their plans.
		Subscription fees: Generate revenue from monthly or annual subscription plans. Product sales: Earn a commission on products sold through the platform. Premium features: Offer additional features or services at a premium price. Partnerships: Collaborate with brands and suppliers for sponsored content, affiliate marketing, or exclusive product offerings. Advertising: Display targeted ads to users based on their preferences and browsing behaviour. Custom design services: Provide personalized design consultations or services for an additional fee.
	CUSTOMER REACH STRATEGY	Marketing Channels: • Content marketing: Create high-quality blog posts, articles, and social media content related to home decor trends, tips, and inspiration. • Social media: Utilize platforms like Instagram, Pinterest, and Facebook to showcase products, engage with customers, and run targeted advertising campaigns. • Search engine optimization (SEO): Optimize the website for relevant keywords to improve search engine rankings and attract organic traffic. • Email marketing: Build an email list and send targeted campaigns to promote new products, sales, and exclusive offers. • Influencer marketing: Collaborate with home decor influencers and bloggers to reach a wider audience and generate buzz.

COMPETITION		• Public relations: Seek media coverage and press mentions to increase brand awareness and credibility. Customer Acquisition Tactics: • Paid advertising: Use platforms like Google Ads and social media advertising to target specific demographics and interests. • Partnerships: Collaborate with complementary businesses, such as furniture stores, paint companies, and interior design firms. • Referral programs: Incentivize customers to refer friends and family to the platform. • Contests and giveaways: Run contests and giveaways to attract new customers and generate excitement.
	REFERRAL GENERATION STRATEGY	Referral program: Offer incentives to customers who refer new users to the platform. This could include discounts, free products, or exclusive access to limited-edition items. Shareable content: Create shareable content, such as social media posts, blog articles, or infographics, that encourages customers to share the platform with their friends and family. Loyalty program: Implement a loyalty program that rewards repeat customers with points or discounts that can be redeemed for products or services. Partnerships: Collaborate with other businesses that share a similar target audience to cross-promote each other's products or services. Customer satisfaction: Provide exceptional customer service to encourage satisfied customers to refer the platform to others.
	TOP COMPETITORS	Home mate, Home Centre, Maison’s du Monde, Home Box
	OUR COMPETITIVE ADVANTAGE	Home mate (Egypt): • Wide product range: Offers a diverse selection of home decor items to cater to various tastes and styles. • Local focus: Specializes in Egyptian and Middle Eastern home decor trends and products. • Easy-to-use platform: Provides a user-friendly interface for browsing and shopping.

	Home centre (Egypt): • Competitive pricing: Offers affordable home decor products without compromising on quality. • Strong brand reputation: Well-established in the Egyptian market with a reputation for reliability and customer satisfaction. • Wide store network: Has a physical presence in multiple locations across Egypt, making it convenient for customers to shop both online and in-store. Maison’s du Monde (Egypt): • Unique and stylish products: Offers a curated selection of trendy and unique home decor items. • International brand: Benefits from the global reach and expertise of the Maison’s du Monde brand. • High-quality materials: Uses premium materials and craftsmanship in its products.
METRICS SUCCESS MILESTONE MARKER 1	Success Milestone 1: Platform Launch and AI Integration • Goal: Successfully launch the home decoration platform with a functioning product catalog and AI recommendation engine. • Key Tasks: ◦ Develop and deploy the website with a catalog of curated decoration products. ◦ Implement the AI recommendation system to offer personalized suggestions based on user inputs. ◦ Ensure a seamless user interface (UI) and user experience (UX) to enable easy browsing and purchasing. ◦ Create a marketing campaign to attract initial users and partners. • Success Indicator: Achieving the first 1,000 active users on the platform within the first three months of launch.
SUCCESS MILESTONE MARKER 2	Success Milestone 2: Customer Engagement and Product Expansion • Goal: Increase customer engagement and expand the product offering based on user feedback. • Key Tasks:

- Add new products and categories to the catalog based on market demand and user behaviour.
- Improve AI algorithms for more accurate and personalized recommendations.
- Engage users with personalized communication, including email recommendations, product suggestions, and discounts.
- Develop partnerships with designers and décor brands to expand product listings.
- Success Indicator: Increase user retention rate by 20% and expand the catalog by 50% within six months of launch.

INTERNAL FACTORS

STRENGTHS (+)

Innovative AI Feature: Unique selling point that offers personalized decoration recommendations based on users' preferences, traits, and place details, setting the platform apart from traditional home décor websites.

Personalization: High level of customization and personalization, improving user satisfaction and engagement.

Diverse Product Catalog: A curated range of products from multiple brands, giving customers a wide variety of choices in one place.

User-Friendly Platform: An easy-to-navigate interface with integrated AI to simplify the shopping and decision-making process.

Scalability: The platform can easily expand to include more products, features, and services, offering growth opportunities.

WEAKNESSES (–)

High Development Costs: Building an AI-driven platform requires significant investment in technology, which could strain initial budgets.

Complex AI Implementation: Creating an accurate AI recommendation system may be complex, and poor initial suggestions could lead to user dissatisfaction.

Dependency on AI: Over-reliance on AI recommendations might alienate users who prefer a more hands-on approach to decoration.

limited Brand Awareness: As a new entrant, the brand lacks recognition and may take time to build a trusted reputation in the market.

Niche Appeal: The AI-driven approach may cater to a niche market, limiting its appeal to a broader audience.

EXTERNAL FACTORS

OPPORTUNITIES (+)

Growing Demand for Home Décor: The home décor market is expanding, particularly with more people investing in their homes, offering a ripe opportunity for a personalized solution.

Partnerships with Designers and Brands: Potential to form strategic partnerships with furniture and décor brands, interior designers, and influencers to enhance credibility and expand offerings.

AI Evolution: Advancing AI technology allows for continuous improvement in recommendations, user experience, and features, creating a competitive edge.

Expansion into B2B: The platform could target businesses looking to furnish office spaces, providing growth opportunities in the commercial sector.

International Growth: Once established, the platform can scale globally, tapping into international home décor markets and preferences.

THREATS (–)

Competition: High competition from established home décor websites, interior design apps, and e-commerce giants like Amazon, Wayfair, and IKEA.

Technological Challenges: Potential for technical issues or AI inaccuracies could negatively affect user experience and reduce customer trust.

Economic Downturns: The home décor market may be sensitive to economic conditions, with customers cutting back on spending during recessions.

Privacy Concerns: Collecting and using personal data for AI recommendations could raise privacy concerns among users, requiring robust data protection and transparency.

Market Saturation: The home décor space may become crowded with similar platforms offering personalized experiences, making differentiation more challenging.

Chapter 3: Analysis To The System

3.1 Gather our information (Esaltare Dekor)

In this chapter, we provide a comprehensive analysis of the system developed for this graduation project. The goal of the analysis is to offer a clear and thorough understanding of the system's structure, components, and underlying design principles. It serves as the foundation for the system's implementation and future evaluation. Through this analysis, we aim to present the rationale behind the design decisions, highlight the challenges encountered during the process, and explain how the system fulfills both functional and non-functional requirements.

The first step in the analysis process is the identification and definition of the system's requirements. These requirements are divided into two main categories: functional and non-functional. Functional requirements refer to the core functionalities that the system must provide to fulfill its intended purpose, detailing what the system should do, such as user interactions, data processing, and outputs. Non-functional requirements, on the other hand, define how the system should perform, covering aspects such as performance, scalability, security, and usability. Non-functional requirements ensure that the system is not only functional but also efficient, secure, and user-friendly. Defining these requirements at the outset ensures that the development process remains focused on the end goals, helping to avoid scope creep and ensuring that the system meets both user needs and technical constraints.

Once the system's requirements are clearly outlined, the next step is to define the system design and architecture. This involves understanding the overall structure of the system, its main components, and how they interact with one another. The design serves as a blueprint for the development process and provides a clear framework for implementing the required functionalities. A key element of the system design is determining the software architecture, which defines the organization of the system's components and their interactions. For example, if the system is a web-based application, the architecture might follow a client-server model, where the client interacts with a server that processes requests and returns data. Moreover, modularization is often a fundamental aspect of the system design, dividing the system into smaller, manageable modules, each responsible for a specific function. These modules communicate with each other through defined interfaces, ensuring that the overall system remains flexible and maintainable. During this phase of analysis, attention is also given to the data flow within the system, identifying how data is input into the system, processed, stored, and output. Understanding the data flow helps in optimizing performance,

improving the user experience, and ensuring that data is handled securely and efficiently.

To develop a system that is both effective and innovative, it is important to evaluate existing systems that serve similar purposes. By studying these existing solutions, we can identify their strengths and weaknesses and compare them to the proposed system. This analysis helps in understanding the shortcomings of current systems and using that knowledge to build a more robust and efficient solution. For instance, if the system is designed to manage data storage, evaluating other data management tools or platforms in the market could reveal issues such as poor scalability, security vulnerabilities, or limited user experience. By understanding these limitations, the project can incorporate features that address these problems. Additionally, evaluating existing systems helps to align the system with best practices and industry standards, ensuring it follows proven methods for solving similar challenges.

The selection of appropriate tools and technologies is another critical aspect of system design. In this chapter, we discuss the chosen technologies in detail, explaining why they were selected and how they contribute to the overall system. This includes both hardware and software technologies, such as programming languages, frameworks, databases, and third-party libraries or services. For example, if the system is a web application, we justify the choice of front-end frameworks like React or Angular and back-end technologies such as Node.js or Django. Factors like ease of use, community support, and scalability all play a significant role in these decisions. Moreover, considerations like security and performance are key in choosing the database management systems, such as opting for SQL or NoSQL depending on the data requirements. In some cases, the project may also involve the use of cloud services or APIs to enhance functionality or ensure scalability. A thorough evaluation of these tools and technologies allows for an informed decision-making process and ultimately results in a system that meets its intended goals.

No system development process is without challenges, and it is important to anticipate and address potential obstacles during the analysis phase. There are often various challenges and constraints that can affect the system's design and implementation, including budget limitations, time constraints, technological barriers, and organizational constraints. For example, a project may face budget limitations that restrict access to certain tools or technologies, forcing the team to find alternative solutions that meet the system's requirements within the available resources. Time constraints might also lead to prioritizing certain features over others, ensuring that the most critical aspects of the system are developed first.

Additionally, technological challenges, such as integrating with legacy systems or handling large amounts of data efficiently, may require specialized expertise or custom solutions. By anticipating these challenges during the analysis phase, the development team can plan strategies to mitigate these risks and proceed with confidence.

Finally, outlining how the system will be evaluated after implementation is an essential part of the analysis. Evaluation is crucial to determine whether the system meets the defined requirements and performs effectively in real-world conditions. The evaluation process may involve performance testing, where the system is tested for speed, reliability, and scalability under various conditions. User feedback will also play a significant role in evaluating usability and overall user experience. Furthermore, security testing will ensure that the system adheres to privacy and security standards. This comprehensive evaluation ensures that the system is thoroughly tested before being deployed and provides valuable insights for future improvements.

In conclusion, this chapter has provided a detailed analysis of the system, focusing on its requirements, design, technologies, and the challenges faced during development. Through this analysis, the foundation has been laid for the subsequent phases of development and testing. Understanding these aspects ensures that the system is built efficiently, meets user needs, and can be evaluated effectively after implementation. By addressing the system's functional and non-functional requirements, evaluating existing solutions, and selecting appropriate tools and technologies, the development process can proceed with a clear roadmap. Moreover, anticipating potential challenges and defining a comprehensive evaluation strategy ensures that the project is prepared for successful implementation and deployment.

As part of the system analysis for this **Esaltare Dekör** project, we conducted primary research through an **interview** and a **questionnaire** to gather valuable insights from stakeholders and potential users. This research aimed to gain a deeper understanding of the needs, preferences, and expectations that our target users have for the system, as well as to identify any challenges they might face. Primary research like this is essential to ensure that the system we are developing aligns with real-world user needs and solves the problems it is designed to address.

The **interview** was conducted with key stakeholders, including potential end-users and experts in the field, in order to collect qualitative data about their experiences, pain points, and expectations for the system. Through this direct interaction, we were able to obtain in-depth feedback on the requirements and

functionalities that were considered most important, as well as understand any concerns or limitations that users might encounter. Interviews provide a unique opportunity to explore complex issues in detail, allowing for open-ended responses that can offer rich insights into user behavior and preferences.

Alongside the interview, a **questionnaire** was designed to collect quantitative data from a broader group of potential users. The questionnaire aimed to gather structured feedback from a larger sample size, enabling us to identify common trends, preferences, and user needs. The questions were carefully crafted to address key aspects of the system, including its usability, functionality, and overall user experience. By collecting responses from a diverse range of participants, we were able to ensure that the system would be suitable for a wide audience, while also gathering feedback on specific areas that could be improved or further developed.

The combination of both the interview and the questionnaire allowed us to triangulate the data, providing a more comprehensive understanding of user needs. The interview provided rich, qualitative insights that helped us understand the underlying motivations and concerns of users, while the questionnaire provided statistical data that allowed us to quantify trends and preferences. Together, these research methods created a well-rounded picture of the expectations for the system and highlighted key areas for focus during the design and development phases.

Furthermore, the findings from both the interview and questionnaire have been instrumental in shaping the system's requirements and features. By involving users and stakeholders early in the process, we ensured that the system would be user-centered, addressing real-world needs and challenges. This approach also allowed us to identify potential issues early on and address them proactively, increasing the likelihood of the system's success once implemented.

In the following sections, we will review the specific questions asked during the interview and the questionnaire, highlighting key insights that influenced the design and development of the system. This review provides a deeper understanding of the feedback received and how it guided our decision-making process. The data collected from both sources has been invaluable in refining the system's features and ensuring that it meets the needs and expectations of its users.

3.1.1 Interview

Objective:

- **To gain in-depth insights and detailed responses.**
- **To have a dynamic and interactive conversation.**
- **To cover a broad range of topics in detail.**

Target-Specific Questions:

- 1. Can you describe the main goals and objectives of Esaltare Dekör AI-powered e-commerce platform?**
 - Esaltare Dekör AI-powered e-commerce platform aims to provide a highly personalized and efficient shopping experience by leveraging AI for tailored recommendations, dynamic pricing, inventory management, and enhanced customer support.
- 2. How does this app differentiate itself from other online furniture stores or competitors?**
 - Esaltare Dekör AI-powered e-commerce platform differentiates itself by offering a highly personalized shopping experience through advanced AI technologies that tailor recommendations, optimize pricing, and enhance customer support, setting it apart from other online furniture stores.
- 3. What inspired the integration of multi-vendor support into your platform?**
 - The integration of multi-vendor support into Esaltare Dekör platform was inspired by the desire to create a diverse and comprehensive marketplace, allowing multiple sellers to offer a wide range of products, thereby enhancing customer choice and boosting the platform's overall value.
- 4. What features are available in the vendor dashboard to help manage their stores?**
 - The vendor dashboard on Esaltare Dekör platform provides comprehensive tools for product, order, and inventory management, along with sales analytics and customer interaction features, all designed to streamline store operations.
- 5. How do vendors manage their product listings, inventory, and pricing?**
 - Vendors manage their product listings, inventory, and pricing on Esaltare Dekör platform through a comprehensive dashboard that provides tools for real-time updates, automated inventory tracking, and dynamic pricing adjustments.
- 6. What specific AI technologies and tools are integrated into the web app?**
 - Esaltare Dekör platform integrates machine learning, natural language processing, predictive analytics to enhance personalization, efficiency, and security.
- 7. How does AI enhance the user experience on your e-commerce platform?**
 - AI enhances the user experience on Esaltare Dekör e-commerce platform by providing personalized recommendations, optimizing pricing, and improving customer support through advanced technologies.

Technical and Operational Questions:

8. Which e-commerce platform will be used as the foundation for the app?

- Esaltare Dekör AI-powered e-commerce platform is built on Shopify, leveraging its robust features and flexibility to create a seamless shopping experience.

9. How will the app's user interface (UI) be designed to create a seamless and intuitive shopping experience?

- The app's user interface (UI) will be designed with a focus on simplicity, consistency, and accessibility, ensuring an intuitive and seamless shopping experience for all users.

10. What technologies and frameworks will be used for the front-end and back-end development?

- Esaltare Dekör' platform will use **React** for the front-end, **Laravel** and **PHP** for the back end, and integrate **machine learning** and **natural language processing (NLP)** technologies Type chapter title (level 3) 3 to enhance functionality and user experience.

11. How will the app handle product variations (e.g., color, size, material, logo)?

- The app will handle product variations by allowing vendors to create and manage different options (e.g., color, size, material, logo) through a user-friendly interface that supports dropdown menus and filtering, ensuring customers can easily select their preferred variants.

12. How will the app integrate with payment gateways and shipping providers?

- The app will integrate with payment gateways and shipping providers through APIs, enabling seamless transaction processing and real-time shipping updates.

13. What security measures will be implemented to protect against cyber threats and data breaches?

- Esaltare Dekör platform will implement multi-layered security measures, including encryption, firewalls, multi-factor authentication (MFA), intrusion detection systems (IDS), and regular security audits to protect against cyber threats and data breaches.

Strategic and Business Questions:

14. How does AI fit into Esaltare Dekör long-term business strategy?

- AI is central to Esaltare Dekör long-term business strategy, driving innovation, enhancing customer experiences, and optimizing operations to maintain a competitive edge and support sustainable growth.

15. What competitive advantages does Esaltare Dekör gain from using AI in its e-commerce platform?

- Esaltare Dekör gains competitive advantages from using AI by offering superior customer experiences, streamlining operations, and staying ahead of market trends, which collectively drive sales and boost customer engagement.

Customer and Market Insights Questions:

16. What marketing strategies will be used to promote the app and attract new customers?

- Esaltare Dekör will use a mix of SEO, PPC advertising, social media marketing, influencer partnerships, and email campaigns to promote the app and attract new customers.

3.1.2 Questionnaire

Objective

- **Identify User Needs:** Understand the specific needs, preferences, and expectations of the platform's users.
- **Define Functional Requirements:** Determine the essential features and functionalities that the platform must include to meet user needs.
- **Gather Detailed Specifications:** Collect detailed information about how each feature should work, including user interactions and system behaviors.
- **Evaluate User Experience:** Assess how users interact with similar platforms and what improvements they would like to see.
- **Prioritize Features:** Identify which features are most important to users and should be prioritized during development.
- **Identify Technical Requirements:** Gather information about the technical specifications, such as device compatibility, performance expectations, and integration needs.
- **Ensure Security and Compliance:** Understand any specific security and privacy requirements to ensure the platform complies with relevant regulations.
- **Budget and Timeline**

General Questions:

1. **What is the primary purpose of the Esaltare Dekör platform?**
 - The primary purpose of the Esaltare Dekör platform is to provide a personalized and efficient online shopping experience for furniture and home decor through advanced AI technologies.
2. **Who are the main **users** of the platform (e.g., customers, employees, partners)?**
 - The main users of the Esaltare Dekör platform are customers, vendors, and employees, each interacting with the platform to shop, sell, and manage operations respectively.
3. **What are the key features you would like to see on the platform?**
 - Key features of the Esaltare Dekör platform include personalized recommendations, dynamic pricing, efficient inventory management, 24/7 customer support, and robust security measures.

User Experience:

4. **How should users register and log in to the platform?**
 - Users should register and log in to the Esaltare Dekör platform through a simple process involving email authentication, ensuring quick and secure access.
5. **What kind of user profiles and personalization options should be available?**

- User profiles on Esaltare Dekör platform should include personalized content and preferences, allowing users to manage their details, view order history, and receive tailored recommendations based on their behavior and interactions.

6. What are the main tasks users will perform on the platform?

- The main tasks users will perform on the Esaltare Dekör platform include browsing and purchasing products, managing orders, and interacting with customer support.

Content Management:

7. What types of content will be displayed on the platform (e.g., product listings, blog posts, videos)?

- The Esaltare Dekör platform will display various types of content, including product listings, blog posts, videos, user reviews, and interactive media to engage and inform customers.

8. How should users be able to search and filter content?

- Users should be able to search and filter content on Esaltare Dekör platform using intuitive search bars and dynamic filters that allow for real-time updates and precise refinement of search results.

9. What tools are needed for content creation and management?

- For content creation and management on Esaltare Dekör AI-powered e-commerce platform, you need a combination of tools for writing, design, project management, SEO, and analytics.

Interaction and Engagement:

10. How should users interact with each other and with the platform (e.g., comments, reviews, messaging)?

- Users should interact with each other and the platform through **comments, reviews, and messaging** to foster engagement, provide feedback, and enhance the overall user experience.

11. What kind of support and customer service features are needed?

- For effective support and customer service, you need features like **24/7 live chat, a comprehensive knowledge base, ticketing system, and multi-channel support** (including email, phone, and social media) to ensure timely and efficient assistance.

12. Are there any gamification elements or rewards you would like to include?

- For Esaltare Dekör AI-powered e-commerce web app, consider including **loyalty points systems, progress bars, badges, leaderboards, and interactive quizzes** to enhance user engagement and retention.

Technical Requirements:

13. What devices and browsers should the platform be compatible with?

- The platform should be compatible with **all major devices (desktops, laptops, tablets, smartphones) and browsers (Chrome, Firefox, Safari, Edge, Opera)** to ensure a seamless user experience across different environments.

14. What are the security and privacy requirements for the platform?

- The platform must ensure **data encryption, secure user authentication, regular security audits, and robust access controls** to protect user data and maintain privacy.

Performance and Scalability:

16. What are the expected traffic and usage patterns for the platform?

- The platform should be designed to handle **high traffic volumes during peak shopping periods, with consistent usage throughout the year, and scalability to accommodate growth.**

17. How should the platform handle high traffic and ensure performance?

- To handle high traffic and ensure performance, the platform should implement **load balancing, autoscaling, caching, and use a content delivery network (CDN)** to distribute traffic efficiently and maintain fast response times.

18. What are the requirements for data storage and backup?

- For data storage and backup, ensure **regular updates, comprehensive data capture, offsite storage, and easy access for restoration** to protect against data loss and ensure business continuity.

Additional Features:

19. Are there any specific design or branding guidelines that need to be followed?

- For Esaltare Dekör AI-powered e-commerce platform, follow **consistent branding guidelines including logo usage, color palette, typography, imagery, and tone of voice** to ensure a cohesive and recognizable brand identity.

20. What are the budget and timeline for developing the platform?

- For Esaltare Dekör AI-powered e-commerce platform, the budget can range from **\$30,000 to \$135,000+ per year** depending on the size and complexity of the project, with a timeline of **1 to 6+ months** for development.

3.1.3 Classification of the requirements

Functional Requirements for Esaltare Dekör:

1. **User Authentication and Authorization:**
 - Users must be able to create accounts, log in, and log out securely.
 - Role-based access control to ensure appropriate access levels for users and administrators.
2. **Product Catalog Management:**
 - Ability to add, update, and delete products.
 - Categorization and tagging of products for easy navigation and search.
3. **Personalized Recommendations:**
 - AI algorithms to provide personalized product recommendations based on user behavior and preferences.
 - Display of recommended products on the homepage and product pages.
4. **Search Functionality:**
 - Advanced search options with filters for price, category, brand, etc.
 - search suggestions and auto-complete features.
5. **Shopping Cart and Checkout:**
 - Users can add, update, and remove items from the shopping cart.
 - Secure and user-friendly checkout process with multiple payment options.
6. **Wishlist and Favorites:**
 - Users can add products to a Wishlist for future reference.
 - Ability to share wish lists with friends and family.
7. **Promotions and Discounts:**
 - Support for promotional codes, discounts, and special offers.
 - AI-driven personalized promotions based on user behavior and preferences.
8. **Social Media Integration:**
 - Integration with social media platforms for sharing products and reviews.
 - Social login options for quick and easy account creation.
9. **Order Management:**
 - Users can view order history, track orders, and receive notifications about order status.
 - Admins can manage orders, process returns, and handle customer inquiries.
10. **Multi-Vendor Support:**
 - Platform to support multiple vendors with individual dashboards.
 - Vendor-specific analytics and reporting.
11. **Content Management System (CMS):**
 - Admin interface for managing website content, including blogs, banners, and product descriptions.
 - Support for multimedia content like images and videos.
12. **Loyalty Programs:**
 - Implementation of loyalty programs to reward repeat customers.
 - AI-driven suggestions for loyalty rewards based on user activity.
13. **Customer Service:**
 - AI-driven chatbots to assist users with queries and provide support.
 - Integration with customer service platforms for human support when needed.
14. **Advanced Filtering Options:**
 - Enhanced filtering options for users to narrow down product searches.
 - AI-driven filter suggestions based on user preferences.
15. **User Activity Tracking:**
 - Tracking user activity to provide insights into user behavior.
 - AI-driven recommendations for improving user engagement.

16. Inventory Management:

- Real-time inventory tracking and updates.
- AI-driven demand forecasting optimizes stock levels.

17. User Reviews and Ratings:

- Users can leave reviews and ratings for products.
- Moderation tools for managing user-generated content.

18. Analytics and Reporting:

- Dashboard for tracking sales, user behavior, and other key metrics.
- AI-driven insights and reports to help with decision-making.

Non-Functional Requirements for Esaltare Dekör:

1. Performance:

- The app should be loaded within 2 seconds for optimal user experience.
- The system should handle at least 1000 concurrent users without performance degradation.

2. Scalability:

- The architecture should support horizontal and vertical scaling to accommodate growth.
- The system should be able to handle increased load during peak times, such as sales events.

3. Security:

- Data encryption for user information and transactions.

4. Data Integrity:

- Ensuring data accuracy and consistency across the platform.
- Regular data validation and integrity checks.

5. Interoperability:

- Ability to integrate with third-party systems and APIs.
- Support for various data formats and communication protocols.

6. Extensibility:

- Designing the system to be easily extendable for future features and enhancements.
- Modular architecture to support plug-and-play components.

7. Energy Efficiency:

- Optimizing the platform to minimize energy consumption.
- Implementing green computing practices where possible.

8. User Training and Support:

- Providing comprehensive user guides and tutorials.
- Offering customer support through multiple channels, including chat, email, and phone.

9. Version Control:

- Implementing version control for all software components.
- Maintaining a clear version history for tracking changes and updates.

10. Incident Management:

- Establishing procedures for handling system incidents and outages.
- Regular drills and simulations to ensure readiness for real incidents.

11. Auditability:

- Keeping detailed logs of all system activities for audit purposes.
- Ensuring compliance with audit requirements and standards.

12. **Usability:**

- Intuitive and user-friendly interface.
- Accessibility features to ensure usability for all users, including those with disabilities.

13. **Reliability:**

- The system should have an uptime of 99.9%.
- Regular backups and disaster recovery plans in place.

14. **Maintainability:**

- Modular codebase to facilitate easy updates and maintenance.
- Comprehensive documentation for developers and administrators.

15. **Compatibility:**

- Cross-browser compatibility (Chrome, Firefox, Safari, Edge).
- Responsive design to ensure usability on various devices (desktops, tablets, smartphones).

16. **Compliance:**

- Adherence to legal and regulatory requirements in all operating regions.
- Regular updates to ensure ongoing compliance with new regulations.

17. **Monitoring and Logging:**

- Continuous monitoring of system performance and user activity.
- Detailed logging for troubleshooting and audit purposes.

18. **Sustainability:**

- Implementing practices to ensure the long-term sustainability of the platform.
- Regularly reviewing and updating the system to keep it current with technological advancements.

User Requirements:

To gather user requirements for Esaltare Dekör' AI-powered e-commerce platform in more detail, consider the following:

1. **Stakeholder Interviews:** Conduct in-depth interviews with key stakeholders to understand their specific needs, goals, and expectations. This helps in aligning the platform's features with business objectives.
2. **Surveys and Questionnaires:** Use detailed surveys to collect data from potential users about their preferences, pain points, and desired features. This quantitative data can guide the development process.
3. **User Stories and Use Cases:** Develop comprehensive user stories and use cases to capture detailed requirements from the user's perspective. This helps in understanding how different users will interact with the platform.
4. **Workshops and Brainstorming Sessions:** Organize collaborative workshops and brainstorming sessions with stakeholders and team members to identify and prioritize requirements. This fosters a shared understanding and collective decision-making.
5. **Competitor Analysis:** Conduct a thorough analysis of competitors to identify successful features and functionalities that it could be beneficial to include in your platform. This helps in staying competitive and innovative.
6. **Prototyping and Feedback:** Create interactive prototypes of the platform and gather feedback from users to refine and validate requirements. This iterative process ensures that the final product meets user expectations.
7. **AI Tools:** Leverage AI-powered tools to automate the requirement gathering and analysis process. These tools can help in generating comprehensive requirements, identifying duplicates, and ensuring accurate documentation and completeness.

Business Requirements:

Gathering business requirements for Esaltare Dekör AI-powered e-commerce platform involves several key steps to ensure a comprehensive understanding of the project's needs and goals. Here's a structured approach:

1. Stakeholder Identification and Engagement

- **Identify Stakeholders:** List all individuals and groups who have an interest in the project, such as business owners, project managers, developers, and end-users.
- **Engagement Methods:** Use interviews, surveys, and workshops to gather their input. This helps in understanding their needs, expectations, and any potential concerns.

2. Requirement Elicitation

- **Techniques:** Utilize brainstorming sessions, focus groups, and user stories to gather detailed requirements. These methods encourage open discussion and diverse perspectives.
- **AI-Powered Tools:** Leverage AI tools to facilitate natural language understanding and interactive dialogues. These tools can help in accurately capturing business needs by interpreting and processing complex requirements.

3. Requirement Documentation

- **Structured Format:** Document the requirements in a Business Requirements Document (BRD). This should include:
 - **Project Objectives:** Clear goals and expected outcomes.
 - **Scope:** Boundaries of the project.
 - **Business Requirements:** Detailed needs and functionalities.
 - **Key Stakeholders:** Individuals or groups involved.
 - **Constraints:** Limitations and restrictions.
 - **Cost-Benefit Analysis:** Evaluation of the project's financial viability.

4. Validation and Verification Requirement

- **Validation:** Ensure the requirements are complete, feasible, and aligned with business goals by reviewing them with stakeholders.
- **AI Models:** Use AI to analyze the requirements for inconsistencies, conflicts, or missing information. AI can provide recommendations for improvement, enhancing the quality of the requirements.

5. Continuous Requirement Refinement

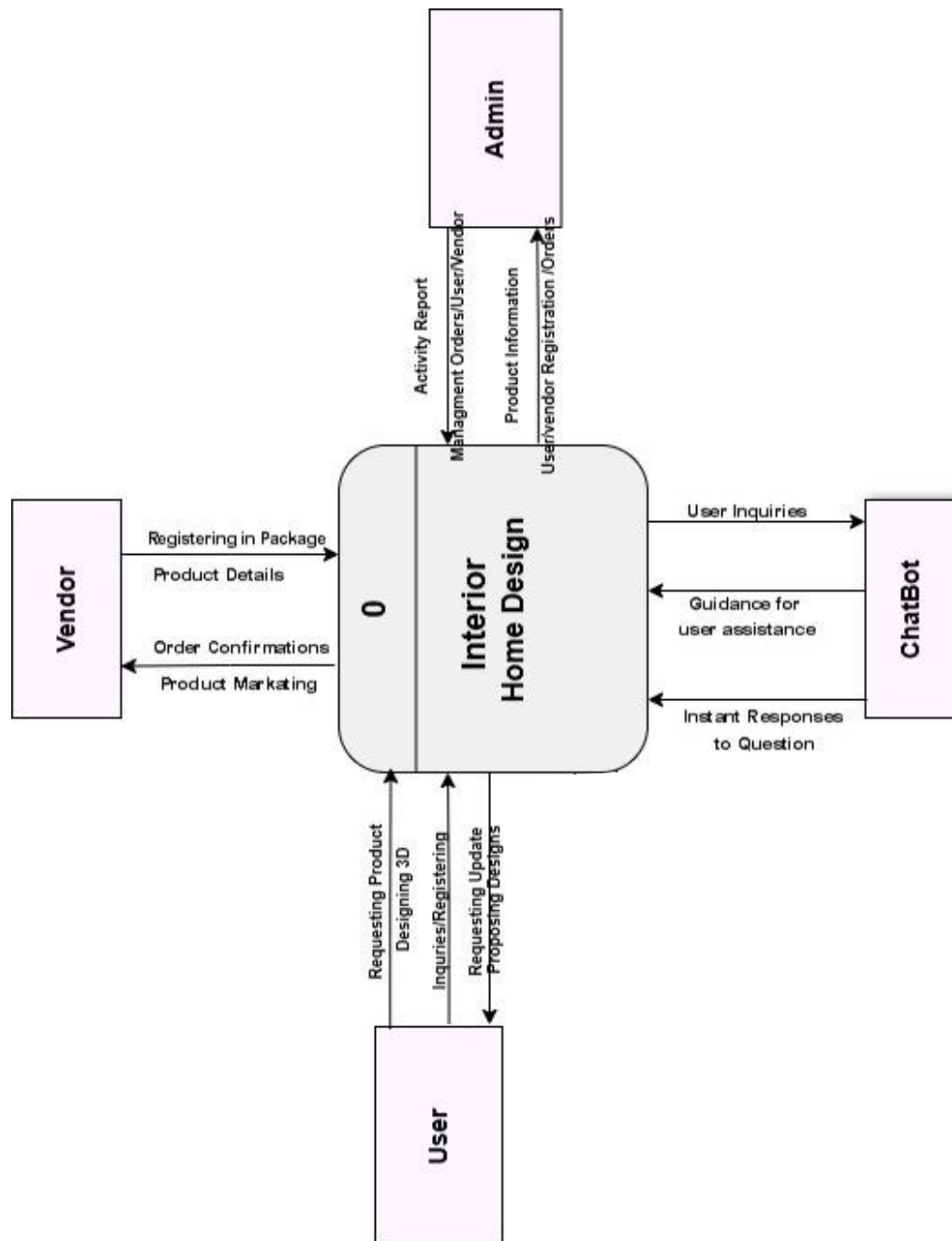
- **Iterative Process:** Continuously refine the requirements based on stakeholder feedback and changing business needs. This iterative approach helps in adapting to new insights and evolving project dynamics.
- **AI Adaptability:** Utilize AI's ability to learn and adapt over time to keep the requirements up-to-date and relevant.

3.2 Esaltare Dekor Requirements Structuring

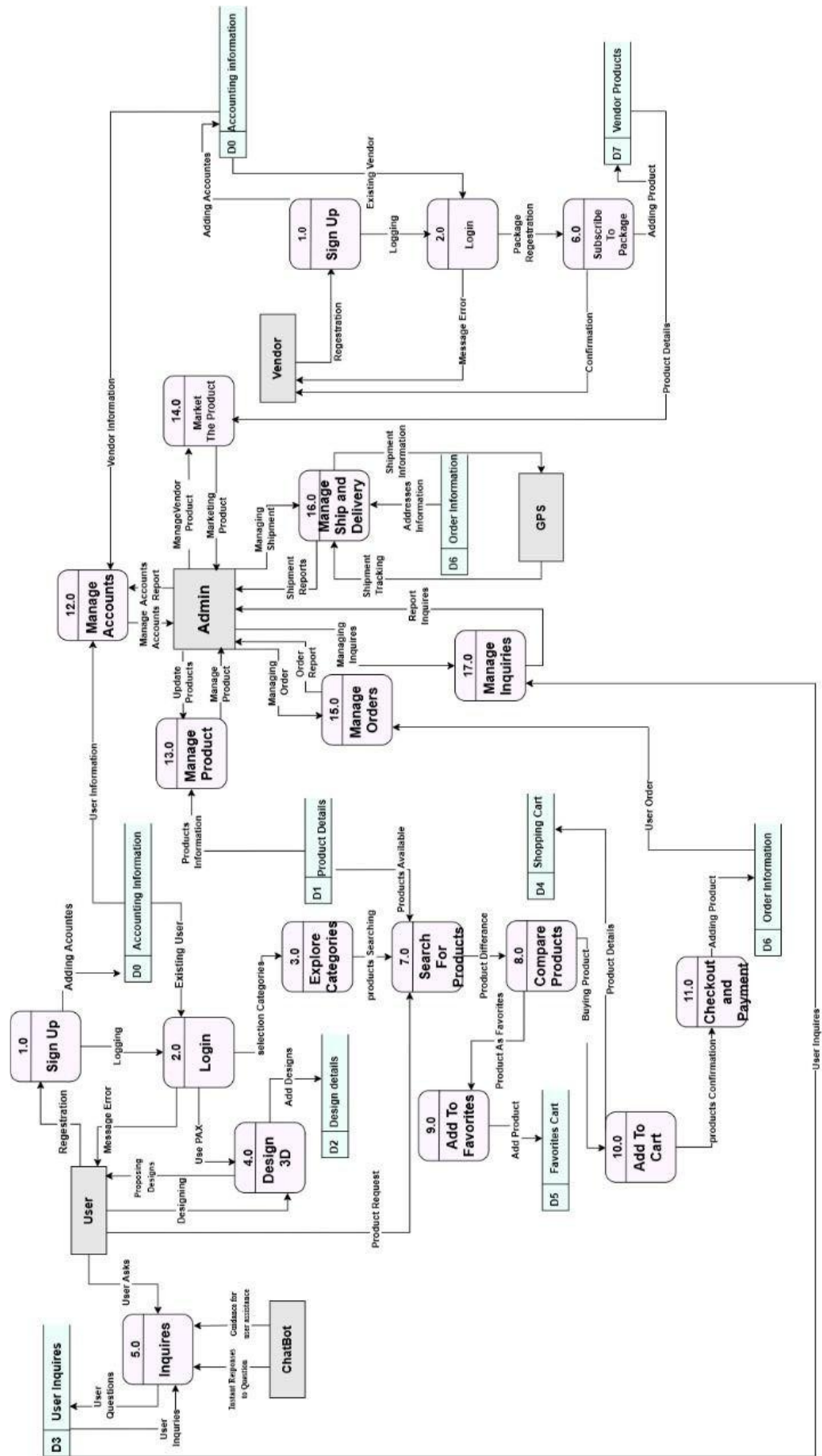
3.2.1 Process Modeling

3.2.1.1 DFD and Context Diagram

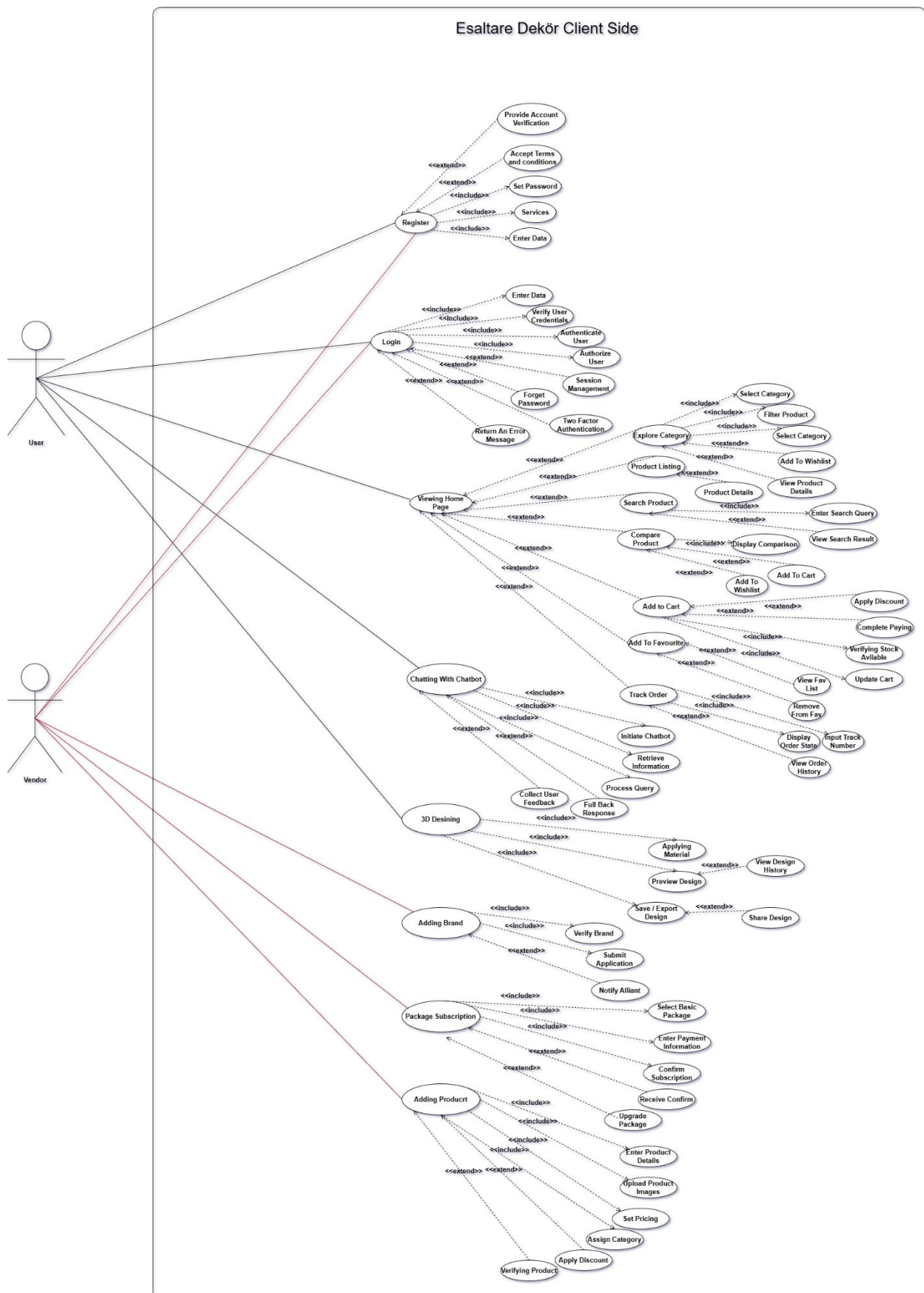
Context Diagram



Data Flow Diagram



3.2.1.2 Use Case Diagrams and Analysis



The **Esaltare Dekör** Client-Side system, as depicted in its use case diagram, is a comprehensive platform designed to seamlessly integrate interior design services with a robust e-commerce experience. The system caters to two primary user groups: registered Users, who engage with the full suite of features, and Visitors, who have access to a subset of functionalities geared towards exploration and discovery. At the heart of the system lies a sophisticated account management module, allowing Visitors to register and become Users, while registered Users can securely log in and manage their personalized profiles.

This includes updating personal details, contact information, design preferences, saved addresses, and potentially stored payment methods, laying the groundwork for a tailored user journey. The platform's browsing and searching capabilities are meticulously designed to facilitate the discovery of desired products and designs. Users can navigate through predefined categories, apply granular filters based on criteria like price range, style, material, and size, and utilize a powerful search bar to pinpoint specific items or keywords. This multifaceted approach ensures that users can efficiently locate precisely what they envision for their interior design projects.

Product interaction is a key element, enabling users to delve into detailed product descriptions, specifications, high-resolution images, and available options. They can add items to their shopping carts for purchase, curate wish lists for future consideration and leverage a comparison feature to evaluate multiple products side-by-side. A standout feature of the Esaltare Dekör system is its robust design collaboration functionality.

Users can create new design projects, potentially utilizing integrated tools, templates, or a comprehensive design library. The platform fosters collaboration by allowing users to seamlessly share their designs with interior designers, friends, family, or other stakeholders. Viewing shared designs, providing feedback, and incorporating suggestions are streamlined, promoting iterative design improvements. The innovative "10-Degree Viewing" feature takes visualization to the next level, potentially offering 360-degree views, interactive 3D models, augmented reality experiences that immerse users in the design. Furthermore, the system facilitates the integration of current design trends, providing access to a curated library of trending elements or suggestions.

Direct communication with interior designers is facilitated through an integrated chat feature, enabling real-time interaction, discussions about project requirements, inquiries about products or services, and requests for consultations. This direct line of communication fosters a collaborative and personalized design experience.

The Esaltare Dekör system also provides a seamless purchasing and subscription experience. Users can navigate a secure checkout process, applying discount codes or promotional offers to maximize value. The platform supports package subscriptions, offering recurring design services, consultations, product deliveries, or bundled combinations. Users can easily access detailed information about available subscription packages and manage their active subscriptions, including renewals, cancellations, and upgrades.

While primarily a client-side system, the platform allows authorized users, such as administrators, to manage the product catalog. This includes adding new products with comprehensive information, assigning them to appropriate categories, applying discounts, defining design categories, and verifying product details before publication.

Visitors to the platform have access to browsing and searching functionalities, allowing them to explore the available products and designs. They can also access general information about the company and its services, gaining insights into Esaltare Dekör offerings.

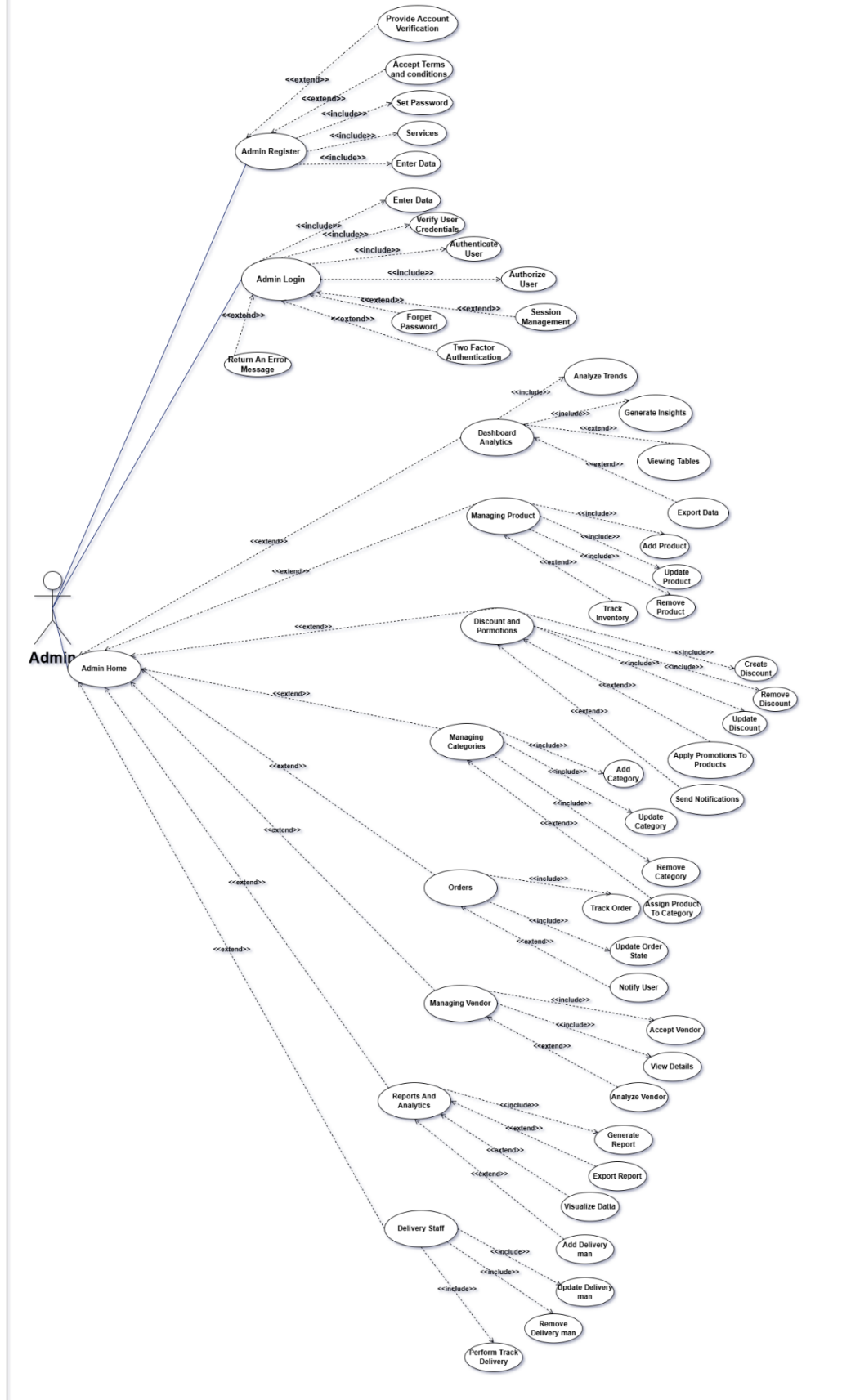
The use case diagram, while comprehensive, suggests potential enhancements. A more detailed breakdown of user roles and permissions would clarify access levels and functionalities.

Explicitly outlining supported payment gateway integrations would provide transparency regarding transaction processing. Incorporating order tracking functionality would significantly enhance the user experience by providing real-time updates on order status.

Expanding customer support beyond the designer chat feature to include FAQs, a help desk, or contact forms would further improve accessibility and responsiveness. Finally, ensuring mobile responsiveness across various devices is crucial for maximizing accessibility in today's mobile-driven world. The Esaltare Dekör Client-Side system, with its comprehensive suite of features, offers a powerful and user-friendly platform for interior design services and product sales. By seamlessly integrating design collaboration, e-commerce functionalities, and direct communication with designers, the system empowers users to bring their interior design visions to life.

The platform's focus on personalization, collaboration, and streamlined transactions positions it as a valuable tool for both individuals seeking design assistance and interior design professionals. Continuous refinement and expansion of features, based on user feedback and evolving industry trends, will further solidify Esaltare Dekör's position in the market.

Esaltare Dekor Admin Side



The Esaltare Dekör Admin Side system, as depicted in its use case diagram, provides a comprehensive suite of administrative functionalities designed to empower the administrator in managing all facets of the Esaltare Dekör platform. The system revolves around the central actor, the Admin, who interacts with various modules to maintain, operate, and analyze the platform's performance. The Admin's journey begins with secure account management, encompassing login (potentially with two-factor authentication for enhanced security), logout, password resets, and account verification.

The system diligently handles instances of incorrect data entry, ensuring data integrity. Beyond basic account management, the Admin possesses extensive control over the platform's core components. Service management capabilities allow Admin to add new services, edit existing ones, delete outdated offerings, and efficiently search for specific services within the system. User management functionalities enable the Admin to oversee user accounts, including viewing detailed user information, editing user profiles, and authorizing user access. This granular control over user accounts ensures proper access management and platform security.

Product management forms a crucial aspect of the Admin's responsibilities. The Admin can add new products to the catalog, update existing product information, delete discontinued or obsolete products, and meticulously manage product stock levels. This comprehensive product management module ensures an accurate and up-to-date product catalog, crucial for the platform's e-commerce operations. The Admin also manages discounts and coupons, creating new promotional offers, updating existing discounts, and deleting expired or irrelevant coupons. This functionality empowers Admin to implement targeted marketing strategies and incentivize customer purchases. Category management allows Admin to organize products into logical categories, adding new categories, updating existing ones, deleting categories as needed, and potentially applying promotions or sending notifications related to specific categories. This structured approach to product categorization enhances the user experience by facilitating easy navigation and product discovery.

Order management is another key area of administrative control. The Admin can update order details, assign pending orders to appropriate staff members, and view comprehensive order information.

This module ensures efficient order processing and fulfillment, contributing to customer satisfaction. Messaging management allows the Admin to manage communication within the platform, including viewing message details and potentially acting on messages, such as replying to inquiries or marking messages as read.

This centralized messaging system streamlines communication and facilitates efficient issue resolution.

The system provides robust reporting and analytics capabilities, empowering Admin to analyze platform visits, generate various reports on key metrics, export data for further analysis, and access specific pre-defined reports. These analytical tools provide valuable insights into platform usage, performance, and trends, enabling data-driven decision-making.

Delivery staff management is another essential function, allowing the Admin to manage delivery personnel, including adding new staff members, updating their information, deleting inactive staff accounts, and viewing delivery staff details. This module ensures efficient management of the delivery network, crucial for timely order fulfillment. Financial management functionalities enable the Admin to perform financial calculations and generate financial reports. This module provides crucial financial insights into the platform's performance, facilitating financial planning and management.

The use case diagram, while comprehensive, provides a high-level overview of the system's functionalities. Further details regarding specific data fields, validation rules, and user interface elements would be necessary for a complete understanding of the system's implementation. However, the diagram effectively illustrates the broad range of administrative capabilities provided by the Esaltare Dekör Admin Side system. From user and product management to order processing, financial reporting, and delivery staff oversight, the system offers a centralized platform for managing all aspects of the Esaltare Dekör platform. The Admin, through interaction with these various modules, plays a critical role in ensuring the smooth operation, growth, and success of the Esaltare Dekör platform. The system's design emphasizes efficiency, control, and data-driven decision-making, empowering the Admin to effectively manage the complex interplay of users, products, orders, and finances that constitute the Esaltare Dekör ecosystem.

The continuous refinement and enhancement of these administrative tools, based on evolving business needs and technological advancements, will be crucial for maintaining the platform's competitive edge and ensuring its long-term success. The Esaltare Dekör Admin Side system, as depicted in this diagram, provides a solid foundation for effective platform administration, enabling the Admin to maintain a high level of control and visibility across all aspects of the business.

Use Case Analysis and Scenarios

Use Case 1:	Visiting the E-commerce Website
Use Case ID	UC001
Use Case Name	Visit E-commerce Website
Actors	Kareem (Customer)
Preconditions	Kareem has a device with internet access.
Postconditions	Kareem is on the Esaltare Dekör website.
Normal Flow	1. Kareem enters the website URL or searches for it via a search engine. 2. The Esaltare Dekör homepage loads on Kareem's device.
Alternative Flows	1. Kareem enters an incorrect URL. (System redirects to a 404 page or displays an error message.) 2. The website is temporarily unavailable. (System displays a maintenance page or an error message.)
Exceptions	- Internet connectivity issues prevent the website from loading.

Use Case 2:	Explore Product Categories
Use Case ID	UC002
Use Case Name	Explore Product Categories
Actors	Kareem (Customer)
Preconditions	Kareem is on the Esaltare Dekör website.
Postconditions	Kareem has viewed different product categories.
Normal Flow	1. Kareem navigates to the "Categories" section of the website. 2. Kareem browses through the available categories (e.g., Living Room, Bedroom, Kitchen).
Alternative Flows	- Kareem uses the website's search functionality to find a specific category.
Exceptions	- The website's category section is temporarily unavailable.

Use Case 3:	Search for a Product
Use Case ID	UC003
Use Case Name	Search for a Product
Actors	Kareem (Customer)
Preconditions	Kareem is on the Esaltare Dekor website.
Postconditions	Kareem is presented with search results related to his query.
Normal Flow	1. Kareem uses the search bar to enter keywords related to the product he's looking for (e.g., "living room sofa"). 2. The system displays a list of relevant products based on the search query.
Alternative Flows	1. Kareem misspells the product name. (System suggests corrections or displays results for similar products.) 2. No products match Kareem's search query. (The system displays a "no results" message or suggests alternative products.)
Exceptions	- The website's search functionality is temporarily unavailable.

Use Case 4:	View Product Details
Use Case ID	UC004
Use Case Name	View Product Details
Actors	Kareem (Customer)
Preconditions	Kareem is on the Esaltare Dekor website and has viewed search results or product categories.
Postconditions	Kareem has viewed detailed information about a specific product.
Normal Flow	1. Kareem clicks on a specific product from the search results or category listings. 2. The system displays the product details page, including images, description, specifications, price, and customer reviews.
Alternative Flows	- Kareem navigates to a different product from the product details page.
Exceptions	- The product details page fails to load.

Use Case Name	Compare Products
Use Case ID	UC005
Actors	Kareem (Customer)
Preconditions	Kareem is on the Esaltare Dekör website and has viewed product details.
Postconditions	Kareem has compared the features and details of multiple products.
Normal Flow	1. Kareem selects multiple products he wishes to compare. 2. Kareem uses the "compare" feature to view a side-by-side comparison of the selected products, including their features, prices, and customer reviews.
Alternative Flows	- Kareem decides not to compare any more products and returns to browsing.
Exceptions	- The website's comparison feature is temporarily unavailable.

Use Case Name	Add Product to Favorites
Use Case ID	UC006
Actors	Kareem (Customer)
Preconditions	Kareem is on the Esaltare Dekör website and has viewed product details.
Postconditions	The selected product is added to Kareem's favorites list.
Normal Flow	1. Kareem clicks the "Add to Favorites" button on the product details page. 2. The system adds the product to Kareem's favorites list.
Alternative Flows	- Kareem removes the product from his favorites list.
Exceptions	- The website's favorites functionality is temporarily unavailable.

Use Case Name	Add Product to Cart
Use Case ID	UC007
Actors	Kareem (Customer)
Preconditions	Kareem is on the Esaltare Dekör website and has viewed product details.
Postconditions	The selected product is added to Kareem's shopping cart.
Normal Flow	1. Kareem clicks the "Add to Cart" button on the product details page. 2. The system adds the product to Kareem's shopping cart.
Alternative Flows	1. Kareem modifies the quantity of the product in his cart. 2. Kareem removes the product from his cart.
Exceptions	- The website's shopping cart functionality is temporarily unavailable.

Use Case Name	Create Account/Login
Use Case ID	UC008
Actors	Kareem (Customer)
Preconditions	Kareem has added a product to his shopping cart and is proceeding to checkout.
Postconditions	Kareem is logged in to his account or has created a new account.
Normal Flow	1. Kareem is prompted to either log in or create an account. 2. If Kareem is a new user, he provides his personal information (name, email, password) to create an account. 3. If Kareem is an existing user, he enters his email and password to log in.
Alternative Flows	1. Kareem chooses to log in using social media accounts (e.g., Google, Facebook). 2. Kareem forgets his password and uses the "forgot password" option to reset it.
Exceptions	- The account creation or login process fails due to incorrect information or system errors.

Use Case Name	Complete Purchase
Use Case ID	UC009
Actors	Kareem (Customer)
Preconditions	Kareem is logged in and has items in his shopping cart.
Postconditions	Kareem has successfully purchased the items in his cart.
Normal Flow	1. Kareem proceeds to the checkout page. 2. Kareem enters his shipping and payment information. 3. Kareem confirms the order details and completes the purchase.
Alternative Flows	1. Kareem uses a different shipping address. 2. Kareem uses a different payment method. 3. Kareem cancels the purchase.
Exceptions	1. The payment transaction fails. 2. The system encounters an error during order processing.

Use Case Name	Track Order/Get Support
Use Case ID	UC010
Actors	Kareem (Customer)
Preconditions	Kareem has completed a purchase.
Postconditions	Kareem has received updates about his order and has access to post-sales support.
Normal Flow	1. Kareem receives order confirmation and tracking information. 2. Kareem can track his order's progress through the website or via email updates. 3. If Kareem faces any issues with the purchased product, he can contact customer support through various channels (e.g., chat, email).
Alternative Flows	- Kareem contacts customer

Use Case Name	Assist with Product Purchase
Use Case ID	UC011
Actors	Ziad (Customer), AI Chatbot
Preconditions	Ziad has visited the website and is looking for help with purchasing a dining table.
Postconditions	Ziad receives personalized recommendations for a dining table based on his preferences and has the option to continue or refine his search.
Normal Flow	1. Ziad opens the chatbot window on the website. 2. The AI chatbot greets Ziad with a message: "Hello Ziad! How can I assist you today?" 3. Ziad selects the option "Help with making a purchase decision." 4. The AI chatbot asks Ziad: "To provide the best recommendation, can you tell me your preferred home decor style?" 5. Ziad responds: "I prefer a modern, minimalist style." 6. The AI chatbot asks additional questions to gather more information: "What is the size of the room where you want to place the table? Do you have a budget in mind?" 7. Ziad replies: "The room is large, and my budget is around 5000." 8. The AI chatbot asks: "What color preferences do you have for the table? Do you prefer neutral colors like white and gray, or something darker or lighter?" 9. Ziad answers: "I prefer neutral colors, like white and gray." 10. The AI chatbot presents several dining table options that match Ziad's preferences: Modern dining table (large size, gray color, Price: 4500) Sleek modern dining table (large size, white color, Price: 4900) 11. The AI chatbot offers further assistance: "Would you like to explore more options or need more information on any of these tables?" 12. Ziad decides to check the white table option. 13. The AI chatbot provides further details about the white table: "This table is made of high-quality wood with a sleek modern design. It's perfect for larger spaces and easy to maintain." 14. Ziad confirms that he is interested in the white table. 15. The AI chatbot asks: "Would you like to proceed with the purchase, or do you need any more help?" 16. Ziad decides to think about it and exits the chat.

	17. The AI chatbot offers a final message: “If you need more assistance or want to revisit your options, I’m here to help anytime!”
Alternative Flows	1. If Ziad needs additional information or prefers to explore more options: Ziad selects the option “Show more options.” The AI chatbot provides a new set of recommendations based on Ziad’s initial preferences. 2. If Ziad faces any issues or questions about the products: - Ziad asks for more details about a specific product. - The AI chatbot provides additional information such as product material, dimensions, and delivery options.

Use Case Name	3D Furniture Design
Use Case ID	UC012
Actors	Baraa (Customer)
Preconditions	Baraa has accessed the website and is looking to design furniture using a 3D design tool.
Postconditions	Baraa has created a custom furniture design, saved it for future reference, and can share it with others for feedback.
Normal Flow	Browse the 3D Furniture Design Tool: 1. Baraa visits the website and navigates the 3D furniture design tool. Goal: Access to start designing custom furniture. Explore 3D Component Library: 2. Baraa finds the 3D component library, which contains a variety of furniture items such as sofas, tables, and cabinets. Goal: Browse through a wide range of 3D furniture components to create a personalized design. Drag and Drop Components: 3. Baraa starts dragging and dropping components from the library into the design space. Goal: Enable users to visually and interactively create their own furniture designs by adding elements to the workspace. Customize Design: 4. Baraa can resize, rotate, and adjust the position of the components within the design space to achieve the ideal design. Goal: Provide a comprehensive customization experience, allowing Baraa to feel like they are designing their own furniture. Save or Share Design:

	5. After completing the design, Baraa can save the design to their personal account or share it with friends and family for feedback. Goal: Make it easy for customers to save and retrieve their designs and share them with others for social interaction.
Alternative Flows	If Baraa wants to make changes to the design: 1. Baraa can always return to the design tool, modify the size, color, or position of the components, and update the design in real-time. The system allows for continuous adjustments and immediate visual feedback on changes. If Baraa encounters difficulty: 1. Baraa can access help or tutorial options within the interface to guide them in using the 3D tool or explore how to make specific changes (e.g., resizing or rotating components).

Use Case Name	Register Vendor Brand on Esaltare Dekör Platform
Use Case ID	UC014
Actors	"Dekor Plus" (Vendor Brand), Admin (Platform Admin)
Preconditions	"Dekor Plus" is a new vendor brand looking to join Esaltare Dekör and list their products.
Postconditions	"Dekor Plus" is successfully registered on the platform, can list their products, and benefit from marketing campaigns by Esaltare Dekör.
Normal Flow	Brand Registration: 1. "Dekor Plus" team registers via a form on the website to submit a request to list their products and subscribe to a specific marketing package (e.g., Basic Package that promotes up to 5 products). Goal: Provide a smooth and secure registration process for emerging brands. Review and Approval by Admin: 2. The admin reviews the brand's registration and approves it based on the criteria. Goal: Ensure that the brand meets the platform's standards before being allowed to sell and market products. Adding Products: 3. After approval, "Dekor Plus" is granted access to a dashboard where they can add their products with full details such as: - High-quality images, product descriptions, and prices.

	Goal: Simplify the process of adding products while ensuring they meet quality standards for customer visibility. Marketing via the Website: 4. Esaltare Dekör markets "Dekör Plus" products through promotional campaigns on the website to drive traffic and attract customers. Goal: Increase the visibility of emerging brand products and attract more customers. Customer Purchase: 5. A customer visits the platform and makes a purchase of a product from "Dekör Plus." Goal: Facilitate a seamless purchase experience for the customer. Order Fulfillment: 6. Esaltare Dekör manages the shipping and delivery process to ensure the product reaches the customer. Goal: Simplify the purchasing and shipping process while ensuring customer satisfaction.
Alternative Flows	If the registration form is incomplete or invalid:

Use Case Name	System Admin (Admin Role)
Use Case ID	UC015
Actors	Shaimaa (Admin)
Preconditions	Shaimaa has admin access to the platform and is responsible for overseeing the website's operation, content management, customer support, and security.
Postconditions	Shaimaa has successfully monitored the website's performance, updated content, assisted customers, and ensured the security of the platform.
Normal Flow	Monitoring System Performance: 1. Shaimaa starts her day by reviewing the admin dashboard to monitor system performance, checking for any technical issues or alerts. Goal: Ensure the platform is running smoothly with no technical issues. Managing Content: 2. Shaimaa inspects and updates website content, such as adding new products, updating prices, and posting promotional offers. Goal: Keep the website content fresh and up to date, offering customers the latest products and deals. Order Monitoring and Coordination:

	3. Shaimaa monitors the status of orders and coordinates with the shipping and delivery team to ensure that orders are being delivered to customers on time. Goal: Improve customer experience and ensure timely deliveries for customer satisfaction. Providing Customer Support: 4. Shaimaa receives customer inquiries and technical support requests via email or chatbot and provides quick and efficient solutions to any issues or questions. Goal: Offer immediate and effective customer support, resolving problems and addressing concerns promptly. Analyzing Website Data: 5. Shaimaa reviews website analytics, such as the number of visitors, conversion rates, and customer feedback, to identify areas for improvement. Goal: Use data to optimize the user experience and increase sales on the platform. Ensuring Website Security: 6. Shaimaa checks that all security measures are up to date, running necessary tests to protect the site's data and sensitive information. Goal: Ensure the security and protection of the platform, safeguarding customer information and transactions. Shipment Tracking: 7. Shaimaa opens the shipment tracking link and uses the GPS tracking system to find the current location of shipments. Goal: Provide accurate and real-time information about the shipment's location, ensuring customers are informed about their orders.
Alternative Flows	1. Shaimaa is alerted by the system and takes immediate action to resolve the issue (e.g., contacting the technical team or fixing a broken feature). 2. Shaimaa may inform customers about any delays or issues if necessary.

Chapter 4: EDAI Model

4.1 Chatbot overview

In the rapidly evolving field of artificial intelligence and machine learning, the integration of advanced systems to solve specialized problems has become increasingly prominent. This chapter delves into our innovative project that integrates a chatbot with Gemini, focusing specifically on interior design requests. The primary objective of this project is to harness the capabilities of large language models (LLMs), prompt engineering, and agent-based systems to provide users with precise and relevant interior design solutions.

Our project leverages the strengths of Gemini, an AI platform with expertise in interior design, and combines it with cutting-edge techniques in natural language processing and prompt engineering. By doing so, we aim to enhance user experience, ensuring that interior design queries are addressed with accuracy and contextual relevance. This chapter will provide an in-depth look at the architecture, design, and implementation of our system, highlighting the innovative approaches we have taken to achieve our goals.

The integration process begins with the user's interaction with the chatbot, which serves as the primary interface for making interior design requests. These requests are then processed through a series of steps involving data concatenation, response filtration, prompt restructuring, and database querying. Each of these steps is meticulously designed to optimize the system's performance and ensure that users receive the best possible responses.

Our approach is built on the foundation of three core concepts:

Prompt Engineering: This technique involves structuring the input prompts given to the LLMs in a way that guides them towards generating the desired output. Prompt engineering ensures that the responses are relevant and accurate, tailored to the specific needs of the user, and This refers to crafting well-structured input for the chatbot, so it can respond more accurately and effectively. The better the prompt (or instruction), the more relevant and helpful the chatbot's response will be. For instance, giving specific details, context, or constraints can help guide the chatbot toward providing better answers.

Large Language Models (LLMs): These models are trained on vast datasets, enabling them to understand and generate human-like responses. By utilizing LLMs, we ensure that our chatbot can comprehend complex language patterns and provide detailed and coherent answers to interior design queries and

Agent-Based Systems: Our system incorporates modular agents that handle various tasks, such as data filtration, response restructuring, and database querying. This agent-based approach enhances the flexibility and scalability of the system, allowing for seamless integration and efficient processing. These systems consist of multiple independent components (agents) that work together to solve complex problems. In the context of a chatbot, agents can be specialized for specific tasks, such as answering technical questions, conducting sentiment analysis, or managing the conversation flow. Each agent contributes to creating a more precise and well-rounded response.

Retrieval-Augmented Generation (RAG): This approach combines traditional language models with external information retrieval. When the model needs to answer a query, it first searches a database or knowledge base for relevant information and then uses that to generate a more informed and accurate response. This method helps chatbots deliver answers based on real-time data and specialized knowledge.

Natural Language Processing (NLP)

The chatbot uses NLP techniques to parse, understand, and generate human-like responses to user queries.

Example: "The system employs tokenization, named entity recognition (NER), and sentiment analysis to process user inputs."

Let's break each concept down in detail, along with practical applications and implementation strategies.

4.2 Prompt Engineering in Our System

Prompt engineering is a critical component of our project, playing a pivotal role in optimizing the interaction between the chatbot and the large language models (LLMs). By carefully crafting and structuring prompts, we guide the LLMs to generate precise and contextually relevant responses to user queries. Here's how prompt engineering is implemented in our system:

Understanding User Requests: When a user interacts with the chatbot, they provide an interior design request. This request can vary widely in complexity and detail, from simple questions about color schemes to more intricate queries about room layout and furniture recommendations. The first step in prompt engineering is to accurately capture and understand the user's intent.

Constructing the Initial Prompt: Based on the user's request, the system constructs an initial prompt that encapsulates the key elements of the query. This involves identifying the core components of the request, such as the specific design aspect (e.g., lighting, furniture, color), any constraints or preferences mentioned by the user, and any additional context that might influence the response.

Data Concatenation and Enhancement: To enrich the prompt, the system may concatenate relevant data and functionality. This can include integrating user preferences, historical data, and other contextual information that can enhance the quality of the response. By providing the LLM with a comprehensive and detailed prompt, we ensure that the response generated is tailored to the user's needs.

Response Filtration and Refinement: Once the initial prompt is sent to Gemini, it processes the request and returns a response. This response undergoes a filtration process to ensure it meets the required criteria for accuracy and relevance. Any extraneous or irrelevant information is filtered out, and the core response is retained for further refinement.

Restructuring the Prompt: Based on the filtered response, the prompt is restructured to fine-tune the input for the LLMs. This step involves adjusting the wording, format, and structure of the prompt to guide the model towards generating the most accurate and contextually appropriate output. The goal is to create a prompt that is clear, concise, and aligned with the user's intent.

Generating the Final Response: With the restructured prompt, the system queries the LLMs to generate the final response. This response is then processed by the backend to retrieve relevant products and solutions from the database. The final output is presented to the user through the chatbot interface, ensuring a seamless and efficient user experience.

Benefits of Prompt Engineering

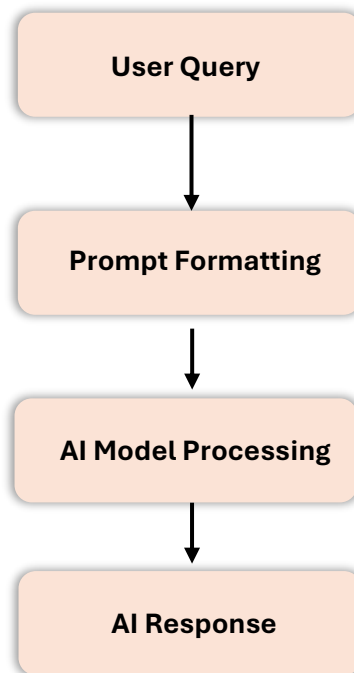
Accuracy: By carefully crafting prompts, we ensure that the LLMs generate responses that are accurate and relevant to the user's query.

Contextual Relevance: Prompt engineering allows us to incorporate contextual information, leading to responses that are more personalized and tailored to the user's needs.

Efficiency: Structured and well-defined prompts help streamline the interaction between the chatbot and the LLMs, reducing processing time and improving system performance.

Flexibility: The modular nature of prompt engineering enables us to adapt and refine prompts based on user feedback and system performance, enhancing the overall effectiveness of the system.

Block Diagram of Prompt Engineering



4.3 Retrieval-Augmented Generation (RAG):

is a powerful technique that combines the strengths of information retrieval and text generation to provide accurate, relevant, and contextual appropriate responses. In our interior design project, RAG plays a crucial role in ensuring that the chatbot delivers high-quality design suggestions and product recommendations based on user queries. By leveraging RAG, we can enhance the chatbot's performance and improve the overall user experience.

How RAG Works in Our System

User Request Handling: When a user interacts with the chatbot and makes an interior design request, the system captures the essential elements of the query. For example, a user might ask for suggestions to redesign their living room with a modern style and neutral colors.

Initial Prompt Construction: The system constructs an initial prompt based on the user's request, capturing the key elements and providing a clear context for further processing. This prompt includes details such as the desired design style, color preferences, and any specific requirements mentioned by the user.

Data Retrieval: Using the initial prompt, the system performs information retrieval to gather relevant data from various sources. These sources can include design blogs, product catalogs, user reviews, and other databases. The retrieval process ensures that the system has access to up-to-date and accurate information that can inform the generation of responses.

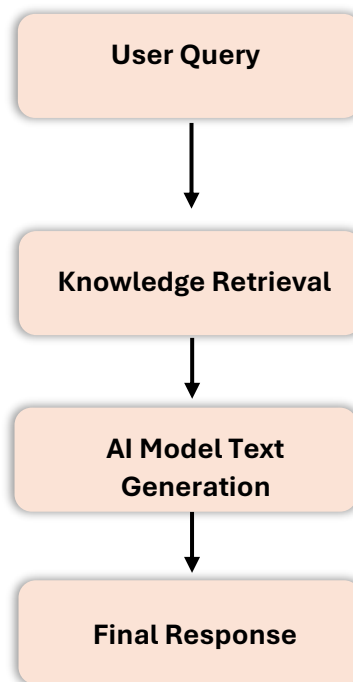
Enhanced Prompt Creation: The retrieved data is then used to enhance the initial prompt, providing additional context and details that can guide the text generation process. This enhanced prompt includes not only the user's original query but also relevant information such as popular design trends, recommended products, and user preferences.

Response Generation: The enhanced prompt is fed into the large language models (LLMs) to generate a coherent and contextually appropriate response. By combining the retrieved data with the generative capabilities of the LLMs, the system can produce responses that are both accurate and relevant to the user's query.

Backend Processing and Product Retrieval: The generated response is further processed by the backend system, which may involve querying the database to retrieve specific products and solutions. This step ensures that the final output includes actionable recommendations and relevant product information.

Final Response Presentation: The final response, which includes design suggestions and product recommendations, is presented to the user through the chatbot interface. The response is tailored to the user's preferences and provides valuable insights and options for their interior design project.

Block Diagram of RAG



4.4 Agent-based systems

are an integral part of our interior design project, enhancing the system's modularity, flexibility, and scalability. In this project, an agent-based system comprises multiple independent agents, each dedicated to a specific task or functionality. These agents collaborate to efficiently process user requests, ranging from data retrieval and prompt engineering to backend processing and response generation. Leveraging agent-based principles allows us to create a robust and dynamic system that delivers accurate and relevant interior design solutions to users.

Components of the Agent-Based System

User Request Handling Agent: This agent is responsible for capturing and interpreting the user's interior design requests. It identifies key components of the query, such as design preferences, style, and specific requirements. The User Request Handling Agent constructs the initial prompt, setting the stage for further processing.

Data Retrieval Agent: The Data Retrieval Agent gathers relevant information from various sources, such as design blogs, product catalogs, and user reviews. This agent ensures that the system has access to accurate and up-to-date data,

enriching the initial prompt with additional context and details. By leveraging retrieved data, the agent enhances the quality of the final response.

Prompt Engineering Agent: This agent plays a critical role in structuring and optimizing the input prompts for the large language models (LLMs). It takes the initial prompt and the retrieved data to create an enhanced prompt that guides the LLMs towards generating precise and contextually relevant responses. The Prompt Engineering Agent ensures that the prompts are well-defined and aligned with the user's intent.

Response Filtration Agent: The Response Filtration Agent processes the responses generated by the LLMs to ensure they are relevant and accurate. This agent filters out any extraneous or irrelevant information, retaining the core response that meets the required criteria. The filtered response is then passed on for further refinement and backend processing.

Backend Processing Agent: This agent is responsible for querying the database to retrieve specific products and solutions based on the filtered response. It ensures that the final output includes actionable recommendations and relevant product information. The Backend Processing Agent facilitates seamless data retrieval and integration, enhancing the system's overall efficiency.

Response Generation Agent: The Response Generation Agent combines the refined prompt and the filtered response to generate the final output. This agent leverages the generative capabilities of the LLMs to produce coherent and contextually appropriate responses. The final output is then processed and presented to the user through the chatbot interface.

User Interaction Agent: This agent manages the presentation of the final response to the user. It ensures that the response is clear, concise, and aligned with the user's preferences. The User Interaction Agent facilitates seamless communication between the user and the system, enhancing the overall user experience.

Benefits of Agent-Based Systems

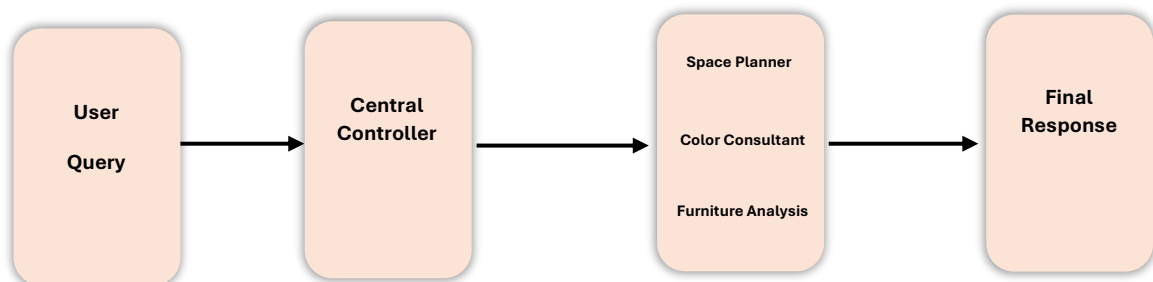
Modularity: Dividing the system into independent agents enhances modularity, allowing for easier management and updates of individual components without affecting the entire system.

Flexibility: Agent-based systems enable the adaptation and refinement of specific functionalities based on user feedback and system performance, improving overall effectiveness.

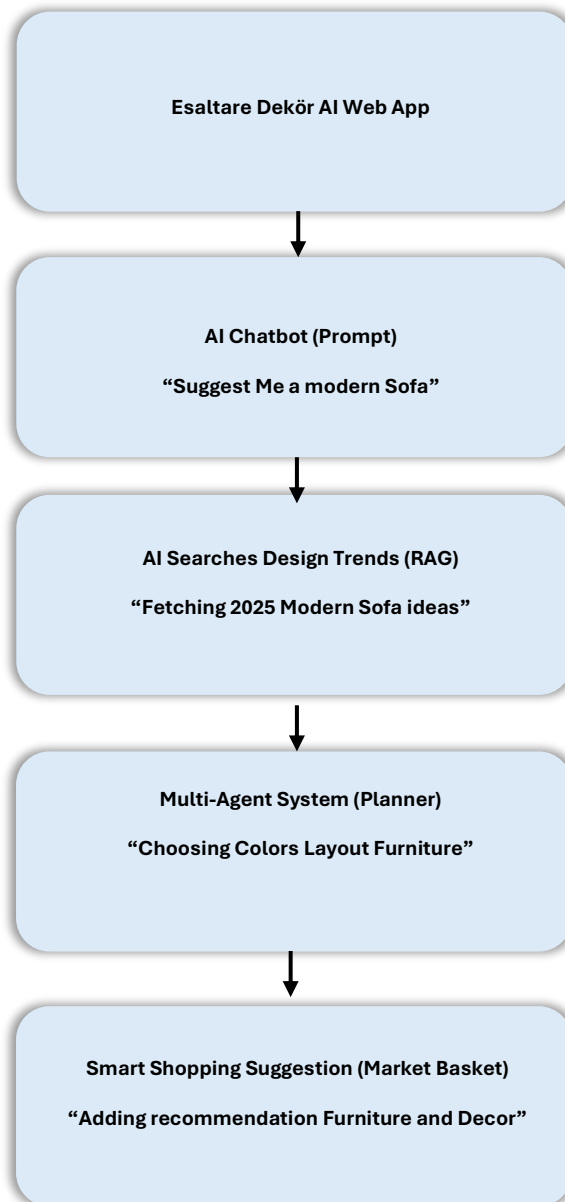
Scalability: The modular nature of agent-based systems allows for efficient scaling, with the ability to add new agents or functionalities as needed to accommodate increased user demand.

Efficiency: Independent agents can process different tasks simultaneously, improving the system's overall efficiency and responsiveness.

Block Diagram of Agent-based systems



4.5 Esaltare Dekor How It All Works Together



Our project focuses on creating a chatbot that integrates with Google's Gemini API to deliver intelligent responses and recommendations for interior design and decoration. This chatbot utilizes a comprehensive database of products to suggest relevant items to users, allowing them to conveniently add these products to their cart or favorites. By combining natural language processing (NLP) with advanced recommendation systems and e-commerce functionalities, the aim is to enhance the user experience by making it seamless and enjoyable.

The chatbot's ability to understand and respond to user queries is powered by NLP, which allows it to interpret and process natural language inputs effectively. This means users can interact with the chatbot using everyday language, making the experience more intuitive and user-friendly. Additionally, the recommendation system is designed to analyze user preferences and provide personalized suggestions, ensuring that the items recommended are tailored to each user's unique tastes and requirements.

The integration with e-commerce functionalities enables users to seamlessly transition from browsing to purchasing within the same platform. Users can view detailed information about the suggested products, compare different items, and make informed decisions before adding them to their cart or favorites. This holistic approach not only streamlines the shopping process but also enhances user satisfaction by providing a comprehensive solution for interior design and decoration needs.

Overall, the project aims to create a chatbot that leverages the power of Google's Gemini API, NLP, and recommendation systems to deliver a sophisticated and user-centric experience. By integrating these technologies, the chatbot will offer intelligent and personalized responses, making it a valuable tool for users looking to improve their interior design and decoration projects. Through this seamless combination of functionalities, the chatbot will provide an efficient, enjoyable, and enriching user experience.

4.6 Key Functionalities

Topic-Based Query Handling:

- The chatbot uses NLP to understand user queries related to interior design and decoration.
- It provides context-aware responses using the Gemini API, which is trained in a vast corpus of design-related knowledge.

Product Recommendation:

- The chatbot suggests products from a database based on user preferences, room type, color schemes, or design themes.
- Recommendations are generated using collaborative filtering or content-based filtering algorithms.

Cart Management:

- Users can add recommended products to their shopping cart directly through the chatbot interface.
- The cart is stored in the database and linked to the user's profile.

Favorites Management:

- Users can save products to their favorites list for future reference.
- The favorites list is stored in the database and accessible across sessions.

User Interaction:

- The chatbot provides a conversational interface for users to interact naturally.
- It supports multi-turn conversations to refine recommendations based on user feedback

4.7 System Architecture

Frontend:

The user interacts with the chatbot through a web-based interface or messaging platform

Backend:

The backend consists of:

- Chatbot Engine: Handles NLP and dialog management.
- Recommendation Engine: Generates product suggestions.
- Database: Stores product details, user profiles, carts, and favorites.
- Gemini API Integration: Provides design-related knowledge and context-aware responses.

Database:

- A relational database (MySQL) store:
- Product information (e.g., name, description, price, image).
- User data (e.g., profiles, carts, favorites).
- Transaction history.

APIs:

- Gemini API: For NLP and knowledge retrieval.
- E-Commerce API: For product browsing, cart management, and checkout.

1- User Query: The user asks a question or requests recommendations (e.g., "Suggest a modern sofa for a small living room").

2- NLP Processing: The chatbot uses NLP to understand the query and extract key details (e.g., "modern sofa," "small living room").

3- Recommendation Generation: The chatbot queries the database and recommendation engine to find suitable products.

4- Response Generation: The chatbot provides a response with product suggestions.

5- Action: The user writes a prompt with a specific action that chatbot can automatically handle.

6- Database action

EDAI Pipeline

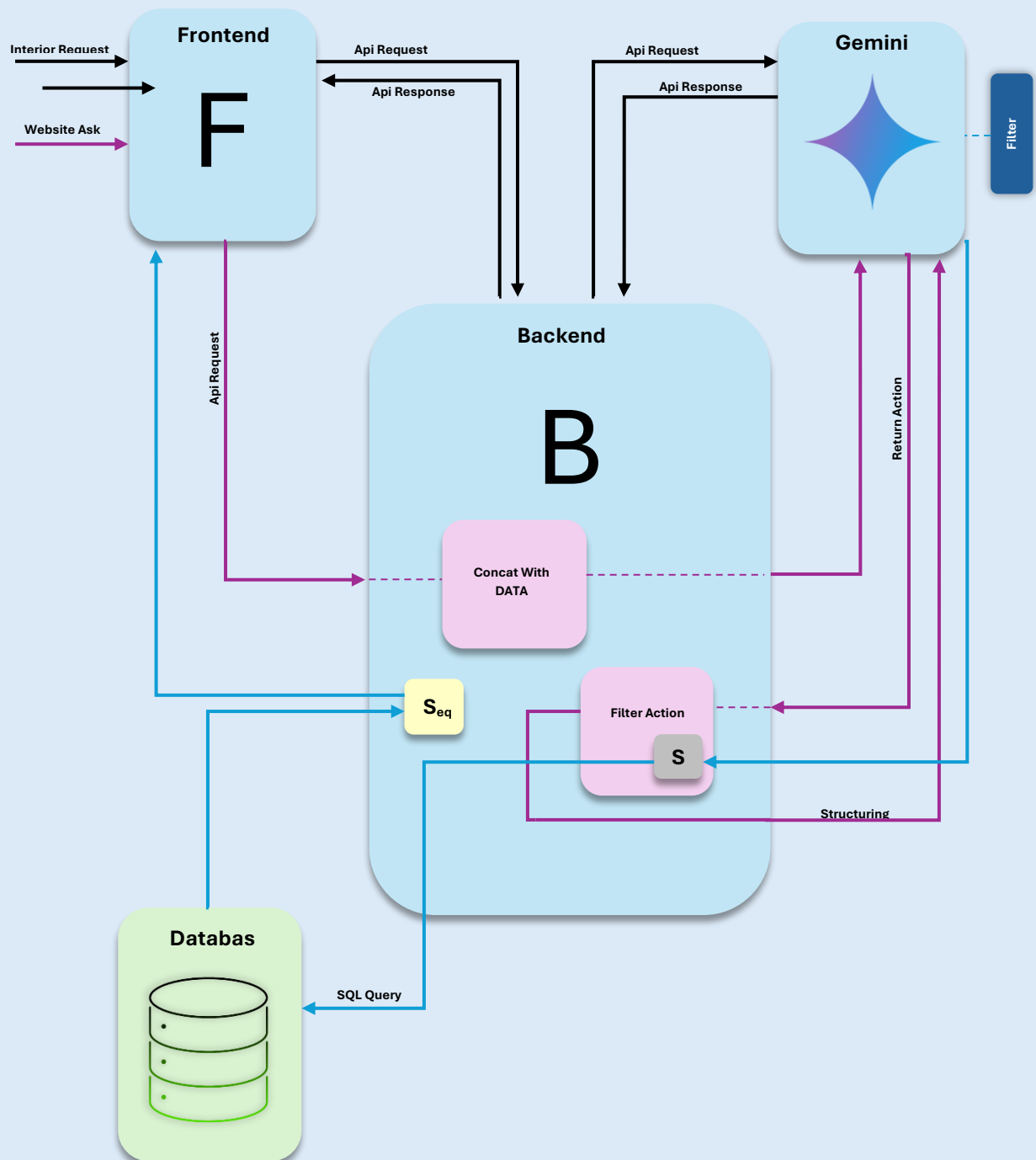


Figure 4-1 EDAI Pipeline

4.8 Personality Analysis

This project involves the development of a personality analysis system designed to assess a user's decoration style preferences through a structured multiple-choice questionnaire. The system classifies users into predefined personality categories based on their responses, helping them discover their ideal interior design style.

Integration with Gemini AI

To enhance the accuracy of the analysis and provide personalized recommendations, Gemini AI is integrated into the system. This integration allows for a more refined interpretation of user responses and ensures that the recommendations are tailored to each individual's unique style preferences. Gemini AI analyzes the responses to the questionnaire, offering intelligent and personalized suggestions.

Personality-Based Design Choices: Understanding how a person's character and lifestyle influence their decorating style.

Interior Style Matching: Connecting users with design themes that suit their tastes and preferences.

Smart Personalization: Using AI to suggest colors, furniture, and decor that align with their personality.

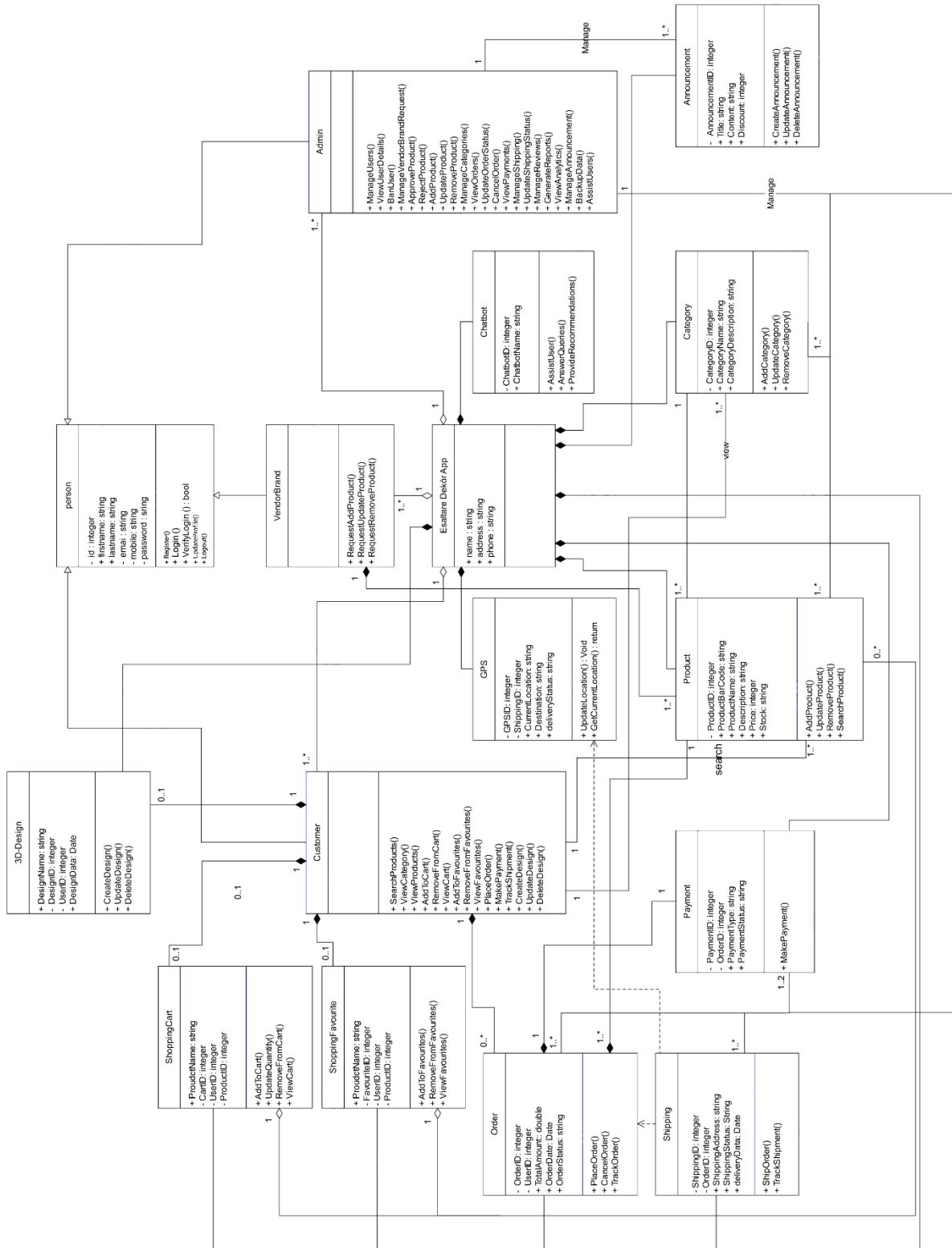
Pattern Recognition: Identifying trends in user responses to predict their ideal home style.

AI-Powered Recommendations: Enhancing suggestions using Gemini AI for accuracy and creativity.

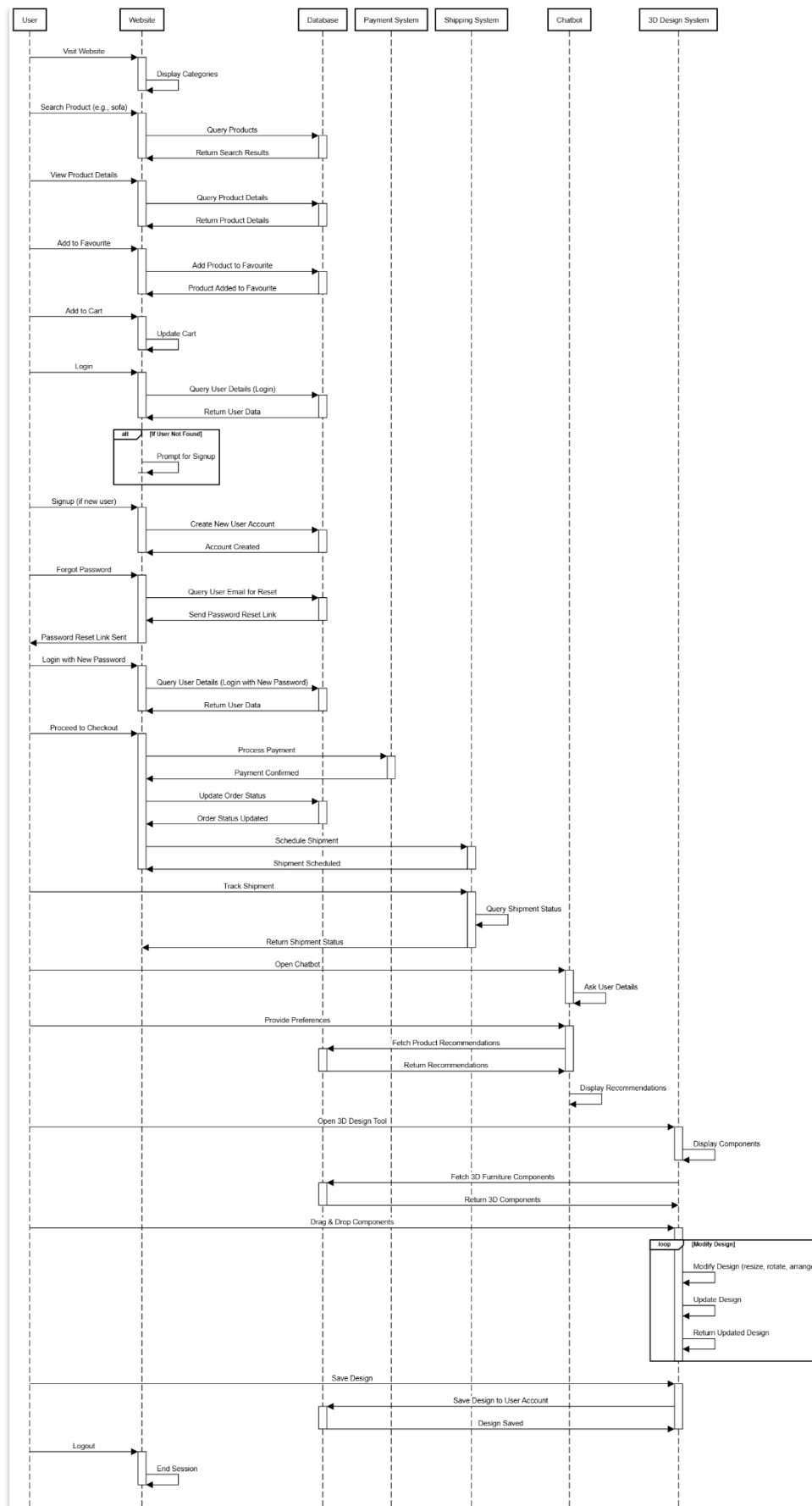
Visual Preference Analysis: Matching users with design elements that reflect their personality.

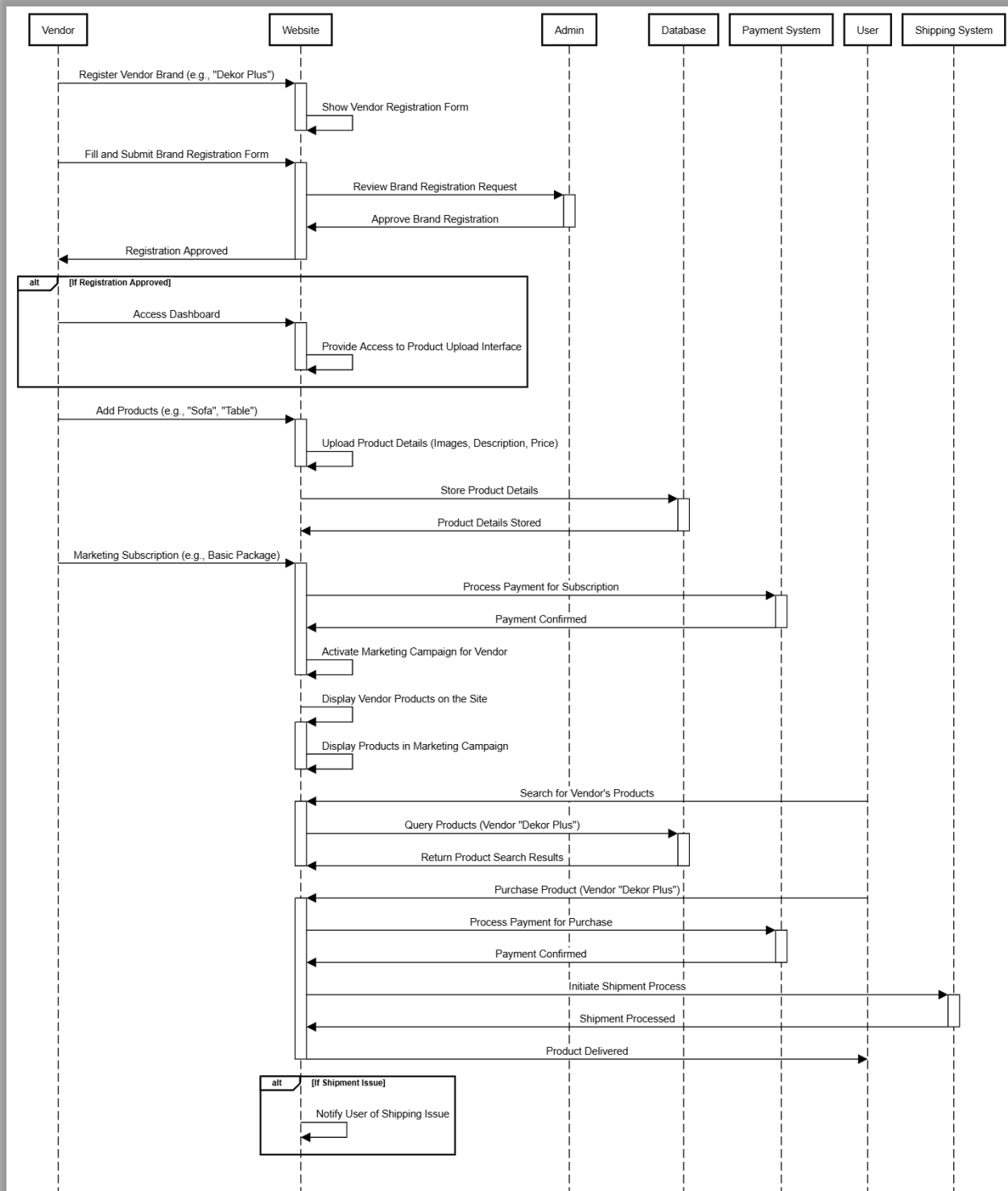
Chapter 5: Design The System

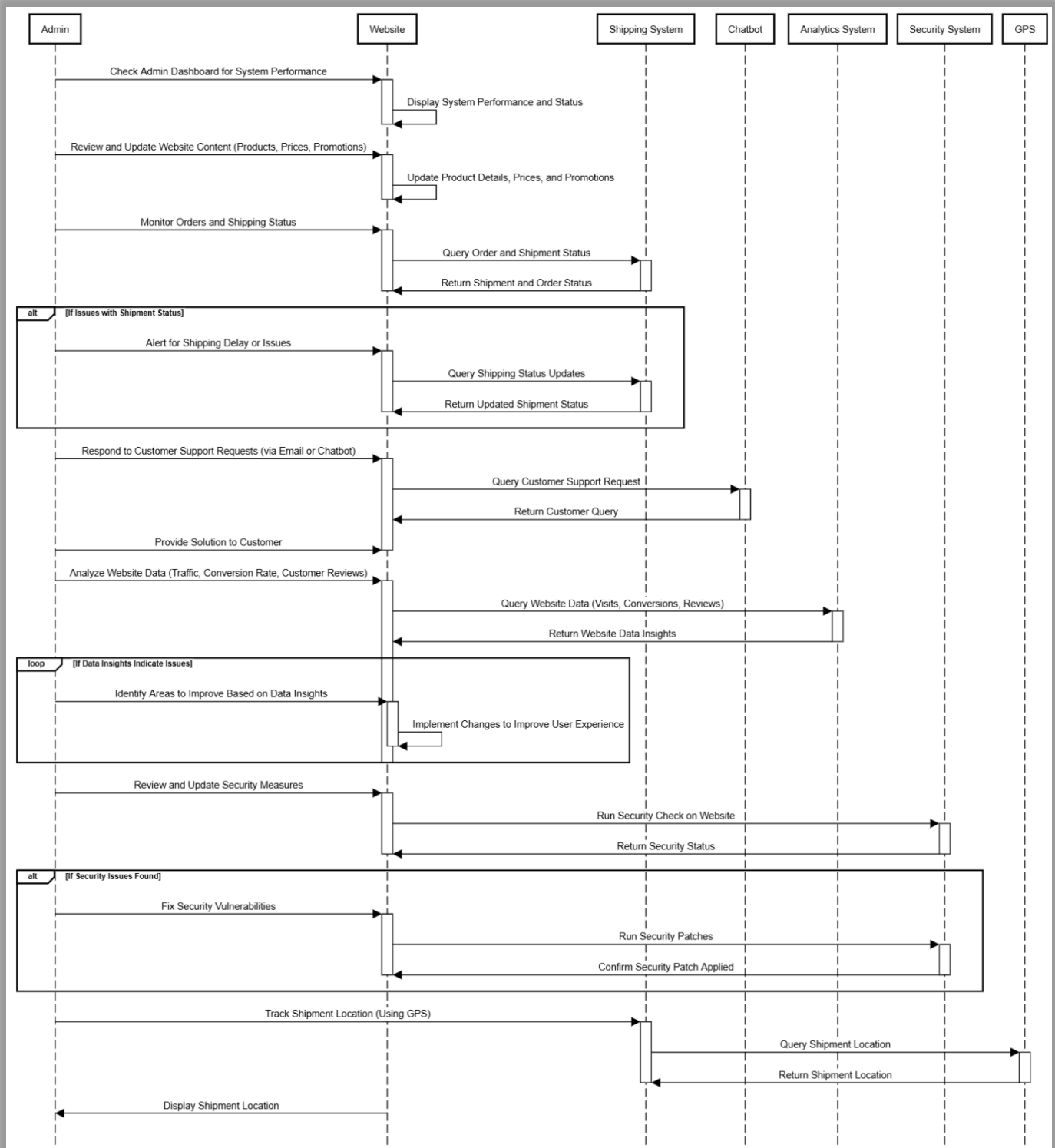
5.1 Class Diagram



5.2 Sequence Diagram







User, Vendor, and Admin. Each of these actors plays a vital role in ensuring the smooth operation of the platform and delivering a seamless experience

1. User Sequence of Actions:

The User begins their journey by visiting the platform. The system displays the available product categories, allowing the user to browse through various options. When the user searches for a specific product, such as a sofa, the website queries the database for relevant products and displays the search results.

After selecting a product, the user views its details, including descriptions, prices, and images. The user can then add the product to their shopping cart, and the system updates the cart to reflect the selection. If the user is logged in, they proceed directly to check out. If not, they are prompted to log in or register for an account.

Once logged in, the user proceeds to make a purchase. The website processes the payment, and if the payment is successful, the order is confirmed, and shipment is scheduled. If payment fails, the user is prompted to provide an alternative payment method, and the system retries the payment.

Upon confirmation of the order, the user receives shipment details and can track the shipment status. Additionally, the user can interact with the Chatbot to receive product recommendations based on preferences or utilize the 3D Design Tool to visualize and customize furniture before purchase.

2. Vendor Sequence of Actions:

The Vendor begins their interaction with the platform by submitting a request to list their brand. This request is reviewed and approved by the Admin. Once approved, the vendor gains access to a control panel, where they can add products to the platform. These products are added with complete details such as high-quality images, descriptions, and pricing information.

After the products are submitted, the Admin reviews and approves them. Once the products are approved, they become available for sale on the platform. The Vendor can monitor the performance of their products, making adjustments to pricing, descriptions, or availability when needed.

The Vendor also engages in marketing campaigns and can track the effectiveness of these promotional activities. When a user purchases one of the vendor's products, the vendor receives a notification, processes the order, and prepares the product for shipment. The Vendor ensures that the product is shipped to the user in a timely manner.

3. Admin Sequence of Actions:

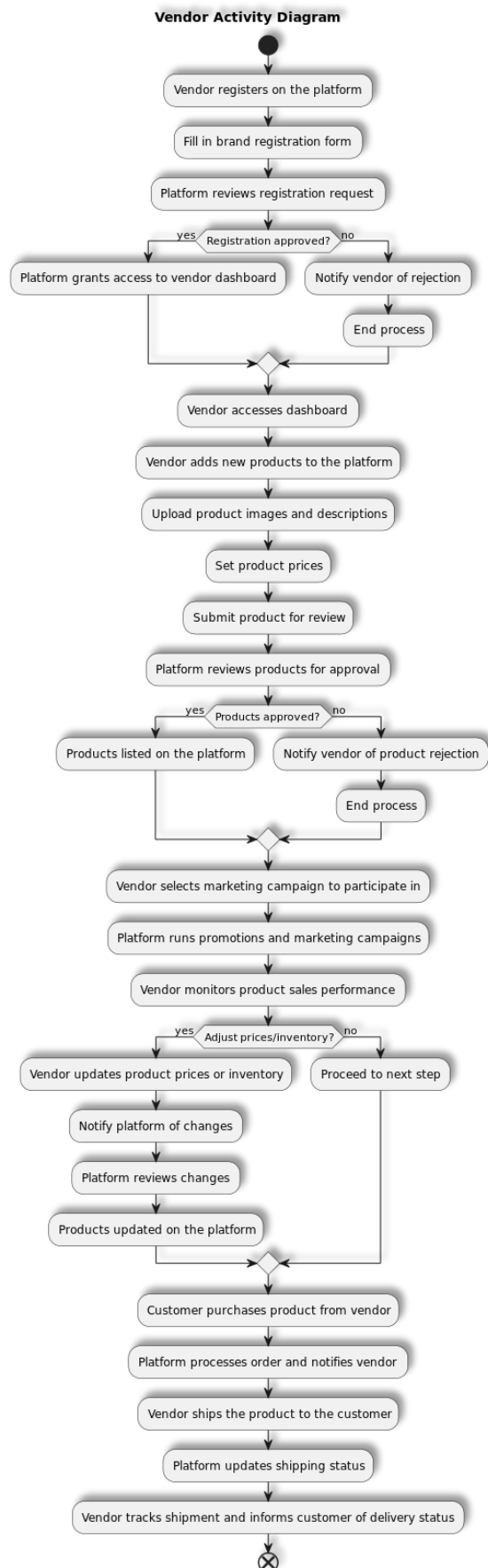
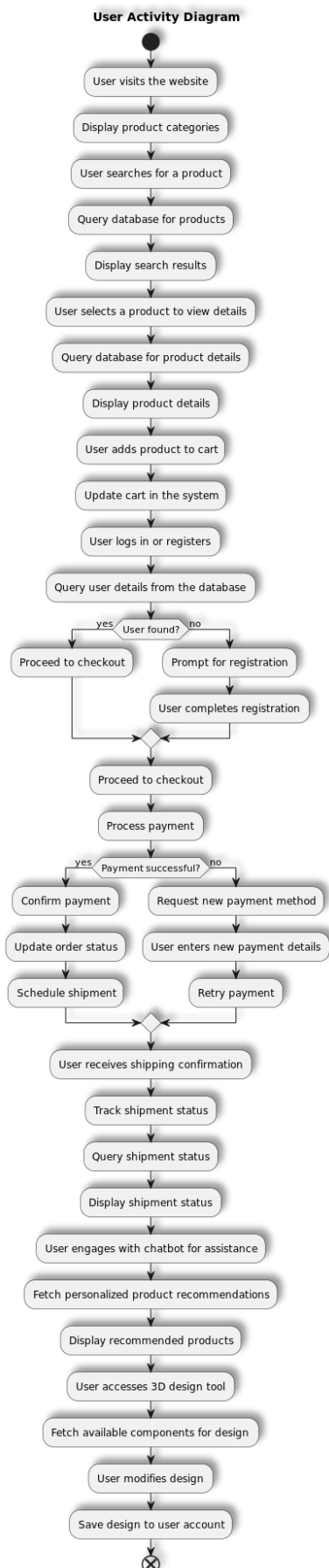
The Admin is responsible for overseeing the entire platform, ensuring it functions smoothly and efficiently. The Admin logs into the platform, starting the day by reviewing the system's health and monitoring performance. The Admin checks for any technical issues, ensuring all components are functioning properly.

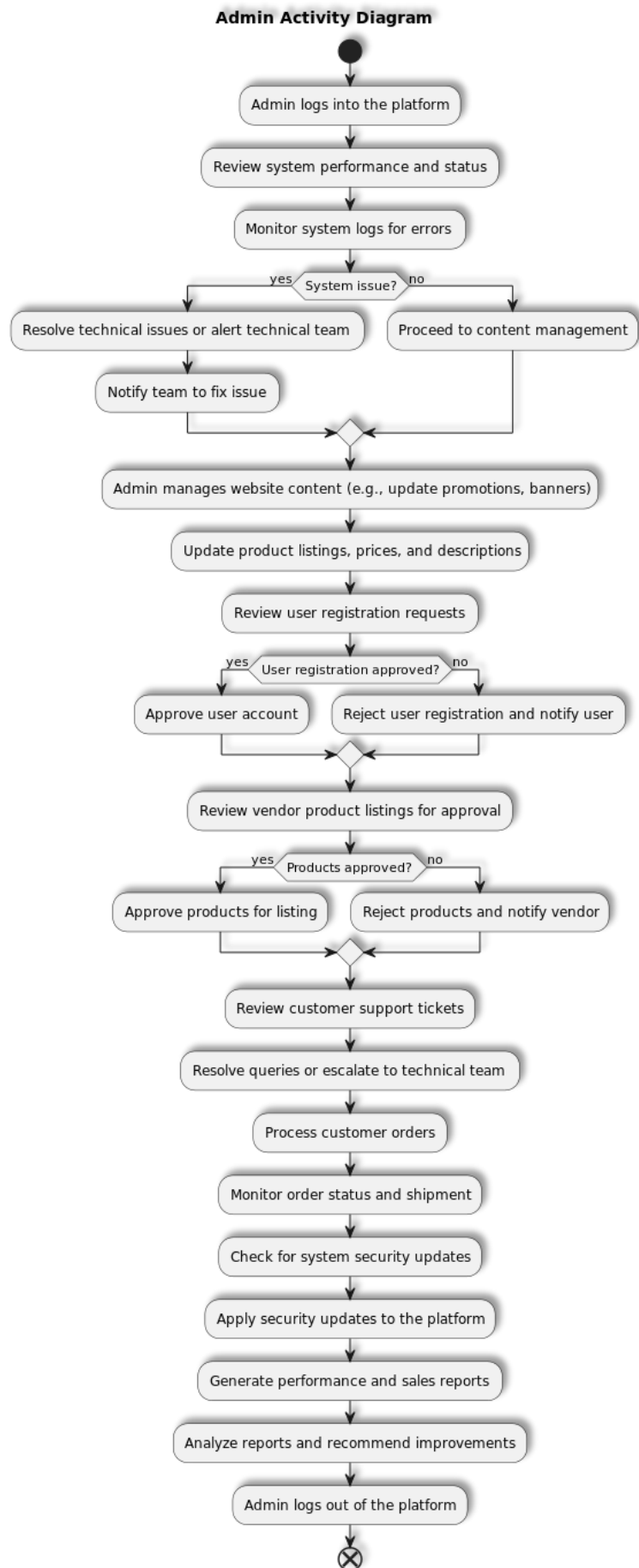
The Admin manages content on the platform, including approving vendor product submissions. When a Vendor submits a product, the Admin reviews the product details and either approves or rejects it based on the platform's standards. The Admin also manages user registrations, ensuring that new users can sign up and existing users can log in successfully.

Additionally, the Admin handles customer support, addressing queries and issues from both users and vendors. The Admin processes and manages orders, ensuring that products are shipped to users on time. They also conduct regular system security checks to safeguard sensitive data and maintain platform integrity.

The Admin generates reports from the platform's data, analyzing traffic, sales, and customer behavior. Using these insights, the Admin implements improvements, such as optimizing product listings and enhancing user experience, to increase sales and platform engagement.

5.3 Activity Diagram





1. User Activity Flow:

The User's activity begins by visiting the website and browsing available product categories. Once they search for a specific product, the system queries the database and displays the relevant results. The user selects a product and views its details. They can then choose to add the product to their cart.

If the user is not logged in, they are prompted to either log in or register. Once logged in, they proceed to the checkout where they enter payment details. The platform processes the payment, and if the payment is successful, the order is confirmed, and shipment is scheduled. If the payment fails, the user is prompted to provide an alternate payment method, and the system retries the payment.

After the purchase is successful, the user receives shipping details and can track the shipment status. Additionally, the user can interact with a Chatbot for product recommendations or use the 3D Design Tool to design customized furniture.

The flow of tasks for the User is linear, with decisions around login and payment determining subsequent actions. If any step fails (like payment failure), the user is prompted to retry, ensuring they can still complete their purchase.

2. Vendor Activity Flow:

The Vendor's activity begins with submitting a request to list their products and brand on the platform. Once the request is reviewed and approved by the Admin, the vendor can log in and access their control panel to add product details. This includes uploading high-quality images, writing product descriptions, and setting prices.

After the products are submitted, the Admin reviews them, and once approved, the products go live on the website for users to browse and purchase. The Vendor can monitor their products' performance, manage pricing, and adjust to ensure competitive pricing or correct errors.

When a product is purchased, the Vendor is notified and prepares the item for shipment. They ensure timely delivery by collaborating with the Shipping System. Throughout the process, the Vendor may also engage in marketing activities or track promotional campaigns to boost visibility and sales.

The Vendor activity flow also follows a linear path. The main decisions are about product approval by the Admin and making adjustments to product listings. There are also feedback loops related to marketing efforts and product performance.

3. Admin Activity Flow:

The Admin is responsible for overseeing all system activities, managing users, vendors, and content. The Admin's activity begins with reviewing the platform's performance and ensuring that everything is running smoothly. They check the health system, monitor for issues, and troubleshoot problems as needed.

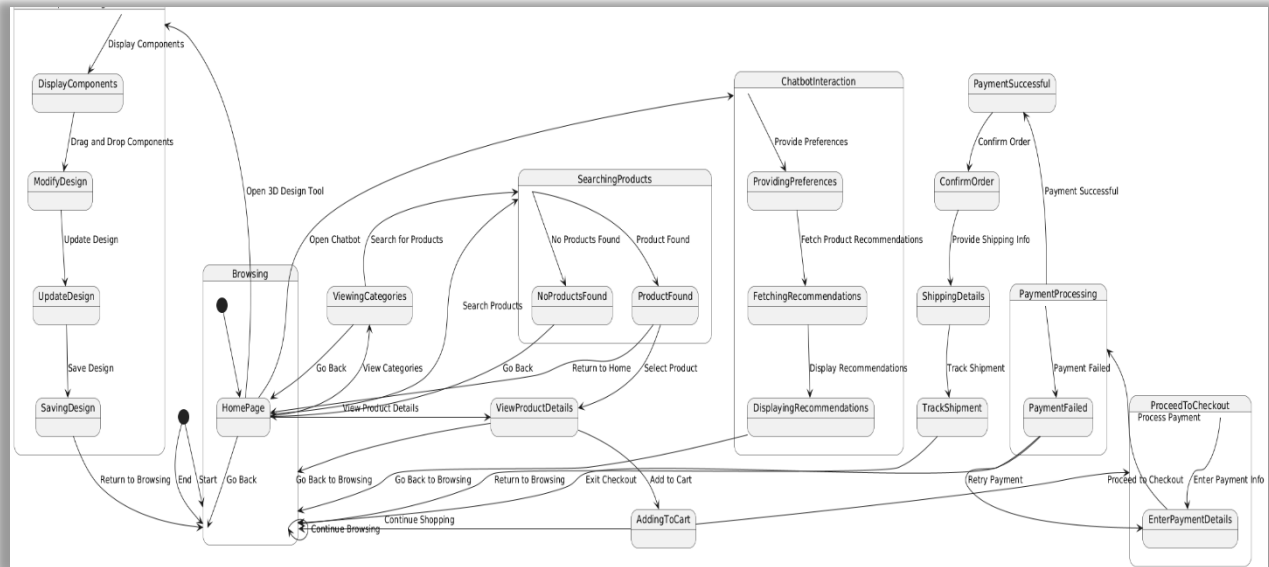
The Admin also manages content, approving or rejecting product listings from Vendors. When vendors submit new products, the Admin evaluates them based on platform guidelines and either approves them for publication or asks for revisions. The Admin is also responsible for managing user registrations and ensuring all users can log in without issues.

Additionally, the Admin provides customer support for both users and vendors. They may be contacted for troubleshooting, issues with orders, or general queries. The Admin monitors orders, ensuring products are shipped on time, and ensures that all security protocols are in place to safeguard the system and user data.

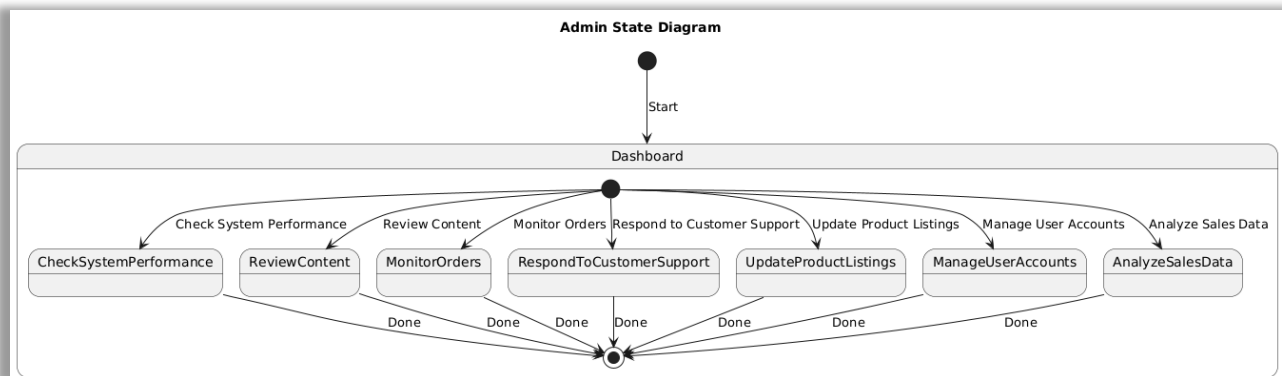
Admin flow is also linear, with decision points focused on approving products, managing users, and maintaining platform security. Loops occur around ongoing system checks and support requests.

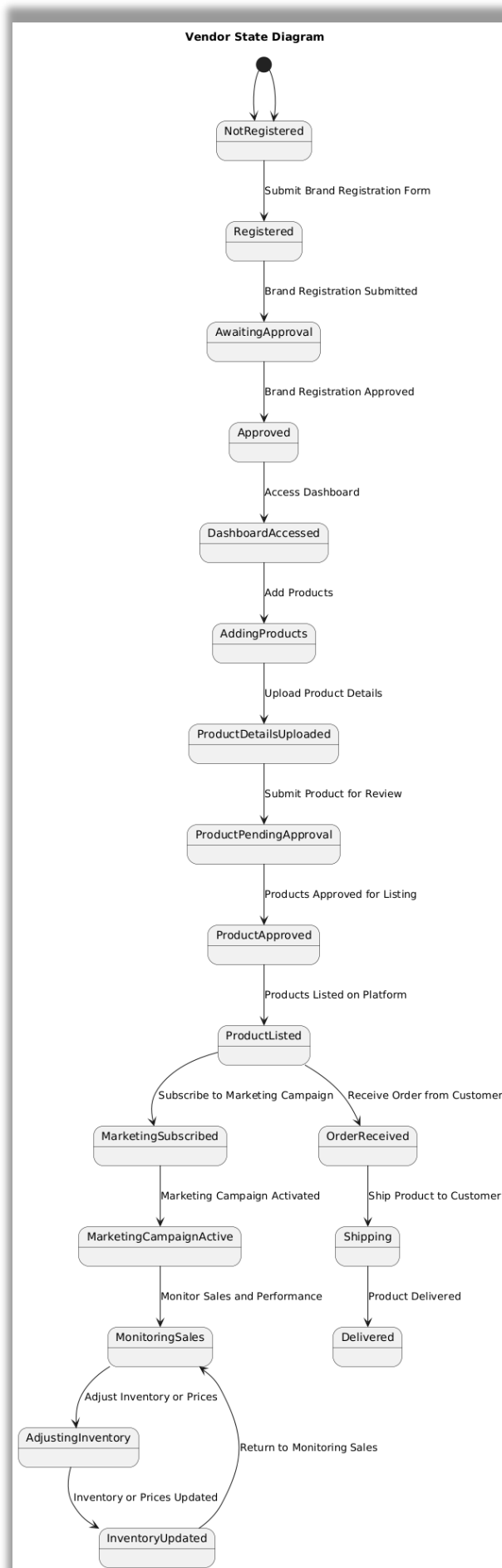
5.4 State Diagram

User State Diagram

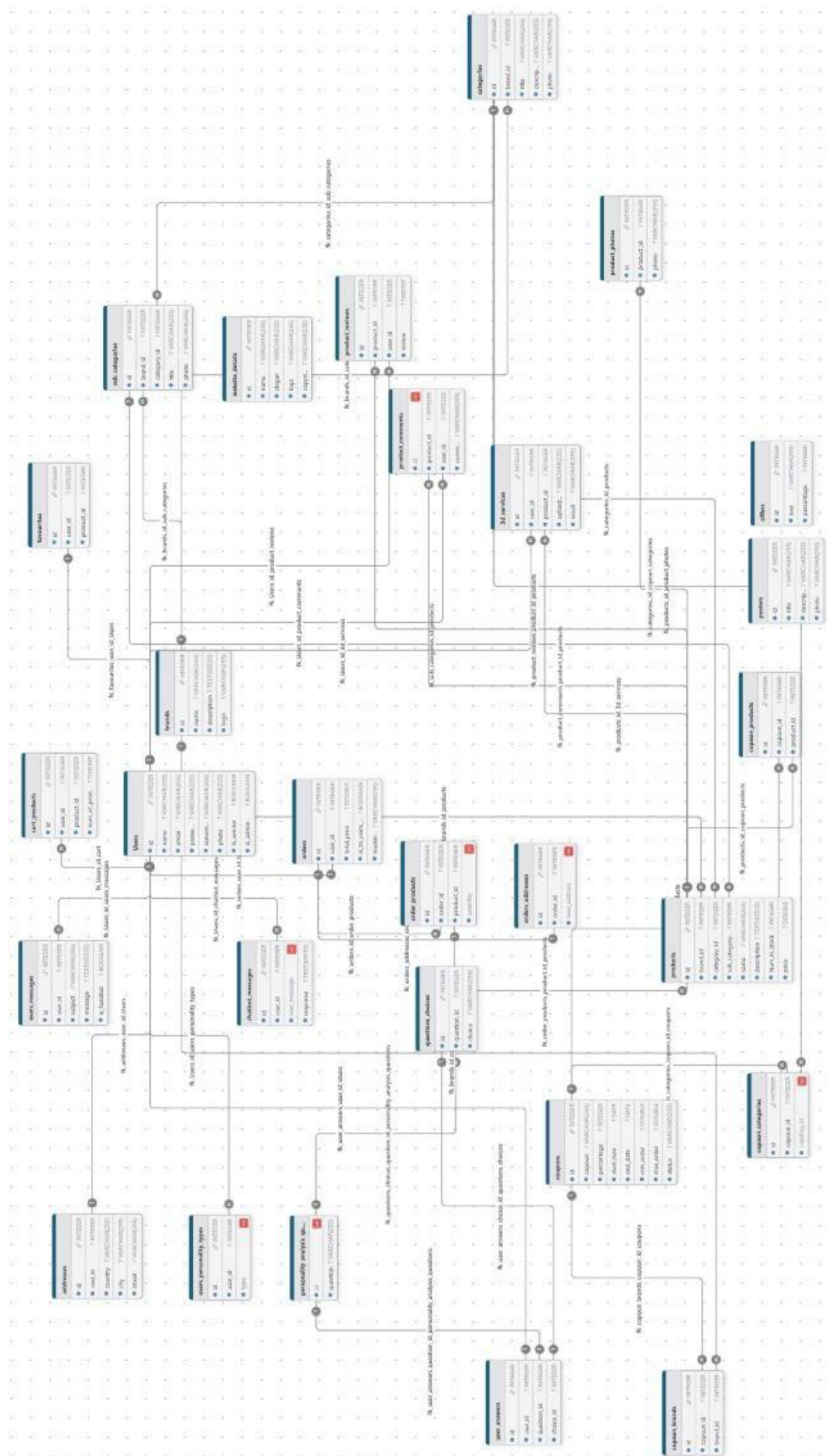


Admin State Diagram

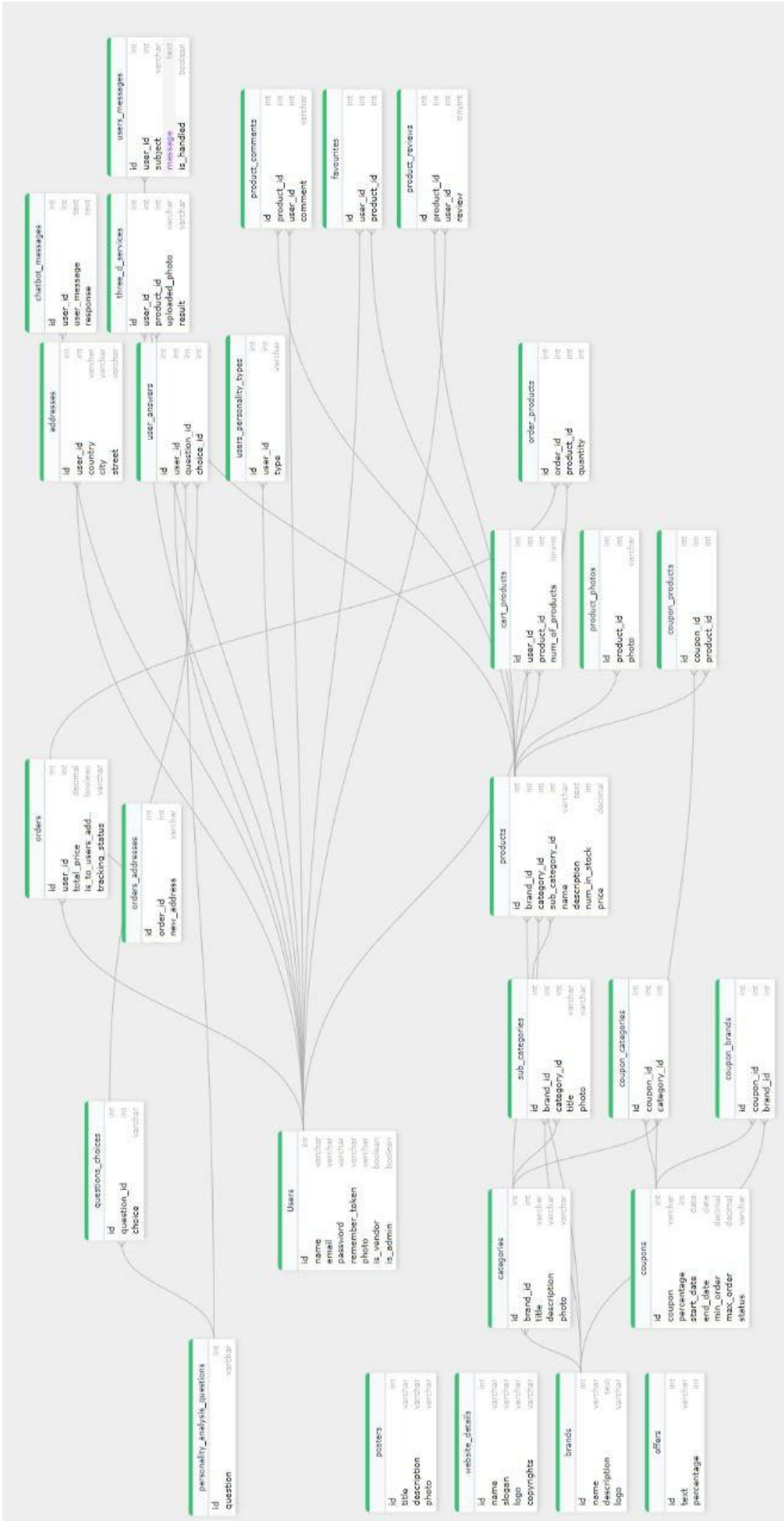




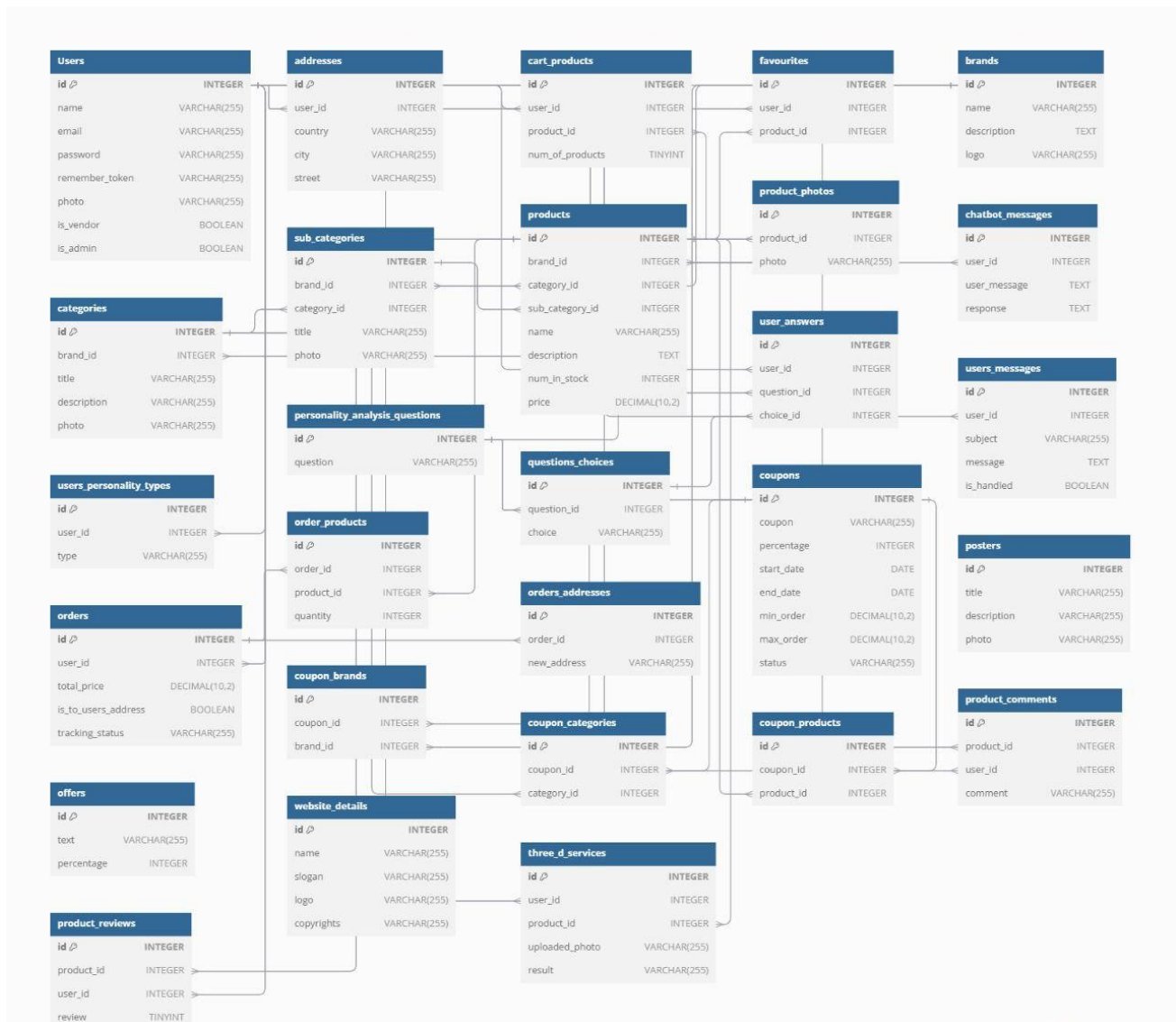
Database Schema



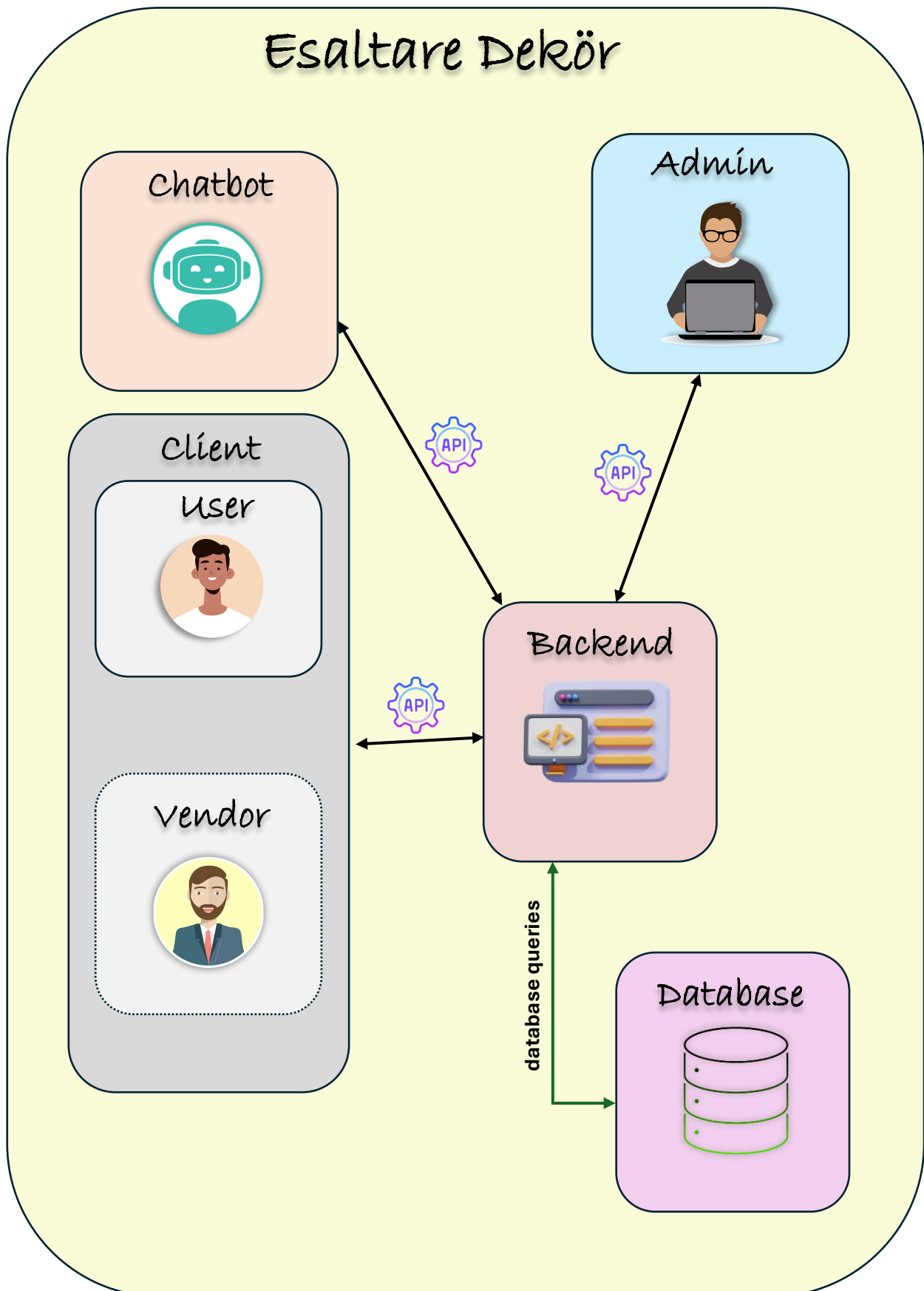
Database Flow



5.5.2 Relational Model or Logical Data Model. (2nd Layer Of Database)



5.6 System Architecture and Structure



5.7 User Interface and Experience Design

5.7.1 Design of The User Interface

A Renaissance in Bloom: The Florentine Silk Weaver's Secret

The year is 1468. Florence, a city bursting with artistic fervor, is abuzz with the whispers of the Renaissance. Amidst the labyrinthine alleys and sun-drenched piazzas, nestled in the heart of the bustling Oltrarno district, resides a modest workshop, the domain of Master Lorenzo, a silk weaver of renowned skill.

Lorenzo's workshop, a sanctuary of creativity, is where magic happens. Threads of the finest silk, dyed in a spectrum of colors gleaned from nature's bounty, dance under his nimble fingers, transforming into tapestries of breathtaking beauty. But Lorenzo holds a secret, a palette of colors passed down through generations, a treasure trove of hues that capture the very essence of Italy – the "Italian Palette."

This palette, inspired by the sun-drenched Tuscan landscape, the azure depths of the Mediterranean, and the earthy tones of ancient ruins, is Lorenzo's most prized possession. It holds the key to his unique artistry, allowing him to create silks that shimmer with life, imbued with the warmth of the Italian sun and the soul of its people.

Our story centers around Lorenzo's young apprentice, Isabella, a girl with a keen eye for color and an insatiable thirst for knowledge. Isabella, an orphan taken in by Lorenzo, is not just learning the craft of silk weaving; she is also being initiated into the secrets of the Italian Palette.

From the warm ochre of the Tuscan earth to the vibrant saffron of sun-ripened fields, from the deep green of cypress trees to the rich crimson of Chianti wine, Lorenzo unveils the stories behind each color, sharing the history and emotions woven into their very fabric.

"These colors, Isabella," Lorenzo would say, his voice raspy with age, "are not just pigments; they are memories, emotions, stories waiting to be told. They are the soul of Italy, passed down through generations, waiting to be woven into new tales."

Isabella, with her innate artistic sensibility, absorbs these lessons like a sponge. She learns to mix and blend the colors, creating subtle nuances that capture the fleeting light of dawn, the hazy warmth of midday, and the fiery hues of sunset.

One day, a commission arrives that could change the course of Isabella's life. The powerful Medici family, patrons of the arts and de facto rulers of Florence, have

requested a silk tapestry of unprecedented grandeur, a piece that would celebrate the beauty of their city and the spirit of the Renaissance.

Lorenzo, sensing his age, decides to entrust this monumental task to Isabella, recognizing her talent and understanding of the Italian Palette. "This is your chance, Isabella," he says, his eyes twinkling, "to show the world the magic you can create with these colors."

Isabella, overwhelmed yet determined, embarks on this ambitious project. She envisions a tapestry that captures the vibrant life of Florence, its bustling markets, its majestic palazzi, its serene gardens, all bathed in the warm glow of the Italian Palette.

She begins by sketching scenes from everyday life, capturing the energy of the city, the laughter of children, the barter in the marketplace, the quiet contemplation in the gardens. Then, she translates these sketches into a symphony of colors, carefully selecting each hue from the Italian Palette to evoke the desired emotion.

For months, Isabella works tirelessly, her nimble fingers weaving stories into the silk, bringing her vision to life. The workshop becomes her world, the rhythmic clatter of the loom her constant companion. The colors of the Italian Palette guide her, whispering tales of the past, inspiring dreams of the future.

As the tapestry nears completion, words spread throughout Florence about the masterpiece being created in Lorenzo's workshop. Curiosity and anticipation grow, reaching the ears of the Medici family themselves.

Finally, the day arrives for the unveiling. The Medici family, along with a throng of dignitaries, artists, and citizens, gather in the piazza outside Lorenzo's workshop. Isabella, her heart pounding with nervous excitement, steps forward and presents her creation.

As the tapestry is unfurled, a collective gasp fills the air. It is a breathtaking spectacle, a panorama of Florentine life, woven in the vibrant hues of the Italian Palette. The colors seem to dance and shimmer, capturing the essence of the Renaissance, the spirit of innovation, the celebration of beauty.

The Medici family is awestruck. They recognize in Isabella's work not just technical mastery, but also a deep understanding of the Italian soul, a connection to the artistic legacy of their city.

Isabella, once an orphaned apprentice, is now hailed as a master weaver, her name whispered with reverence alongside the great artists of the Renaissance. Her tapestry becomes a symbol of Florence, a testament to the power of the Italian Palette, a reminder that true artistry lies not just in skill, but in the ability to capture the heart and soul of a culture.

And so, the story of the Italian Palette continues, woven into the fabric of Florentine history, its colors inspiring generations of artists to come. Isabella's legacy lives on, a testament to the power of a young woman who dared to dream in color, who embraced the heritage of her craft, and who transformed the whispers of the past into a vibrant symphony of the present.

The accolades poured in, commissions multiplied, and Isabella's name became synonymous with Florentine artistry. Yet, amidst the whirlwind of success, Isabella remained grounded, forever grateful to Master Lorenzo and the legacy of the Italian Palette. She understood that her artistry wasn't merely about technical skill, but about the deep emotional connection she forged with the colors.

One crisp autumn evening, as the setting sun painted the Arno River in hues of burnt orange and rose, Isabella sat in her workshop, surrounded by skeins of silk. A new commission lay before her: a series of small panels destined for a private chapel, each meant to evoke a specific virtue. But the initial spark of inspiration eluded her.

She gazed around her workshop, seeking guidance. Her eyes fell upon a worn leather-bound book, Master Lorenzo's most treasured possession – a collection of natural dyes and pigments, each carefully labeled with its origin and traditional significance. Inside, pressed flowers and leaves served as reminders of the natural world's boundless beauty. It was here, within these timeworn pages, that the Italian Palette truly lived.

Isabella carefully opened the book, her fingers tracing the faded script. She lingered on a page filled with swatches of earthy tones: Terra di Siena, Ocre Gialla, Terra Verde. These were the colors of the Tuscan landscape, the rolling hills, the sunbaked earth, the ancient walls whispering stories of generations past. A wave of nostalgia washed over her, memories of her childhood spent exploring the countryside with Lorenzo. This, she thought, is where I will begin.

For the virtue of Patience, she chose the deep, muted green of Terra Verde, reminiscent of the slow, steady growth of olive trees. She paired it with the creamy off-white of Bianco di San Giovanni, evoking the purity and tranquility of a dove's wing. These colors, she felt, spoke of quiet strength, of enduring hope, of the gentle unfolding of time.

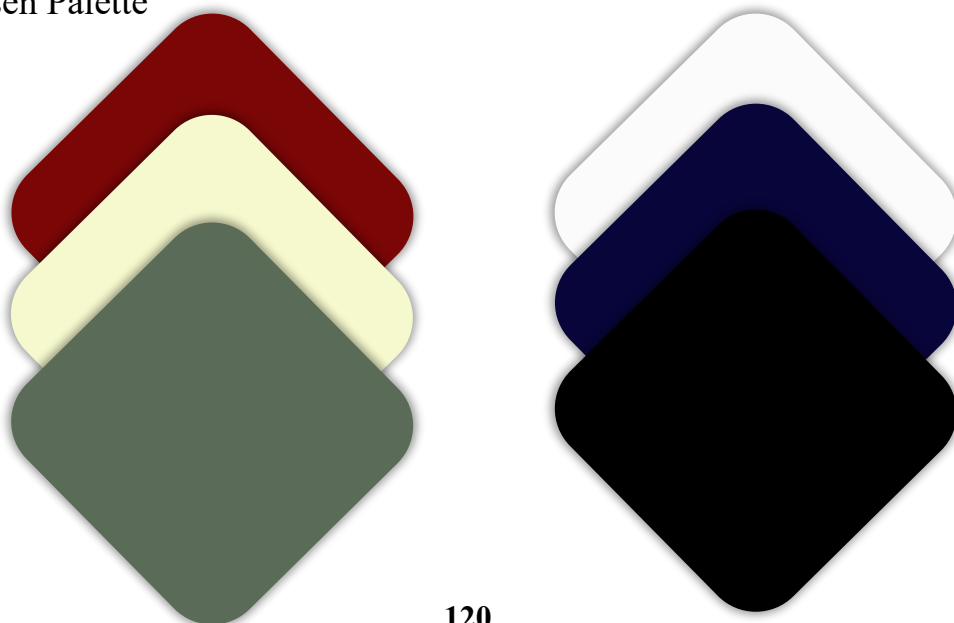
Turning the pages, she discovered a vibrant splash of Giallo Zafferano, the color of saffron, reminiscent of sun-drenched fields and the joyous energy of harvest. This, she knew, was the color of Generosity, its warmth and vibrancy reflecting the abundance of the human spirit. She decided to use it alongside Rosso Chiaro, a light, slightly orange-tinged red, like the blush on a ripe peach, symbolizing the sweetness of giving.

For Humility, she sought a more subdued palette. She chose Grigio di Pietra, the color of weathered stone, a reminder of the enduring strength found in simplicity. She combined it with Blu oltremare chiaro, a soft, muted blue evocative of the vast sky, suggesting the boundless nature of true humility.

And so, for each virtue, Isabella meticulously selected colors from the Italian Palette, drawing inspiration from the natural world, from memories, and from the emotional resonance of each hue. She didn't just choose colors; she chose stories, emotions, and pieces of her own soul.

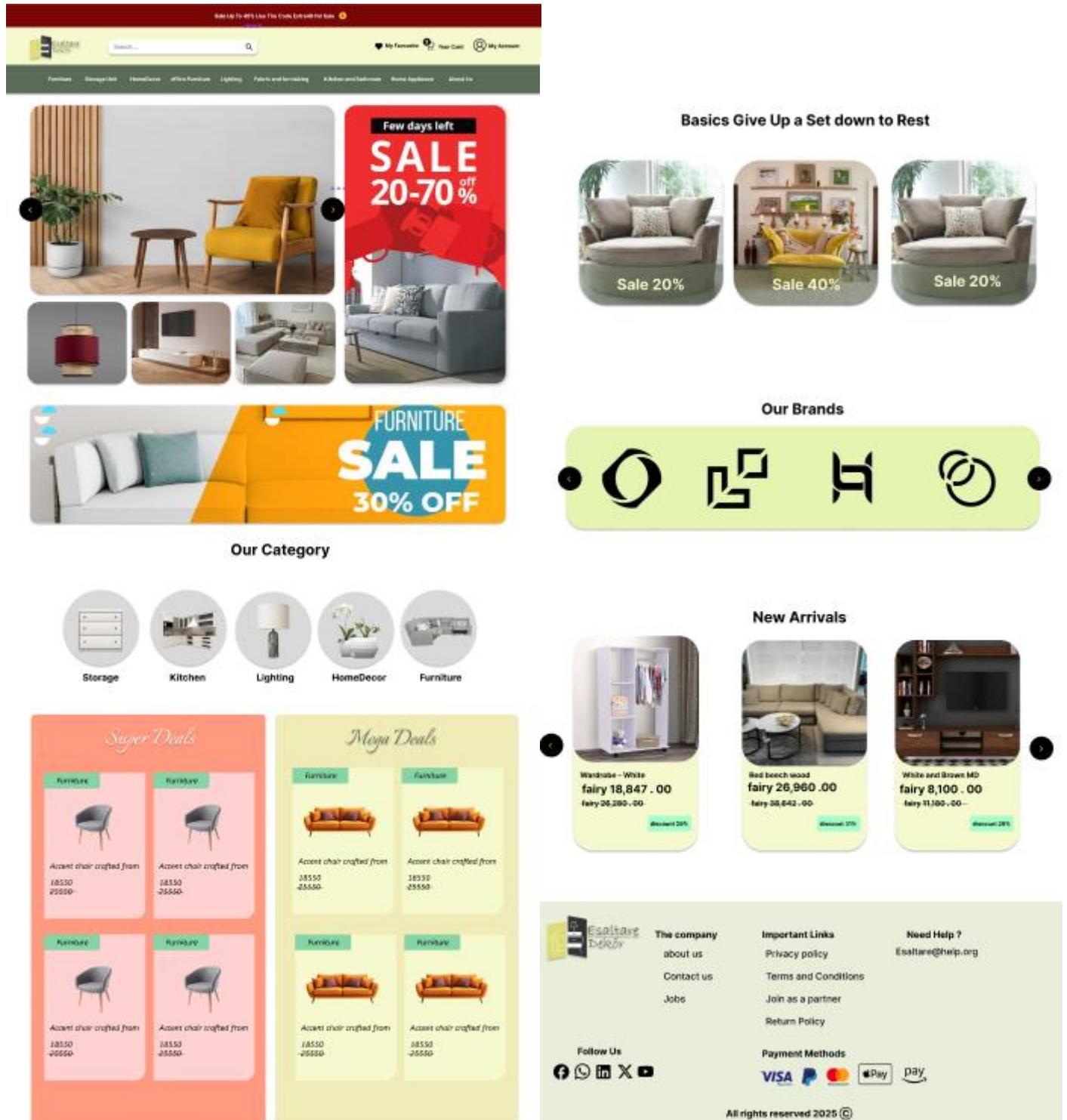
This process of palette selection became an integral part of her creative process. It was a moment of reflection, a dialogue with the past, a connection to the artistic legacy she now carried. The Italian Palette wasn't just a collection of colors; it was a living language, a source of endless inspiration, a key to unlocking the stories waiting to be woven into silk. And with each new commission, Isabella honored that legacy, breathing new life into the timeless hues of her beloved Italy.

The chosen Palette

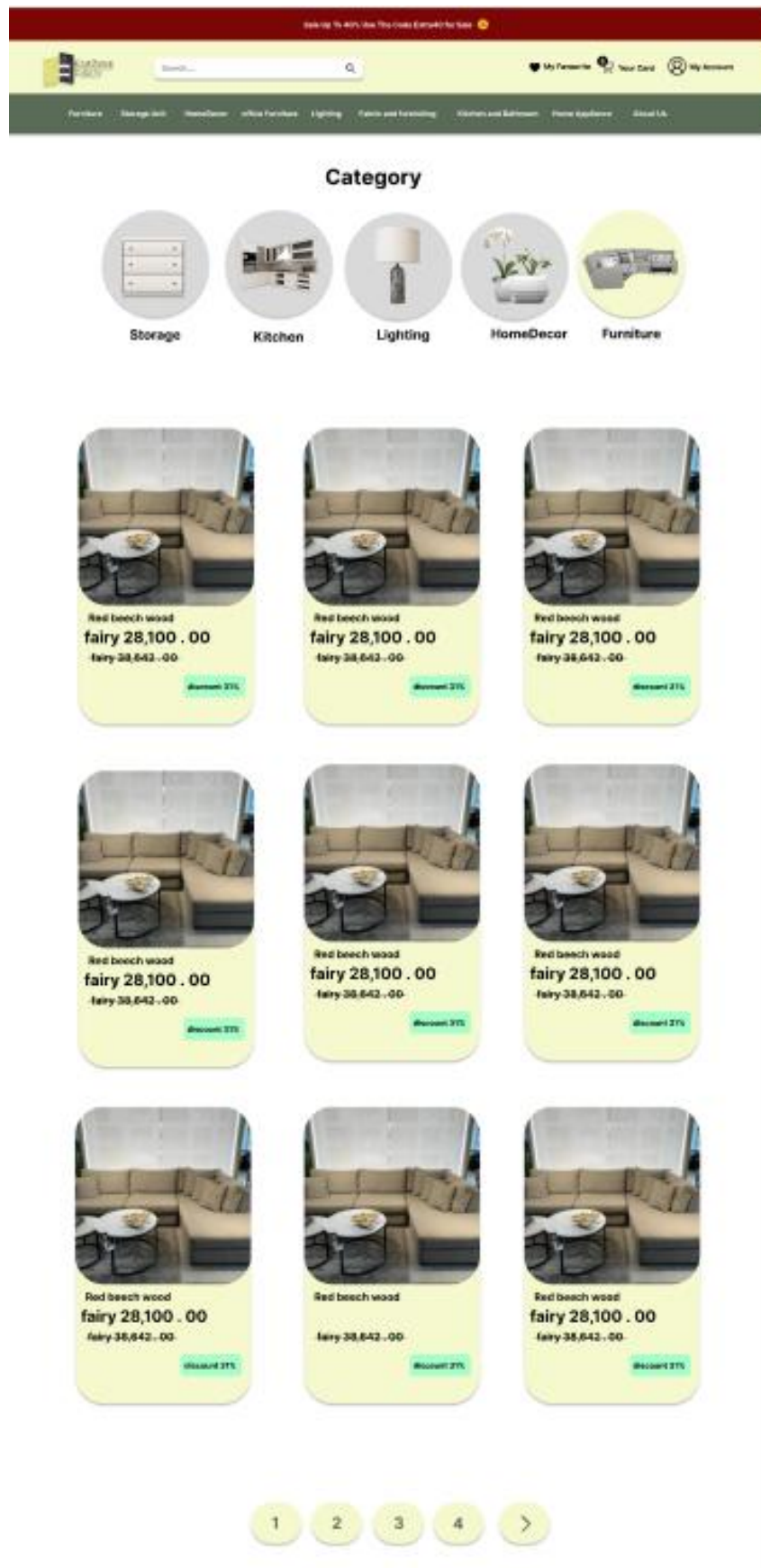


5.7.2 Client User Interface

5.7.2.1 Home Page



5.7.2.2 Category Page



5.7.2.3 Product Page

Save up to 40% like The Good & Beautiful for Sale

Search...

My Favorites

Your Cart

My Account

Home

Storage Unit

Home Decor

Office Furniture

Lighting

Fabric and Carpets

Kitchen and Bathrooms

Home Appliances

About Us

1

2



2

3

4

Corner sofa made of beech wood and linen fabric

★★★★★ 4.7 (3 ratings)

Product at Discount

fairry 28,100 . 00

~~fairry 38,642 . 00~~

discount 27%

You saved 10,542

Delivery during the period from March 1st to March 31st, including holidays and office holidays

Description

- Made of:
- High quality beech wood.
- 33 density sponge.
- Linen
- High quality beech wood.
- 33 density sponge.
- Linen

+

-

Add To Cart

♥

4.7 Based on (11) evaluation

5

4

3

2

1

Highest Rated

Ziad Essam

1/1/2025

Ziad Essam

1/1/2025

Similar Products

1

2



Red beech wood

fairry 28,100 . 00

~~fairry 38,642 . 00~~

discount 27%



Red beech wood

fairry 28,100 . 00

~~fairry 38,642 . 00~~

discount 27%



2

Red beech wood

fairry 28,100 . 00

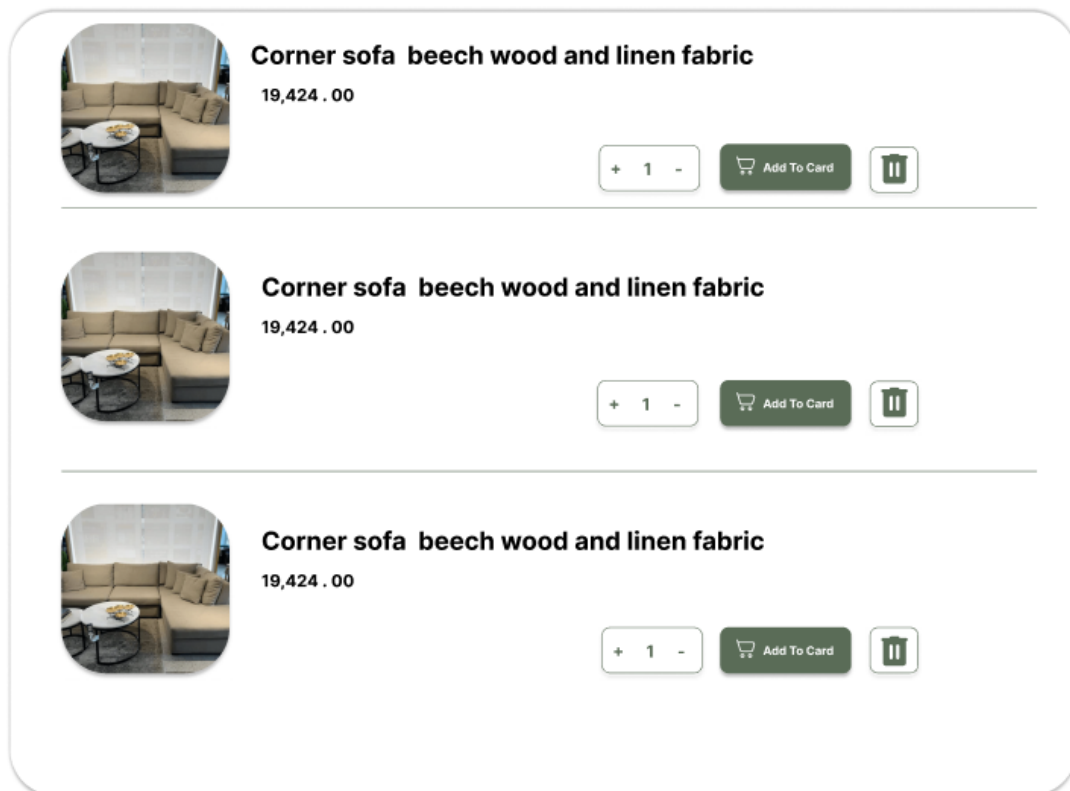
~~fairry 38,642 . 00~~

discount 27%

5.7.2.4 Add To Favorite Page



Your Favorites



5.7.2.5 Add To Card Page

Sale Up To 40% Use The Code Extra40 for Sale



Search.....



 My Favourite

 Your Card

 My Account

Furniture

Storage Unit

HomeDecor

office Furniture

Lighting

Fabric and furnishing

Kitchen and Bathroom

Home Appliance

About Us

Your Card



Corner sofa beech wood and linen fabric

19,424 . 00

+ 1 -





Corner sofa beech wood and linen fabric

19,424 . 00

+ 1 -



Your Address

Enter Coupon

Apply

Application Summary

the total

fairly 77,696 . 00

rival

- fairly 50.00

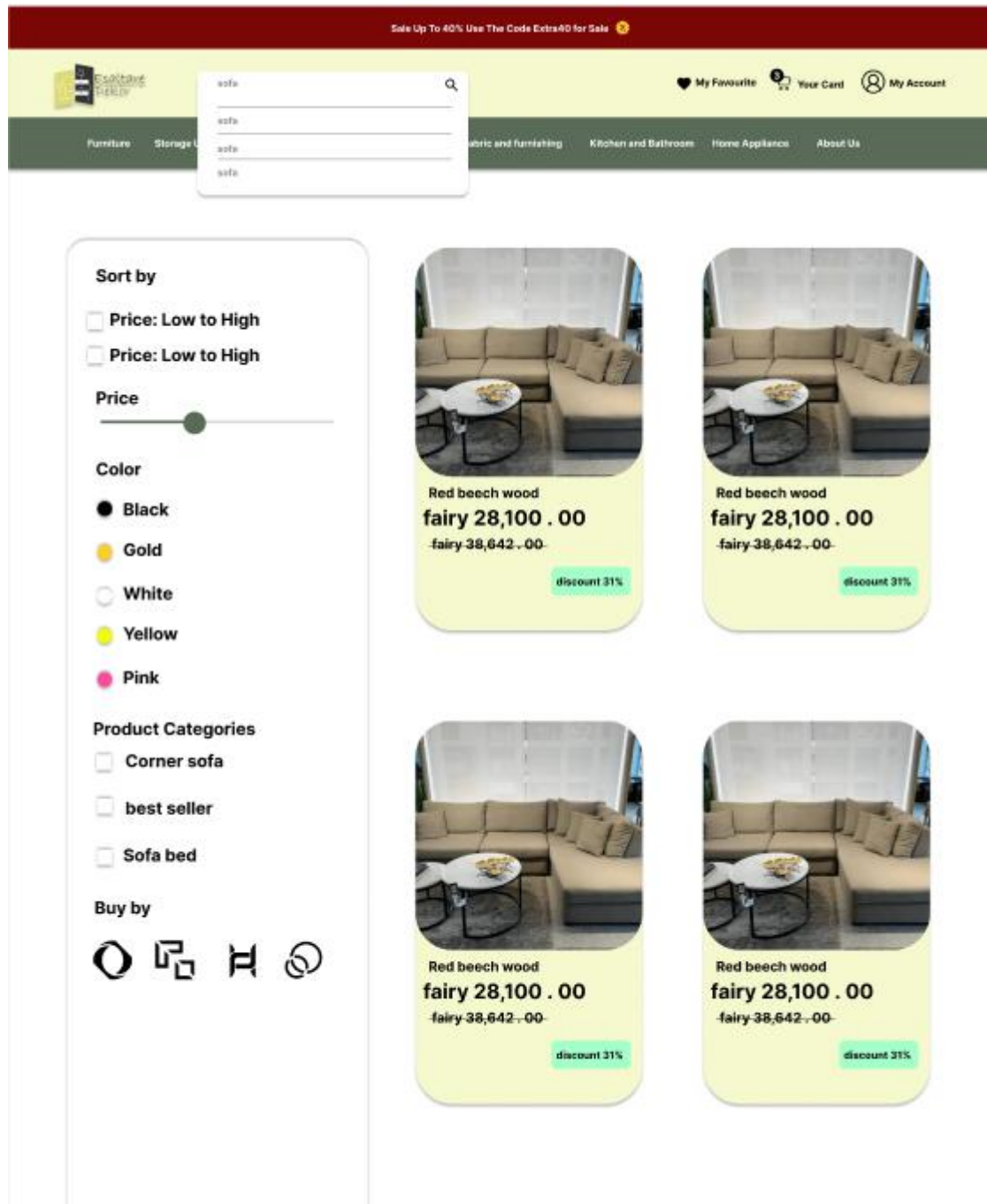
Total

fairly 77,646 . 00

Continue Your Purchase

125

5.7.2.6 Search Product Page



5.7.2.7 Compare Product Page

Sale Up To 60% Use The Code Extra40 for Sale



sofa

My Favourite

Your Card

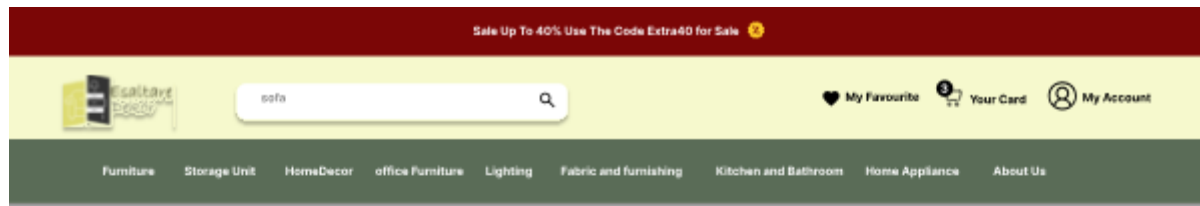
My Account

FurnitureStorage UnitHomeDecoroffice FurnitureLightingFabric and FurnishingKitchen and BathroomHome ApplianceAbout Us

Compare

Image	
	
A two-seater sofa made of high-quality	A two-seater sofa made of high-quality
Price	
18,800.00	25,800.00
Brand	
	
Availability	
In Stock	In Stock

5.7.2.8 User Profile Page



User : ADMGSA



Username

ADMGSA

Email

ADMGSA@gmail.com

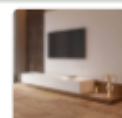
Last Orders



White and Brown MDF TV Unit - 40×40×240 cm

12AHav1246

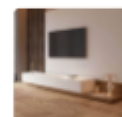
1/2/2025



White and Brown MDF TV Unit - 40×40×240 cm

12AHav1246

1/2/2025



White and Brown MDF TV Unit - 40×40×240 cm

12AHav1246

1/2/2025



White and Brown MDF TV Unit - 40×40×240 cm

12AHav1246

1/2/2025

Track Your Order ►

5.7.2.9 Track Your Order Page

Sale Up To 40% Use The Code Extra40 for Sale



Search...

My Favourite

Your Card

My Account

Furniture

Storage Unit

HomeDecor

office Furniture

Lighting

Fabric and furnishing

Kitchen and Bathroom

Home Appliance

About Us

Delivery By Egypt

Tracking ID: 65498

Arrival: Monday 3 Feb 2024 By 8:00 PM

location: Mit Kenana, Tukh, Qalyubia

Placed

Packing

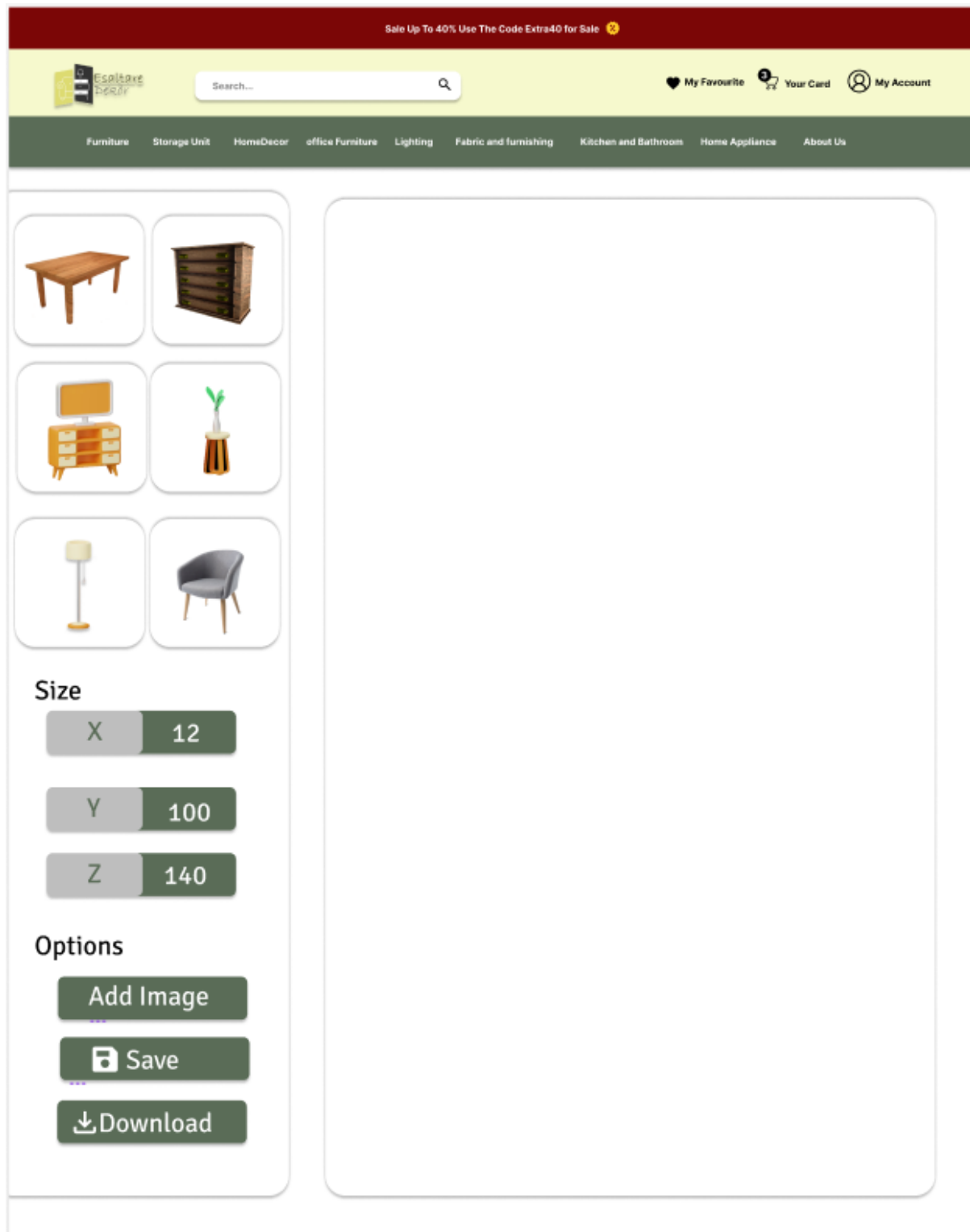
Dispatched

In Transit

Delivered

Updated: 1m Ago

5.7.2.10 3D Design Page



5.7.2.11 Chatbot Page



5.7.2.12 Vendor Packages Page



Welcome to Esaltare Dekor Console Packages
Please Select A Package To Subscribe



5.7.2.13 Vendor Profile Page



My Account

ProfileAdd Product



Vendor Decor Plus

Full Name

Decor

Full Name

Decor@organ.com

Phone Number

015469887452

ZipCode

45698214542

Address

Cairo Egypt

MyPackage ▶

Sign



Product Posted

Product ID	Status	Item Qty	Total
7bF987	Processing	1	17,800
A857523	Processing	1	5,800

5.7.2.14 Add Product Page



My Account

ProfileAdd Product

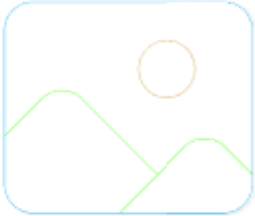
Adding New Product

Upload Image

...

Upload Image

↑



Product Name

Product price

Product Category

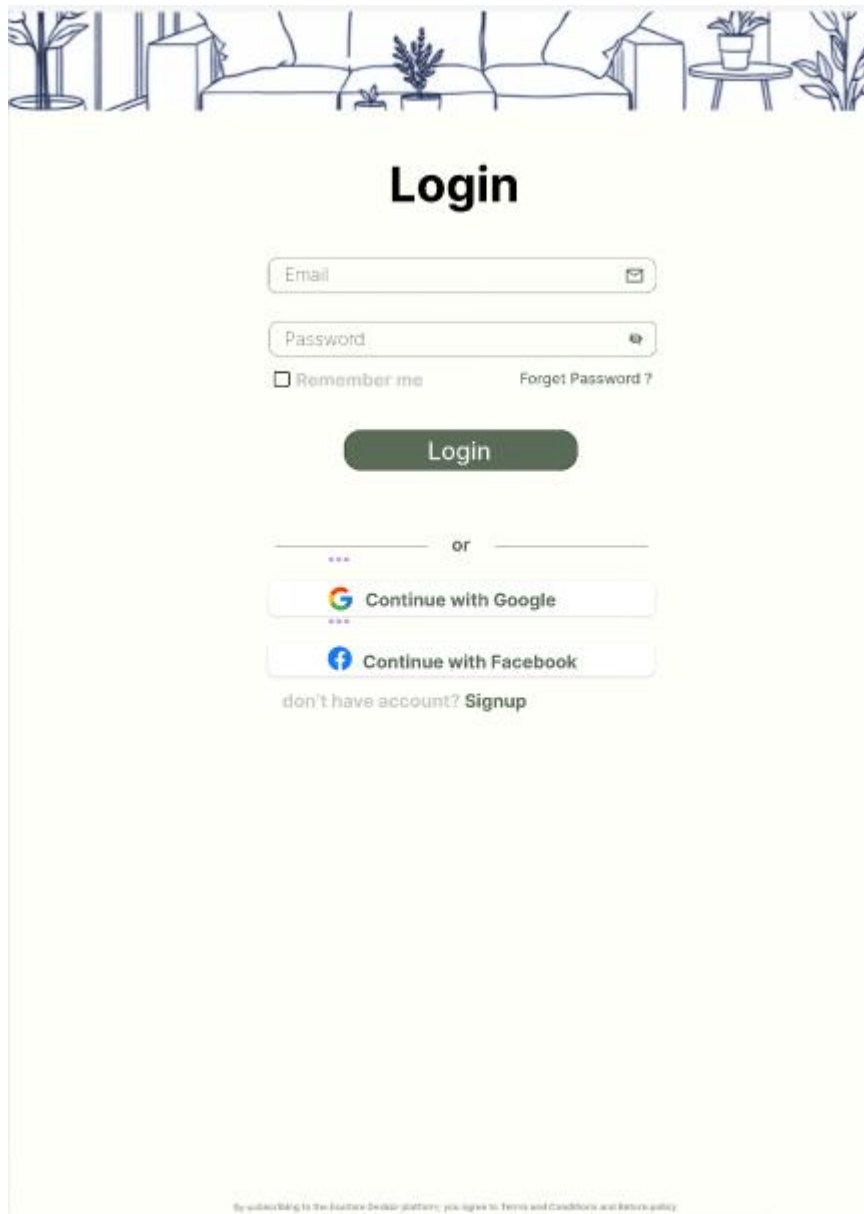
Product Subcategory

Product Description

Add Description....

Add Product

5.7.2.15 Login Page



The image shows a login page with a light yellow background. At the top, there is a decorative header with a line-art illustration of a living room interior, including a sofa, a coffee table, and potted plants. Below the header, the word "Login" is centered in a large, bold, black font. Underneath, there are two input fields: "Email" and "Password", each with a small icon on the right (an envelope for email and a key for password). Below the "Email" field, there is a checkbox labeled "Remember me" and a link "Forget Password?". A dark green "Login" button is centered below these elements. Below the button, the word "or" is centered between two horizontal lines. Underneath, there are two social login buttons: "Continue with Google" (with the Google logo) and "Continue with Facebook" (with the Facebook logo). Below these buttons, there is a link "don't have account? Signup". At the very bottom, there is a small, faint line of text: "By using Biking to the Future Desktop platform, you agree to Terms and Conditions and Privacy policy".

Login

Email

Password

☐ Remember me [Forget Password ?](#)

Login

or

 Continue with Google

 Continue with Facebook

[don't have account? Signup](#)

By using Biking to the Future Desktop platform, you agree to Terms and Conditions and Privacy policy

5.7.2.16 Sign Up Page



Create Account







Create Account

or

 Continue with Google

 Continue with Facebook

Already have an Account ? [Login](#)

By subscribing to the Eastern Dialect platform, you agree to Terms and Conditions and Return policy

5.7.2.17 Forget Password Page

← **Forgot Password**

Please enter your email to reset the password

Your Email

Reset Password

← Check Your Email

we sent a reset link to email@gmail.com
enter 5 digit code that mentioned in the email

Verify Code

← Set a New Password

Create a new password Ensure it different from the previous password

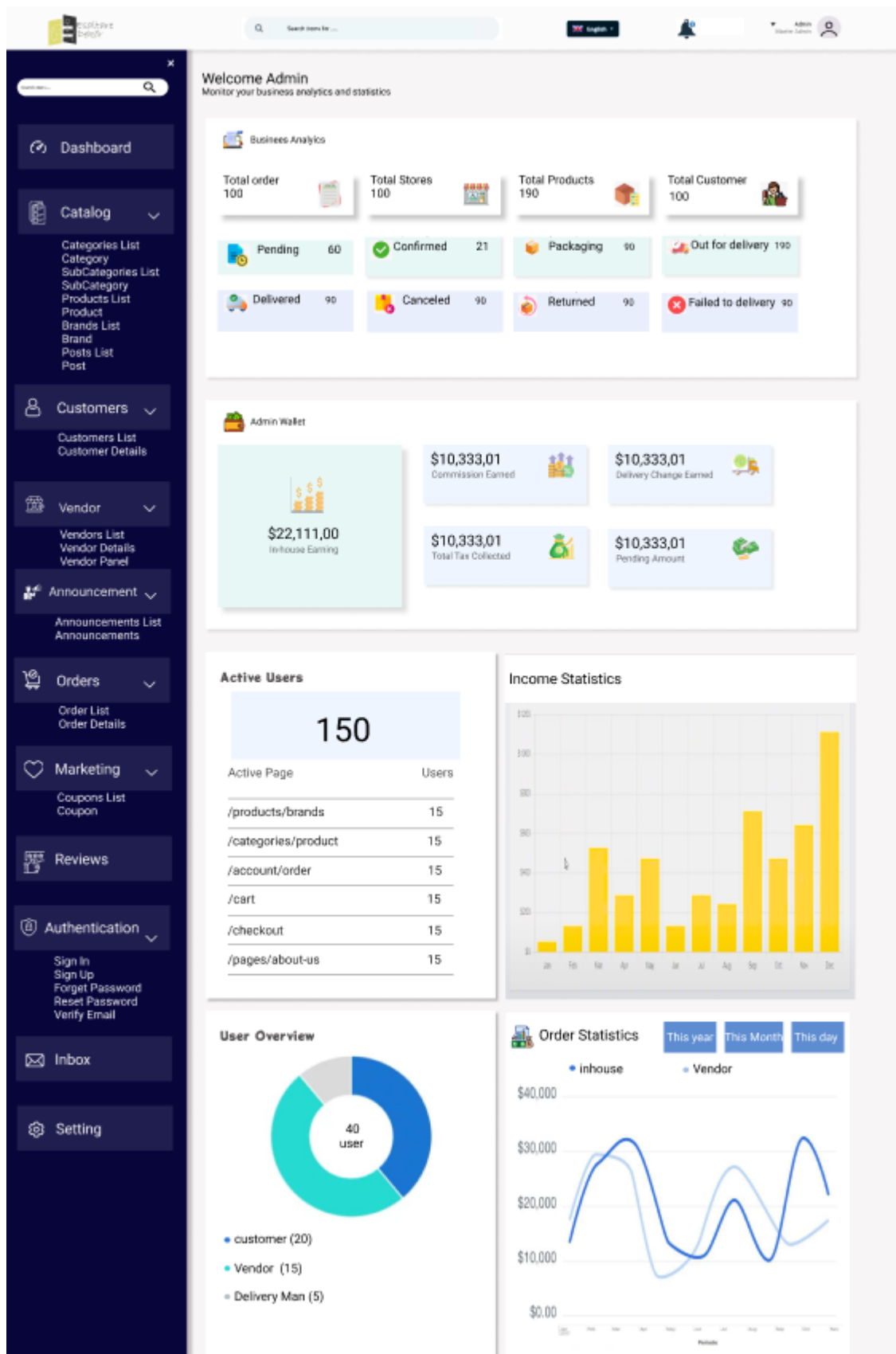


☐ Remember me

Update Password

5.7.3 Admin User Interface

5.7.3.1 Dashboard Page



[English](#)
Admin
Moude Admin

[Home Menu](#)

Dashboard

Catalog ▾

Categories List
Category
SubCategory List
SubCategory
Products List
Product
Brands List
Brand
Posts List
Post

Customers ▾

Customers List
Customer Details

Vendor ▾

Vendors List
Vendor Details
Vendor Panel

Announcement ▾

Announcements List
Announcements

Orders ▾

Order List
Order Details

Marketing ▾

Coupons List
Coupon

Reviews

Authentication ▾

Sign In
Sign Up
Forget Password
Reset Password
Verify Email

Inbox

Setting

Category List

New Category

Export

ID	Category Image	Name	Description	Priority	Category Status	Action
1		Bernhardt Plush Joli Sofa	Bernhardt Furniture is a leader in the direction of furniture design....	5	Active	
2		Bernhardt Plush Joli Sofa	Bernhardt Furniture is a leader in the direction of furniture design....	5	InActive	
3		Bernhardt Plush Joli Sofa	Bernhardt Furniture is a leader in the direction of furniture design....	5	Active	
4		Bernhardt Plush Joli Sofa	Bernhardt Furniture is a leader in the direction of furniture design....	5	InActive	
5		Bernhardt Plush Joli Sofa	Bernhardt Furniture is a leader in the direction of furniture design....	5	Active	
6		Bernhardt Plush Joli Sofa	Bernhardt Furniture is a leader in the direction of furniture design....	5	InActive	
7		Bernhardt Plush Joli Sofa	Bernhardt Furniture is a leader in the direction of furniture design....	5	Active	

<
1
2
3
4
5
6
7
>

Esaltaire Dekor HC. Copyright ZSK@2025

Business Setup

Profile

Home

5.7.2.3 Add Category Page

Esaltare Dekor

Search Item For ...

English

Admin

Admin

Dashboard

Catalog

- Categories List
- Category
- SubCategory List
- SubCategory
- Products List
- Product
- Brands List
- Brand
- Posts List
- Post

Customers

- Customers List
- Customer Details

Vendor

- Vendors List
- Vendor Details
- Vendor Panel

Announcement

- Announcements List
- Announcements

Orders

- Order List
- Order Details

Marketing

- Coupons List
- Coupon

Reviews

Authentication

- Sign In
- Sign Up
- Forget Password
- Reset Password
- Verify Email

Inbox

Setting

Add Category

Basic Information

Category Name

Priority

Description

Write The Description Here

Add Category

Visibility

☒ Published

☐ Scheduled

☐ Hidden

Image

Esaltare Dekor HC. Copyright ZSK@2025

Business Setup

Profile

Home

5.7.3.4 Edit Category

Esaltare Dekor

Search Here for ...

English

Admin

Profile Admin

Dashboard

Catalog

- Categories List
- Category
- SubCategory List
- SubCategory
- Products List
- Product
- Brands List
- Brand
- Posts List
- Post

Customers

- Customers List
- Customer Details

Vendor

- Vendors List
- Vendor Details
- Vendor Panel

Announcement

- Announcements List
- Announcements

Orders

- Order List
- Order Details

Marketing

- Coupons List
- Coupon

Reviews

Authentication

- Sign In
- Sign Up
- Forget Password
- Reset Password
- Verify Email

Inbox

Setting

Edit Category

Basic Information

Category Name

Category 1

Priority

5

Description

Write The Description Here

Edit Category

Visibility

☒ Published

☐ Scheduled

☐ Hidden

Image



Esaltare Dekor HC. Copyright ZSK@2025

Business Setup

Profile

Home

[English](#)

Admin
Master Admin

×

🔍

- Dashboard
- Catalog ▾
 - Categories List
 - Category
 - SubCategory List
 - Products List
 - Product
 - Brands List
 - Brand
 - Posts List
 - Post
- Customers ▾
 - Customers List
 - Customer Details
- Vendor ▾
 - Vendors List
 - Vendor Details
 - Vendor Panel
- Announcement ▾
 - Announcements List
 - Announcements
- Orders ▾
 - Order List
 - Order Details
- Marketing ▾
 - Coupons List
 - Coupon
- Reviews
- Authentication ▾
 - Sign In
 - Sign Up
 - Forget Password
 - Reset Password
 - Verify Email
- Inbox
- Setting

Sub Category List

New SubCategory

📄 Export

ID	Sub Category Image	Name	Category Name	Priority	Sub Category Status	Action
1		Italian class and quality	Bernhardt Plush Joli Sofa	5	Active InActive	✎ 🗑️
2		Italian class and quality	Bernhardt Plush Joli Sofa	5	Active InActive	✎ 🗑️
3		Italian class and quality	Bernhardt Plush Joli Sofa	5	Active InActive	✎ 🗑️
4		Italian class and quality	Bernhardt Plush Joli Sofa	5	Active InActive	✎ 🗑️
5		Italian class and quality	Bernhardt Plush Joli Sofa	5	Active InActive	✎ 🗑️
6		Italian class and quality	Bernhardt Plush Joli Sofa	5	Active InActive	✎ 🗑️
7		Italian class and quality	Bernhardt Plush Joli Sofa	5	Active InActive	✎ 🗑️

«

1

2

3

4

5

6

7

»

Esaltare Dekor HC. Copyright ZSK@2025

⚙️ Business Setup
👤 Profile
🏠 Home

5.7.3.6 Add Subcategory Page

Esaltare Dekor

Search items for ...

English

Admin

Dashboard

Catalog

- Categories List
- Category
- SubCategory List
- SubCategory
- Products List
- Product
- Brands List
- Brand
- Posts List
- Post

Customers

- Customers List
- Customer Details

Vendor

- Vendors List
- Vendor Details
- Vendor Panel

Announcement

- Announcements List
- Announcements

Orders

- Order List
- Order Details

Marketing

- Coupons List
- Coupon

Reviews

Authentication

- Sign In
- Sign Up
- Forget Password
- Reset Password
- Verify Email

Inbox

Setting

Add SubCategory

Basic Information

SubCategory Name

Name

Category Name

Priority

Sub Category Status

Image



Add SubCategory

Esaltare Dekor HC. Copyright ZSK@2025

Business Setup

Profile

Home

5.7.3.7 Edit Subcategory Page

Esaltare Dekor

Search Here...

English

Admin

Dashboard

Catalog

- Categories List
- Category
- SubCategory List
- SubCategory
- Products List
- Product
- Brands List
- Brand
- Posts List
- Post

Customers

- Customers List
- Customer Details

Vendor

- Vendors List
- Vendor Details
- Vendor Panel

Announcement

- Announcements List
- Announcements

Orders

- Order List
- Order Details

Marketing

- Coupons List
- Coupon

Reviews

Authentication

- Sign In
- Sign Up
- Forget Password
- Reset Password
- Verify Email

Inbox

Setting

Edit SubCategory

Basic Information

SubCategory Name

Sub1

Name

Name1

Category Name

Category1

Priority

1

Sub Category Status

Active

Edit SubCategory

Image

Esaltare Dekor HC. Copyright ZSK@2025

Business Setup

Profile

Home

5.7.3.8 Products Page

English 1

Dashboard

Catalog

Categories List
Category
SubCategory List
SubCategory
Products List
Product
Brands List
Brand
Posts List
Post

Customers

Customers List
Customer Details

Vendor

Vendors List
Vendor Details
Vendor Panel

Announcement

Announcements List
Announcements

Orders

Order List
Order Details

Marketing

Coupons List
Coupon

Reviews

Authentication

Sign In
Sign Up
Forgot Password
Reset Password
Verify Email

Inbox

Setting

Product List

New Product

Export

ID	Product Image	Product Name	Description	Category Name	Sub Category	Price	Num In Stock	Product Status	Action
1		Bernhardt Plush Joli Sofa	Bernhardt Furniture is a leader in the direction of furniture design...	Bernhardt Plush Joli Sofa	Bernhardt Plush Joli Sofa	250\$	18	Active InActive	
2		Bernhardt Plush Joli Sofa	Bernhardt Furniture is a leader in the direction of furniture design...	Bernhardt Plush Joli Sofa	Bernhardt Plush Joli Sofa	250\$	18	Active InActive	
3		Bernhardt Plush Joli Sofa	Bernhardt Furniture is a leader in the direction of furniture design...	Bernhardt Plush Joli Sofa	Bernhardt Plush Joli Sofa	250\$	18	Active InActive	
4		Bernhardt Plush Joli Sofa	Bernhardt Furniture is a leader in the direction of furniture design...	Bernhardt Plush Joli Sofa	Bernhardt Plush Joli Sofa	250\$	18	Active InActive	
5		Bernhardt Plush Joli Sofa	Bernhardt Furniture is a leader in the direction of furniture design...	Bernhardt Plush Joli Sofa	Bernhardt Plush Joli Sofa	250\$	18	Active InActive	
6		Bernhardt Plush Joli Sofa	Bernhardt Furniture is a leader in the direction of furniture design...	Bernhardt Plush Joli Sofa	Bernhardt Plush Joli Sofa	250\$	18	Active InActive	
7		Bernhardt Plush Joli Sofa	Bernhardt Furniture is a leader in the direction of furniture design...	Bernhardt Plush Joli Sofa	Bernhardt Plush Joli Sofa	250\$	18	Active InActive	

« 1 2 3 4 5 6 7 »

Esaltare Dekor HC. Copyright ZSK@2025

Business Setup

Profile

Home

5.7.3.9 Add Products Page

Esaltare Dekor

Search Menu

Dashboard

Catalog

Categories List
Category
SubCategory List
SubCategory
Products List
Product
Brands List
Brand
Posts List
Post

Customers

Customers List
Customer Details

Vendor

Vendors List
Vendor Details
Vendor Panel

Announcement

Announcements List
Announcements

Orders

Order List
Order Details

Marketing

Coupons List
Coupon

Reviews

Authentication

Sign In
Sign Up
Forget Password
Reset Password
Verify Email

Inbox

Setting

Search Item for ...

English

Admin

Market Admin

Add Product

Basic Information

Name

Pricing

Price

Category

Description

Write The Description Here

Add Product

Visibility

☒ Published

☐ Scheduled

☐ Hidden

Image

Esaltare Dekor HC. Copyright ZSK@2025

Business Setup

Profile

Home

5.7.3.10 Edit Products Page

Esaltare Dekor

Search

English

Admin

Dashboard

Catalog

- Categories List
- Category
- SubCategory List
- SubCategory
- Products List
- Product
- Brands List
- Brand
- Posts List
- Post

Customers

- Customers List
- Customer Details

Vendor

- Vendors List
- Vendor Details
- Vendor Panel

Announcement

- Announcements List
- Announcements

Orders

- Order List
- Order Details

Marketing

- Coupons List
- Coupon

Reviews

Authentication

- Sign In
- Sign Up
- Forget Password
- Reset Password
- Verify Email

Inbox

Setting

Edit Product

Basic Information

Name

Pricing

Price

Category

Description

Write The Description Here

Edit Product

Visibility

☒ Published

☐ Scheduled

☐ Hidden

Image

Esaltare Dekor HC. Copyright ZSK@2025

Business Setup

Profile

Home

5.7.3.11 Brand Page

Esaltare Dekor

English

Search Items...

Dashboard

Catalog

- Categories List
- Category
- SubCategory List
- SubCategory
- Products List
- Product
- Brands List
- Brand
- Posts List
- Post

Customers

- Customers List
- Customer Details

Vendor

- Vendors List
- Vendor Details
- Vendor Panel

Announcement

- Announcements List
- Announcements

Orders

- Order List
- Order Details

Marketing

- Coupons List
- Coupon

Reviews

Authentication

- Sign In
- Sign Up
- Forget Password
- Reset Password
- Verify Email

Inbox

Setting

Brand List

New Brand

Export

ID	Brand Image	Name	Description	Priority	Brand Status	Action
1		Bonus brand Arflex	This company has more that five decades of experience	5	Active InActive	
2		Bonus brand Arflex	This company has more that five decades of experience	5	Active InActive	
3		Bonus brand Arflex	This company has more that five decades of experience	5	Active InActive	
4		Bonus brand Arflex	This company has more that five decades of experience	5	Active InActive	
5		Bonus brand Arflex	This company has more that five decades of experience	5	Active InActive	
6		Bonus brand Arflex	This company has more that five decades of experience	5	Active InActive	
7		Bonus brand Arflex	This company has more that five decades of experience	5	Active InActive	

<

1

2

3

4

5

6

7

>

Esaltare Dekor HC. Copyright ZSK@2025

Business Setup

Profile

Home

5.7.3.12 Add Brand Page

Esaltare Dekor

Search Menu

Dashboard

Catalog

- Categories List
- Category
- SubCategory List
- SubCategory
- Products List
- Product
- Brands List
- Brand
- Posts List
- Post

Customers

- Customers List
- Customer Details

Vendor

- Vendors List
- Vendor Details
- Vendor Panel

Announcement

- Announcements List
- Announcements

Orders

- Order List
- Order Details

Marketing

- Coupons List
- Coupon

Reviews

Authentication

- Sign In
- Sign Up
- Forget Password
- Reset Password
- Verify Email

Inbox

Setting

Search Pages for ...

English

ADMIN Master Admin

Add Brand

Basic Information

Brand Name

Priority

Brand Status

Description

Write Description Here

Add Brand

Image

5.7.3.13 Edit Brand Page

Esaltare Dekor HC
Search Items for ...
English
Admin

- [Dashboard](#)
- [Catalog](#)
 - Categories List
 - Category
 - SubCategory List
 - SubCategory
 - Products List
 - Product
 - Brands List
 - Brand
 - Posts List
 - Post
- [Customers](#)
 - Customers List
 - Customer Details
- [Vendor](#)
 - Vendors List
 - Vendor Details
 - Vendor Panel
- [Announcement](#)
 - Announcements List
 - Announcements
- [Orders](#)
 - Order List
 - Order Details
- [Marketing](#)
 - Coupons List
 - Coupon
- [Reviews](#)
- [Authentication](#)
 - Sign In
 - Sign Up
 - Forget Password
 - Reset Password
 - Verify Email
- [Inbox](#)
- [Setting](#)

Edit Brand

Basic Information

Brand Name

Priority

Brand Status

Description

Image

[Edit Brand](#)

Esaltare Dekor HC. Copyright ZSK@2025
Business Setup
Profile
Home

5.7.3.14 Posts Page

English

Admin
Admin Admin

Dashboard

Catalog

- Categories List
- Category
- SubCategory List
- SubCategory
- Products List
- Product
- Brands List
- Brand
- Posters List
- Poster

Customers

- Customers List
- Customer Details

Vendor

- Vendors List
- Vendor Details
- Vendor Panel

Announcement

- Announcements List
- Announcements

Orders

- Order List
- Order Details

Marketing

- Coupons List
- Coupon

Reviews

Authentication

- Sign In
- Sign Up
- Forget Password
- Reset Password
- Verify Email

Inbox

Setting

Poster List

New Poster
Export

ID	Poster Image	Title	Description	Priority	Poster Status	Action
1		Bernhardt Plush Joli Sofa	Bernhardt Furniture is a leader in the direction of furniture design....	5	Active InActive	
2		Bernhardt Plush Joli Sofa	Bernhardt Furniture is a leader in the direction of furniture design....	5	Active InActive	
3		Bernhardt Plush Joli Sofa	Bernhardt Furniture is a leader in the direction of furniture design....	5	Active InActive	
4		Bernhardt Plush Joli Sofa	Bernhardt Furniture is a leader in the direction of furniture design....	5	Active InActive	
5		Bernhardt Plush Joli Sofa	Bernhardt Furniture is a leader in the direction of furniture design....	5	Active InActive	
6		Bernhardt Plush Joli Sofa	Bernhardt Furniture is a leader in the direction of furniture design....	5	Active InActive	
7		Bernhardt Plush Joli Sofa	Bernhardt Furniture is a leader in the direction of furniture design....	5	Active InActive	

« 1 2 3 4 5 6 7 »

Esaltare Dekor HC. Copyright ZSK@2025

Business Setup

Profile

Home

5.7.3.15 Add Post Page

The screenshot displays the 'Add Post' page within the Esaltare Dekor HC application. The interface includes a dark sidebar on the left with a search bar and various navigation categories. The main content area is titled 'Add Post' and contains a form for creating a new post. The form is divided into two main sections: 'Basic Information' and 'Image'.

Basic Information

- Post Title:** A text input field.
- Priority:** A text input field.
- Poster Status:** A text input field.
- Description:** A large text area with the placeholder text 'Write Description Here'.

Image

An image upload icon is located next to the 'Basic Information' section.

Footer

The footer contains the text 'Esaltare Dekor HC. Copyright ZSK@2025' and navigation links for 'Business Setup', 'Profile', and 'Home'.

5.7.3.16 Edit Post Page

Esaltare Dekor HC

Search Product/Post...

English

NOTIFY

ESALTARE DEKOR HC

Search Product/Post...

Edit Post

Basic Information

Post Title
Title1

Priority
1

Poster Status
Active

Image

Description
Write Description Here

Edit Poster

Dashboard

Catalog

- Categories List
- Category
- SubCategory List
- SubCategory
- Products List
- Product
- Brands List
- Brand
- Posts List
- Post

Customers

- Customers List
- Customer Details

Vendor

- Vendors List
- Vendor Details
- Vendor Panel

Announcement

- Announcements List
- Announcements

Orders

- Order List
- Order Details

Marketing

- Coupons List
- Coupon

Reviews

Authentication

- Sign In
- Sign Up
- Forget Password
- Reset Password
- Verify Email

Inbox

Setting

Esaltare Dekor HC. Copyright ZSK@2025

Business Setup

Profile

Home

5.7.3.17 Announcement Page

Sideline Esaltare Dekor HC

English
🔔
👤

[Dashboard](#)

[Catalog](#)

[Categories List](#)
[SubCategory List](#)
[SubCategory Products List](#)
[Product Brands List](#)
[Brand Posts List](#)
[Post](#)

[Customers](#)

[Customers List](#)
[Customer Details](#)

[Vendor](#)

[Vendors List](#)
[Vendor Details](#)
[Vendor Panel](#)

[Announcement](#)

[Announcements List](#)
[Announcements](#)

[Orders](#)

[Order List](#)
[Order Details](#)

[Marketing](#)

[Coupons List](#)
[Coupon](#)

[Reviews](#)

[Authentication](#)

[Sign In](#)
[Sign Up](#)
[Forget Password](#)
[Reset Password](#)
[Verify Email](#)

[Inbox](#)

[Setting](#)

Announcement List

Search by Announcement Title

[New Announcements](#)
[Export](#)

ID	Announcement Image	Announcement Title	Percentage	Announcement Status	Action
1		Bernhardt Plush Joli Sofa	30%	Active InActive	
1		Bernhardt Plush Joli Sofa	30%	Active InActive	
1		Bernhardt Plush Joli Sofa	30%	Active InActive	
1		Bernhardt Plush Joli Sofa	30%	Active InActive	
1		Bernhardt Plush Joli Sofa	30%	Active InActive	
1		Bernhardt Plush Joli Sofa	30%	Active InActive	
1		Bernhardt Plush Joli Sofa	30%	Active InActive	

<
1
2
3
4
5
6
7
>

Esaltare Dekor HC. Copyright ZSK@2025

[Business Setup](#)
[Profile](#)
[Home](#)

5.7.3.18 Add Announcement Page

The screenshot displays the 'Add Announcement' page within the Esaltare Dekor HC admin interface. The top navigation bar includes the Esaltare Dekor logo, a search bar, a language selector set to 'English', a notification bell, and a user profile icon labeled 'ADMIN'. The left sidebar contains a search bar and several menu categories: Dashboard, Catalog (with sub-items like Categories List, SubCategory List, Products List, Brand, Posts List, and Post), Customers (with Customers List and Customer Details), Vendor (with Vendors List, Vendor Details, and Vendor Panel), Announcement (with Announcements List and Announcements), Orders (with Order List and Order Details), Marketing (with Coupons List and Coupon), Reviews, Authentication (with Sign In, Sign Up, Forget Password, Reset Password, and Verify Email), Inbox, and Setting.

The main content area is titled 'Add Announcement'. It contains a 'Basic Information' section with three text input fields: 'Announcement Title', 'Percentage(%)', and 'Announcement Status'. To the right of these fields is an 'Image' section featuring a placeholder icon for an image upload. At the bottom of the form is a dark blue button labeled 'Add Announcement'. The footer of the page shows the copyright notice 'Esaltare Dekor HC. Copyright ZSK@2025' and a set of quick links: Business Setup, Profile, and Home.

5.7.3.19 Edit Announcement Page

Esaltare Dekor HC

Search Items...

English

Admin

Admin Panel

Dashboard

Catalog

- Categories List
- Category
- SubCategory List
- SubCategory
- Products List
- Product
- Brands List
- Brand
- Posts List
- Post

Customers

- Customers List
- Customer Details

Vendor

- Vendors List
- Vendor Details
- Vendor Panel

Announcement

- Announcements List
- Announcements

Orders

- Order List
- Order Details

Marketing

- Coupons List
- Coupon

Reviews

Authentication

- Sign In
- Sign Up
- Forget Password
- Reset Password
- Verify Email

Inbox

Setting

Edit Announcement

Basic Information

Announcement Title

Arvouncement1

Percentage(%)

30%

Announcement Status

Active

Image

Edit Announcement

Esaltare Dekor HC. Copyright ZSK@2025

Business Setup

Profile

Home

5.7.3.20 Coupon Page

Esaltare Dekor HC

Account Name: ...

English

Admin

Admin

Dashboard

Catalog

- Categories List
- Category
- SubCategory List
- SubCategory
- Products List
- Product
- Brands List
- Brand
- Posts List
- Post

Customers

- Customers List
- Customer Details

Vendor

- Vendors List
- Vendor Details
- Vendor Panel

Announcement

- Announcements List
- Announcements

Orders

- Order List
- Order Details

Marketing

- Coupons List
- Coupon

Reviews

Authentication

- Sign In
- Sign Up
- Forget Password
- Reset Password
- Verify Email

Inbox

Setting

Coupon List

Search by Type Coupon

New Coupon

Export

ID	Code	Type	Discount	Priority	Start date	End date	Coupon Status	Action
1	M3LEAMZSK	Percentage	25%	5	2025-02-10 00:00	2025-04-10 00:00	Active InActive	 
2	M3LEAMZSK	Fixed amount	25%	5	2025-02-10 00:00	2025-04-10 00:00	Active InActive	 
3	M3LEAMZSK	Percentage	25%	5	2025-02-10 00:00	2025-04-10 00:00	Active InActive	 
4	M3LEAMZSK	Fixed amount	25%	5	2025-02-10 00:00	2025-04-10 00:00	Active InActive	 
5	M3LEAMZSK	Percentage	25%	5	2025-02-10 00:00	2025-04-10 00:00	Active InActive	 
6	M3LEAMZSK	Fixed amount	25%	5	2025-02-10 00:00	2025-04-10 00:00	Active InActive	 
7	M3LEAMZSK	Percentage	25%	5	2025-02-10 00:00	2025-04-10 00:00	Active InActive	 

< 1 2 3 4 5 6 7 >

Esaltare Dekor HC. Copyright ZSK@2025

Business Setup

Profile

Home

5.7.3.21 Add Coupon Page

The screenshot displays the 'Add Coupon' page within the Esaltare Dekor HC admin interface. The left sidebar contains a search bar and a list of navigation items: Dashboard, Catalog (with sub-items: Categories List, Category, SubCategory List, SubCategory, Products List, Product, Brands List, Brand, Posts List, Post), Customers (with sub-items: Customers List, Customer Details), Vendor (with sub-items: Vendors List, Vendor Details, Vendor Panel), Announcement (with sub-items: Announcements List, Announcements), Orders (with sub-items: Order List, Order Details), Marketing (with sub-items: Coupons List, Coupon), Reviews, Authentication (with sub-items: Sign In, Sign Up, Forget Password, Reset Password, Verify Email), Inbox, and Setting.

The main content area is titled 'Add Coupon' and contains the following sections:

- Basic Information:**
 - code:** A text input field containing 'QWERTY12'.
 - Type:** Three radio button options:
 - ☒ Percentages
 - ☐ Fixed Amount
 - ☐ Free Shipping
 - Discount Value:** A text input field containing '200,00'.
 - Usage limit:** A text input field containing '100,00'.
 - A checkbox labeled 'only for registered customers' which is currently unchecked.
 - Add Coupon:** A dark blue button at the bottom of the form.
- Status:** Two radio button options:
 - ☒ Enabled
 - ☐ Disabled
- Schedule:**
 - A small note: 'The start and end dates for the coupon are required.' (Note: The image shows a typo 'The' and 'coupon' instead of 'coupons').
 - Start Date:** A text input field.
 - End Date:** A text input field.

The footer of the page includes the text 'Esaltare Dekor HC. Copyright ZSK@2025' and a set of navigation links: Business Setup, Profile, and Home.

5.7.3.22 Edit Coupon Page

The screenshot shows the 'Edit Coupon' page in a dashboard. The sidebar on the left contains the following navigation items: Dashboard, Catalog (with sub-items: Categories List, Category, SubCategory List, SubCategory, Products List, Product, Brands List, Brand, Posts List, Post), Customers (with sub-items: Customers List, Customer Details), Vendor (with sub-items: Vendors List, Vendor Details, Vendor Panel), Announcement (with sub-items: Announcements List, Announcements), Orders (with sub-items: Order List, Order Details), Marketing (with sub-items: Coupons List, Coupon), Reviews, Authentication (with sub-items: Sign In, Sign Up, Forget Password, Reset Password, Verify Email), Inbox, and Setting.

The main content area is titled 'Edit Coupon' and contains the following sections:

- Basic Information**
 - code:
 - Type:
 - ☒ Percentages
 - ☐ Fixed Amount
 - ☐ Free Shipping
 - Discount Value:
 - Usage limit:
 - ☐ only for registered customers
- Status**
 - ☒ Enabled
 - ☐ Disabled
- Schedule**
 - Start Date:
 - End Date:

At the bottom of the form is a button labeled 'Edit Coupon'.

The footer of the page contains the text 'Esaltare Dekor HC. Copyright ZSK@2025' and navigation links for Business Setup, Profile, and Home.

5.7.3.23 Customer List Page

ESALTARE DEKOR

Search by ...

English

Admin

Admin Login

Customer List

Search by Customer Name

Export

ID	Profile Image	Customer Name	Email	Phone	Registered	Number of Orders	Total Spent	Action
1		Karim	karimoladisaad@gmail.com	01128552462	May 15, 2025	10	10950\$	
2		Karim	karimoladisaad@gmail.com	01128552462	May 15, 2025	10	10950\$	
3		Karim	karimoladisaad@gmail.com	01128552462	May 15, 2025	10	10950\$	
4		Karim	karimoladisaad@gmail.com	01128552462	May 15, 2025	10	10950\$	
5		Karim	karimoladisaad@gmail.com	01128552462	May 15, 2025	10	10950\$	
6		Karim	karimoladisaad@gmail.com	01128552462	May 15, 2025	10	10950\$	
7		Karim	karimoladisaad@gmail.com	01128552462	May 15, 2025	10	10950\$	

<

1

2

3

4

5

6

7

>

Dashboard

Catalog

Categories List
Category
SubCategory List
SubCategory List
Products List
Product
Brands List
Brand
Posts List
Post

Customers

Customers List
Customer Details

Vendor

Vendors List
Vendor Details
Vendor Panel

Announcement

Announcements List
Announcements

Orders

Order List
Order Details

Marketing

Coupons List
Coupon

Reviews

Authentication

Sign In
Sign Up
Forgot Password
Reset Password
Verify Email

Inbox

Setting

Esaltare Dekor HC. Copyright ZSK@2025

Business Setup

Profile

Home

5.7.3.24 Customer Details

Esaltare Dekor

Search Items for ...

English

Admin

Logout

Dashboard

Catalog

- Categories List
- Category
- SubCategory List
- SubCategory
- Products List
- Product
- Brands List
- Brand
- Posts List
- Post

Customers

- Customers List
- Customer Details

Vendor

- Vendors List
- Vendor Details
- Vendor Panel

Announcement

- Announcements List
- Announcements

Orders

- Order List
- Order Details

Marketing

- Coupons List
- Coupon

Reviews

Authentication

- Sign In
- Sign Up
- Forgot Password
- Reset Password
- Verify Email

Inbox

Setting

Customer Details



Karim Hazem

karimziadsaad@gmail.com

02 01120552462

Last Order

7 days ago - #80294

Average Order Value

\$200.01

Registered

2 months ago

Email Market

Subscribed

Notes about customer

Orders

Total spent \$10950 on 7 orders

#320	Today at 6:10 pm	Pending	4 items	1000\$
#320	June 26,2025	Completed	4 items	1000\$
#320	June 26,2025	Completed	4 items	1000\$
#320	June 26,2025	Completed	4 items	1000\$

«

1

2

3

4

5

6

7

»

Address

Karim Hazem

in the address "789 Pine Avenue," "789 Pine Avenue" would go in Address Line 1

Esaltare Dekor HC. Copyright ZSK@2025

Business Setup

Profile

Home

5.7.3.25 Vendor List

Esaltare Dekor

Search by Brand Name

English

Admin

Dashboard

Catalog

- Categories List
- Category
- SubCategory
- SubCategory List
- Products List
- Product
- Brands List
- Brand
- Posts List
- Post

Customers

- Customers List
- Customer Details

Vendor

- Vendors List
- Vendor Details
- Vendor Panel

Announcement

- Announcements List
- Announcements

Orders

- Order List
- Order Details

Marketing

- Coupons List
- Coupon

Reviews

Authentication

- Sign In
- Sign Up
- Forgot Password
- Reset Password
- Verify Email

Inbox

Setting

Vendor List

Search by Brand Name

Export

ID	Profile Image	Vendor Name	Brand Name	Email	Phone	Registered	Number of Products	Total Earn	Action
1		Karim	Bonus brand Arflex	karimoladood@gmail.com	01120552462	May 15,2025	10	10950\$	
2		Karim	Bonus brand Arflex	karimoladood@gmail.com	01120552462	May 15,2025	10	10950\$	
3		Karim	Bonus brand Arflex	karimoladood@gmail.com	01120552462	May 15,2025	10	10950\$	
4		Karim	Bonus brand Arflex	karimoladood@gmail.com	01120552462	May 15,2025	10	10950\$	
5		Karim	Bonus brand Arflex	karimoladood@gmail.com	01120552462	May 15,2025	10	10950\$	
6		Karim	Bonus brand Arflex	karimoladood@gmail.com	01120552462	May 15,2025	10	10950\$	
7		Karim	Bonus brand Arflex	karimoladood@gmail.com	01120552462	May 15,2025	10	10950\$	

<

1

2

3

4

5

6

7

>

Esaltare Dekor HC. Copyright ZSK@2025

Business Setup

Profile

Home

5.7.3.26 Vendor Details

Esaltare Dekor

Search items for ...

English

Admin

Vendor Panel

Dashboard

Catalog

- Categories List
- Category
- SubCategory List
- SubCategory
- Products List
- Product
- Brands List
- Brand
- Posts List
- Post

Customers

- Customers List
- Customer Details

Vendor

- Vendors List
- Vendor Details
- Vendor Panel

Announcement

- Announcements List
- Announcements

Orders

- Order List
- Order Details

Marketing

- Coupons List
- Coupon

Reviews

Authentication

- Sign In
- Sign Up
- Forgot Password
- Reset Password
- Verify Email

Inbox

Setting

Vendor Details

Karim Hazem



Karim Hazem
karimziadsaad@gmail.com
Brand Name
(+20) 1120552462

Last Product
7 days ago - [product name](#)
Average Product Value
\$200.01
Registered
2 months ago
Email Market
Subscribed

Notes about Vendor

Products

Total Earn \$10950 on 20 Product

Product Name	June 26,2025	Pending	Brand Name	1000\$
Product Name	June 26,2025	Reject	Brand Name	1000\$
Product Name	June 26,2025	Hidden	Brand Name	1000\$
Product Name	June 26,2025	Published	Brand Name	1000\$

Address

Karim Hazem
In the address "789 Pine Avenue," "789 Pine Avenue" would go in Address Line 1

< 1 2 3 4 5 6 7 >

Esaltare Dekor HC. Copyright ZSK@2025 [Business Setup](#) [Profile](#) [Home](#)

5.7.3.27 Order Page

Esaltare Dekor HC

Search Here for ...

English

Notifications

Admin

Dashboard

Catalog

- Categories List
- Category
- SubCategory List
- SubCategory
- Products List
- Product
- Brands List
- Brand
- Posts List
- Post

Customers

- Customers List
- Customer Details

Vendor

- Vendors List
- Vendor Details
- Vendor Panel

Announcement

- Announcements List
- Announcements

Orders

- Order List
- Order Details

Marketing

- Coupons List
- Coupon

Reviews

Authentication

- Sign In
- Sign Up
- Forget Password
- Reset Password
- Verify Email

Inbox

Setting

Order List

Search by Type orders

Export

ID	Number	Date	Customer	Paid	Order Status	Items	Total	Action
1	#3201	June 26,2025	Karim	Yes	New	3 items	1000\$	 
2	#3201	June 26,2025	Karim	Yes	New	3 items	1000\$	 
3	#3201	June 26,2025	Karim	No	Pending	3 items	1000\$	 
4	#3201	June 26,2025	Karim	No	Pending	3 items	1000\$	 
5	#3201	June 26,2025	Karim	Partial	Shipped	3 items	1000\$	 
6	#3201	June 26,2025	Karim	Partial	New	3 items	1000\$	 
7	#3201	June 26,2025	Karim	Yes	Shipped	3 items	1000\$	 

< 1 2 3 4 5 6 7 >

Esaltare Dekor HC. Copyright ZSK@2025

Business Setup

Profile

Home

5.7.3.28 Order Details

English

Dashboard

Catalog

Customers

Vendor

Announcement

Orders

Marketing

Reviews

Authentication

Inbox

Setting

Order #B0294

January 3, 2025 at 11:40 PM | Status: PAID (25 JAN 2025) | [View](#) | [Print Invoice](#)

Notes about order

Items

	beech wood dining chair	1	8,200
	beech wood dining chair	1	8,200
Subtotal		16,400	
Store Credit		-200,00	
Shipping		100,00	
Total		16,300	

SZK Customer

You is the first order

Contact person

001 1886045

0000204640@gmail.com

1470720447582

Shipping Address

House 1234 Building 14 Street

Cairo, Egypt

Postal Code 12345

Billing Address

House 1234 Building 14 Street

Cairo, Egypt

Postal Code 12345

Esaltare Dekor HC. Copyright ZSK@2025

Business Setup
 Profile
 Home

5.7.3.29 Review Page

[English](#)

×

Dashboard

Catalog

- Categories List
- Category SubCategory List
- Products List
- Product Brands List
- Brand Posts List
- Post

Customers

- Customers List
- Customer Details

Vendor

- Vendors List
- Vendor Details
- Vendor Panel

Announcement

- Announcements List
- Announcements

Orders

- Order List
- Order Details

Marketing

- Coupons List
- Coupon

Reviews

Authentication

- Sign In
- Sign Up
- Forget Password
- Reset Password
- Verify Email

Inbox

Setting

Review List

ID	Customer Name	Review Title	Product Name	Priority	Review Status	Action
1	Karim	Bernhardt Plush Joli Sofa	Bernhardt Furniture is a leader in the direction of furniture design....	5	Active	
					InActive	
2	Karim	Bernhardt Plush Joli Sofa	Bernhardt Furniture is a leader in the direction of furniture design....	5	Active	
					InActive	
3	Karim	Bernhardt Plush Joli Sofa	Bernhardt Furniture is a leader in the direction of furniture design....	5	Active	
					InActive	
4	Karim	Bernhardt Plush Joli Sofa	Bernhardt Furniture is a leader in the direction of furniture design....	5	Active	
					InActive	
5	Karim	Bernhardt Plush Joli Sofa	Bernhardt Furniture is a leader in the direction of furniture design....	5	Active	
					InActive	
6	Karim	Bernhardt Plush Joli Sofa	Bernhardt Furniture is a leader in the direction of furniture design....	5	Active	
					InActive	
7	Karim	Bernhardt Plush Joli Sofa	Bernhardt Furniture is a leader in the direction of furniture design....	5	Active	
					InActive	

<
1
2
3
4
5
6
7
>

Esaltare Dekor HC. Copyright ZSK@2025

Business Setup
 Profile
 Home

5.7.3.30 Sign Up Page

The screenshot displays the 'Sign Up' page within the Esaltare Dekor HC application. The interface includes a dark sidebar on the left with various navigation options, a top header with a search bar and user profile, and a main content area with a sign-up form.

Sign Up

Fill out the form to create a new account

First Name

Last Name

Email Address

Phone

Password

☐ I agree to the [terms and conditions](#)

Sign Up

OR CONTINUE WITH

Google **Facebook**

Already have an account? [Sign In](#)

Navigation Menu:

- Dashboard
- Catalog
 - Categories List
 - Category
 - SubCategory List
 - SubCategory
 - Products List
 - Product
 - Brands List
 - Brand
 - Posts List
 - Post
- Customers
 - Customers List
 - Customer Details
- Vendor
 - Vendors List
 - Vendor Details
 - Vendor Panel
- Announcement
 - Announcements List
 - Announcements
- Orders
 - Order List
 - Order Details
- Marketing
 - Coupons List
 - Coupon
- Reviews
- Authentication
 - Sign In
 - Sign Up
 - Forget Password
 - Reset Password
 - Verify Email
- Inbox
- Setting

Footer:

Esaltare Dekor HC. Copyright ZSK@2025 Business Setup Profile Home

5.7.3.31 Sign In Page

Sign In
Log in to your account to continue

Email Address

Password

☐ Remember me [Forgot password?](#)

Sign In

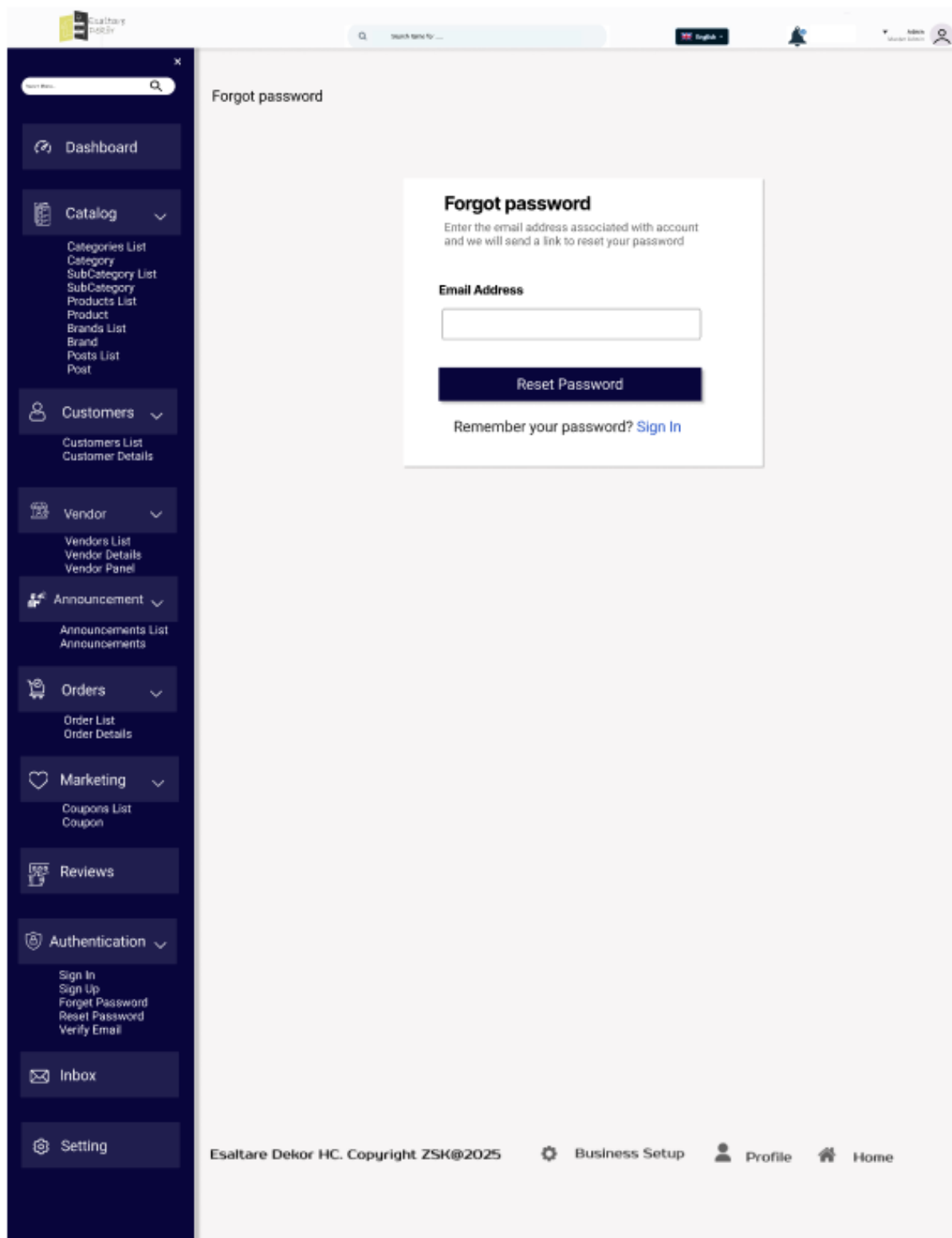
—OR CONTINUE WITH—

Google **Facebook**

Don't have an account? [Sign Up](#)

Esaltare Dekor HC. Copyright ZSK@2025 Business Setup Profile Home

5.7.3.32 Forget Password Page



5.7.3.33 Reset Password Page

The screenshot displays the 'Reset Password' page within the Esaltare Dekor HC admin dashboard. The page features a dark blue sidebar on the left with a search bar and a list of menu items including Dashboard, Catalog, Customers, Vendor, Announcement, Orders, Marketing, Reviews, Authentication, Inbox, and Setting. The main content area is titled 'Reset Password' and contains a white box with the heading 'Set a New Password'. Below the heading, a sub-header reads 'Create a new password Ensure it different from the previous password'. The form includes two input fields: 'New Password' and 'Confirm New Password'. A checkbox labeled 'Remember me' is positioned below the second field. A dark blue button labeled 'Update Password' is located at the bottom of the form. The footer of the page includes the text 'Esaltare Dekor HC. Copyright ZSK@2025' and a row of links: Business Setup, Profile, and Home.

Esaltare Dekor HC

Search Here For ...

English

Admin Master Admin

Reset Password

Set a New Password

Create a new password Ensure it different from the previous password

New Password

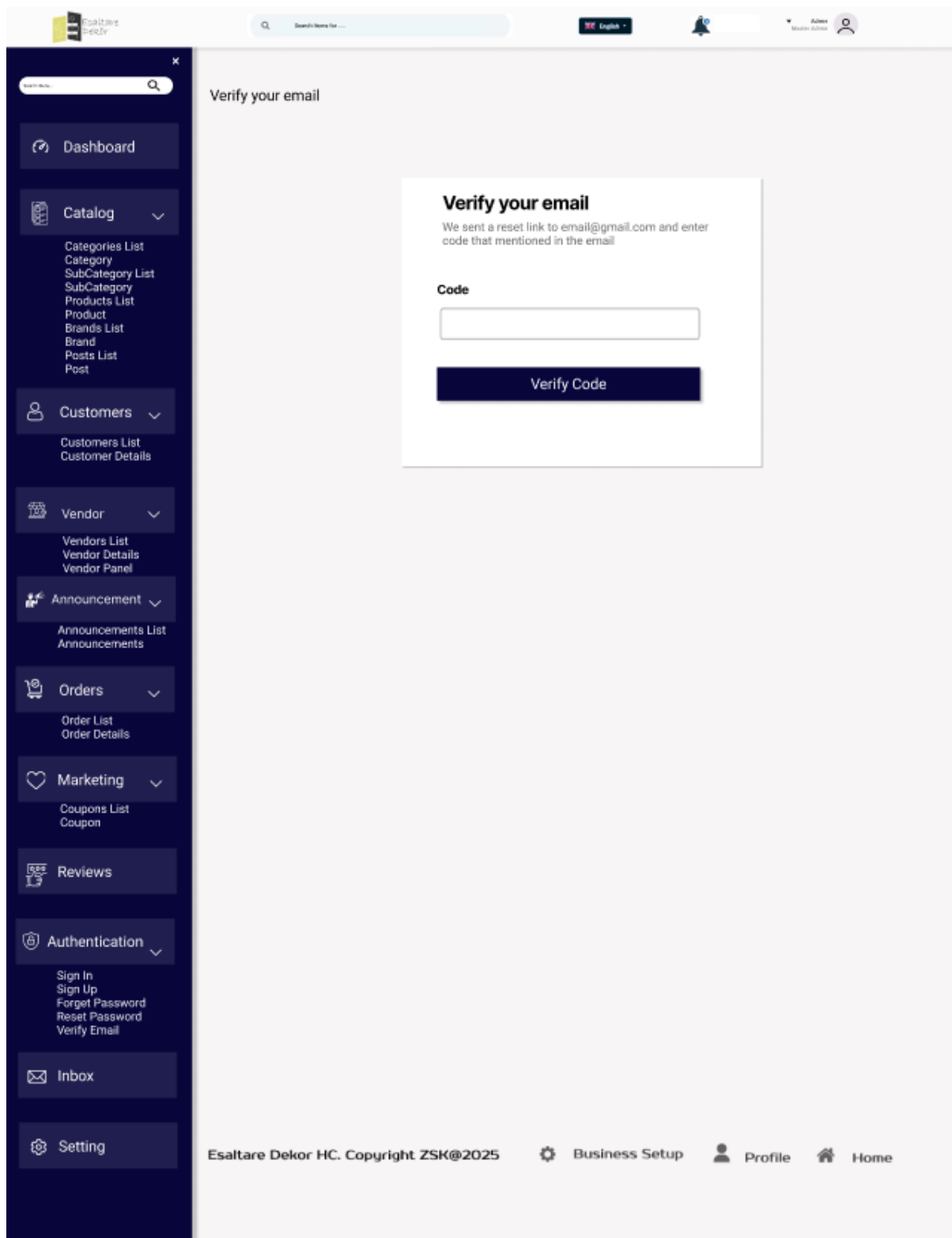
Confirm New Password

☐ Remember me

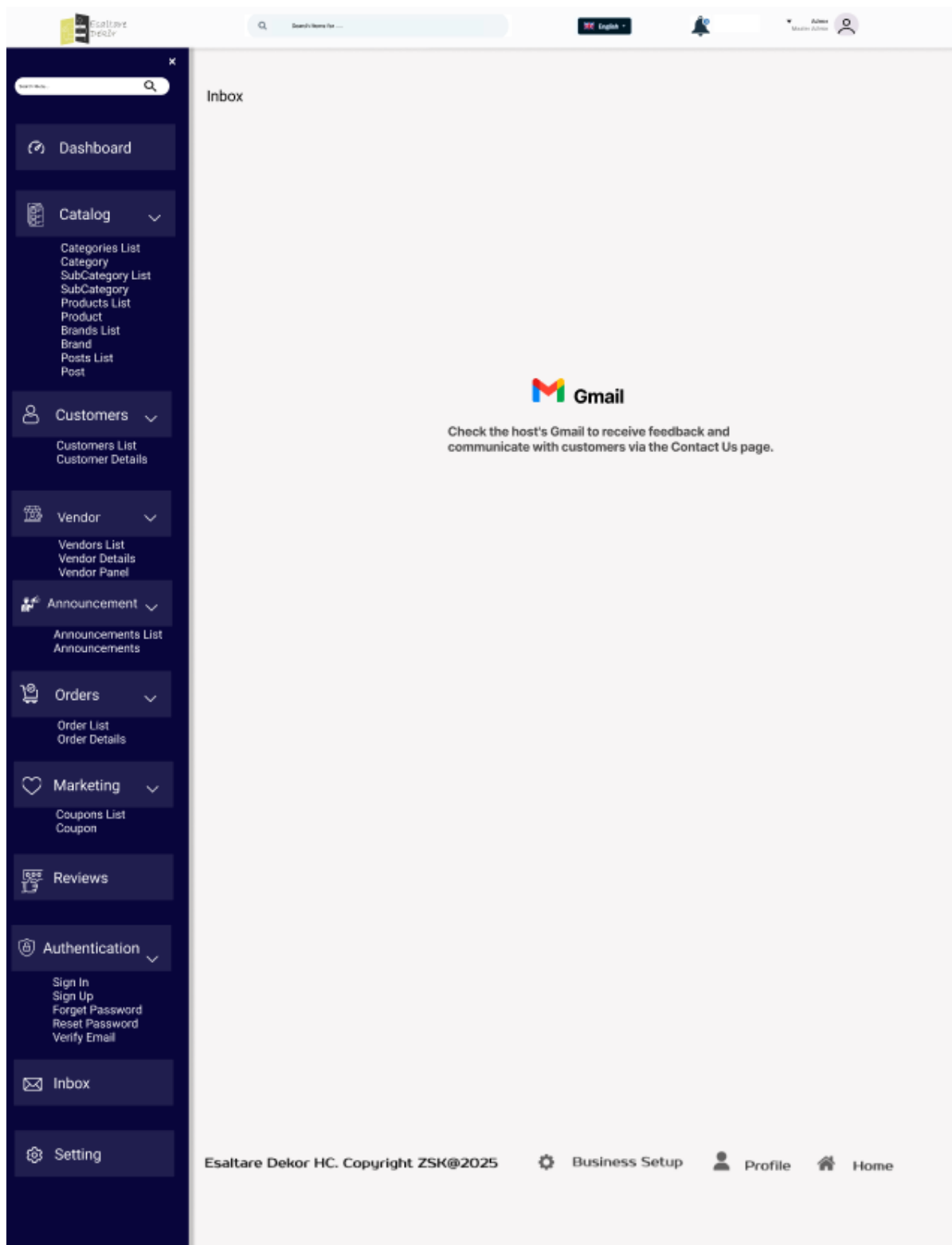
Update Password

Esaltare Dekor HC. Copyright ZSK@2025

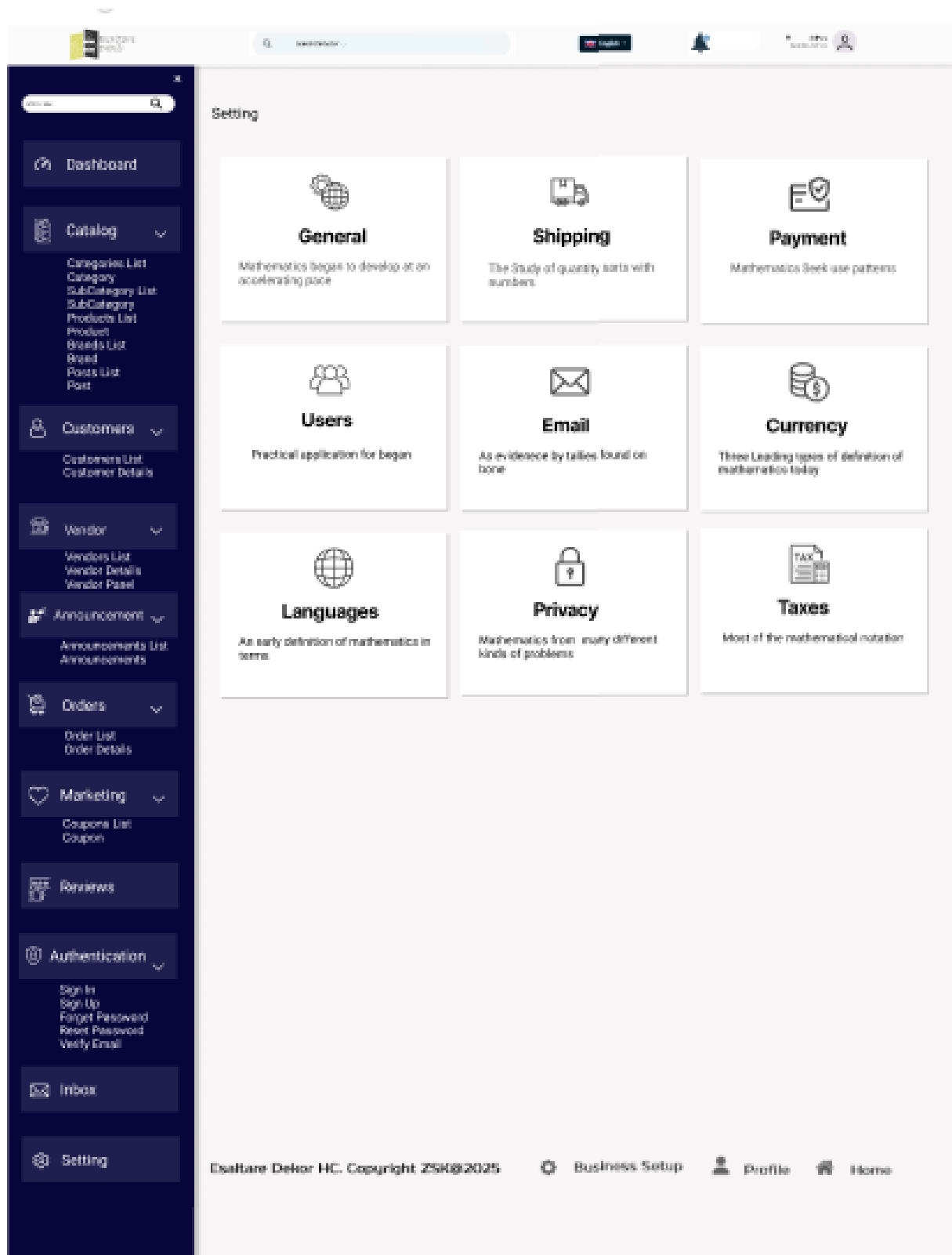
Business Setup Profile Home



5.7.3.34 Inbox Page



5.7.3.35 Setting Page



Chapter 6: Implementation

6.1 Implementation Journey

The implementation of this project was carried out using a systematic and well-planned approach, ensuring that each component was developed using modern and reliable technologies. We divided the implementation into two main parts: the front-end (client side) and the back end (server side), with seamless communication between them through RESTful APIs.

Front-End

For the front-end, we chose **React.js** as the core JavaScript library due to its efficiency, component-based architecture, and wide developer support. To enhance performance and provide a faster development environment, we used Vite as our build tool and development server. **Vite** offered lightning-fast Hot Module Replacement (HMR) and quick start times, which significantly improved productivity. To achieve a visually appealing and responsive user interface, we integrated several design libraries and frameworks. These included:

- **Material UI:** Provided ready-to-use, customizable components following Google's Material Design principles.
- **Bootstrap:** Helped with responsive layout design and simplified grid-based structures.
- **Ant Design:** Added elegant, enterprise-level components for forms, tables, and other UI elements.
- **CSS:** We used custom CSS for fine-tuning styles and ensuring that the UI met our specific design requirements.
- **Redux Toolkit:** Simplified state management by providing a standardized way to write Redux logic, with built-in support for reducers, actions, and async operations.

Combining these tools allowed us to build a highly interactive, modern, and user-friendly interface that adapts well to various screen sizes and devices.

Back-End

The back end was developed using PHP in combination with the Laravel framework. Laravel was selected for its elegant syntax, built-in features (such as routing, middleware, and authentication), and its ability to scale easily with future project needs. It provided a clean MVC (Model-View-Controller) structure, which helped in organizing our codebase efficiently.

For data storage and management, we used **MySQL**, a widely adopted relational database management system. The database schema was designed to ensure data normalization, integrity, and optimized query performance.

To set up the local development environment, we relied on **XAMPP**, which bundled Apache, **MySQL**, and **PHP** in a single package. This allowed us to test the application locally and simulate the production environment during development.

API and Integration

The communication between the front-end and back-end was handled through **RESTful APIs**. These APIs were designed following REST standards to ensure scalability, clarity, and separation of concerns. Each API endpoint was secured and validated to protect against unauthorized access and ensure reliable data exchange.

The API structure enabled modular communication between the UI and the server, making it easier to debug, maintain, and expand the system in future updates. API responses were structured in JSON format for compatibility and simplicity.

6.2 API Documentation

a) Authentication

- **POST** /api/login — Admin login
- **POST** /api/signup — Admin signup
- **POST** /api/user/login — User login
- **POST** /api/user/forgot-password — Forgot password

b) User

- **GET** /api/user/profile — View profile
- **PUT** /api/user/update — Update profile
- **GET** /api/user/orders — Get user orders
- **GET** /api/user/orders/{id} — Get a specific order

c) Products

- **GET** /api/view/products/{id} — View specific product
- **GET** /api/products/{id}/similar — Similar products
- **GET** /api/esaltare/products/filter — Filter products
- **GET** /api/esaltare/categories/{id}/products — Products by category

d) Categories, Brands, Subcategories

GET /api/esaltare/categories — All categories

GET /api/brands — All brands

GET /api/subcategories — All subcategories

POST /api/subcategories — Add subcategory

e) Sales

- **GET** /api/sales — Products on sale
- **DELETE** /api/sales/{id} — Remove product from sale

f) Packages

- **GET** /api/packages — All packages
- **POST** /api/packages — Add a package

g) Compare

- **POST** /api/compare — Add to compare
- **DELETE** /api/compare/{id} — Remove a comparison item
- **DELETE** /api/compare — Clear comparison list

h) Inspirations

- **GET** /api/inspirations — All inspiration photos
- **POST** /api/inspirations — Add photo
- **DELETE** /api/inspirations/{id} — Delete photo

i) Vendor

- **GET** /api/vendor/profile — Vendor profile
- **GET** /api/vendor/package — Vendor package
- **GET** /api/vendor/orders — Vendor orders
- **POST** /api/vendor/add-product — Vendor adds product

j) Home

- **GET** /api/home — Homepage data

q) Search

- **POST** /api/search/all — Full search endpoint

6.3 Frontend Development

The front-end of our application was designed and implemented as a fully integrated and feature-rich user system using React.js and Vite. This setup provided us with a fast, modern development environment and a highly interactive user interface. We focused on building complete experience for both users and vendors, covering all aspects of e-commerce—from product browsing and smart filtering to cart management, wish listing, and product comparison across various categories.

Styling and layout were handled using a combination of CSS, Bootstrap, and Tailwind CSS, giving us full control over responsive design and aesthetic consistency. In addition, we integrated a number of libraries to enhance usability and interactivity, such as Framer Motion for animations, React Slick for carousels, and React Toasty for user notifications.

To support 3D product visualization and a modern immersive experience, we incorporated Three.js through reactiber and react drei, enabling a rich 3D design environment that allows users to engage with products in a three-dimensional space.

We also implemented global state management using Redux Toolkit and Zustand, ensuring smooth data flow across components. Routing and navigation were managed with React Router DOM, while tools like Axios and JS-Cookie were used for API handling and session management.

Additional enhancements include:

- Stripe Integration for payment handling.
- Drag-and-drop support using React DnD.
- Step-based forms for multi-stage user workflows.
- File uploads with React Dropzone.
- 3D environment rendering via Three.js and React Three Fiber.
- Performance optimizations through Vite and modern ESLint configurations.

The combination of these technologies created a robust, modular, and scalable front-end system that is both developer-friendly and user-centric. It lays the foundation for a seamless and immersive e-commerce experience that is not only functional but also visually compelling and future ready.

The admin panel of the application was developed as a fully functional, responsive, and intuitive web dashboard using Create React App (CRA). This environment provided a stable foundation for building a modular, component-based interface. The admin system plays a crucial role in managing and controlling all aspects of the platform, giving administrators the tools they need to oversee operations efficiently.

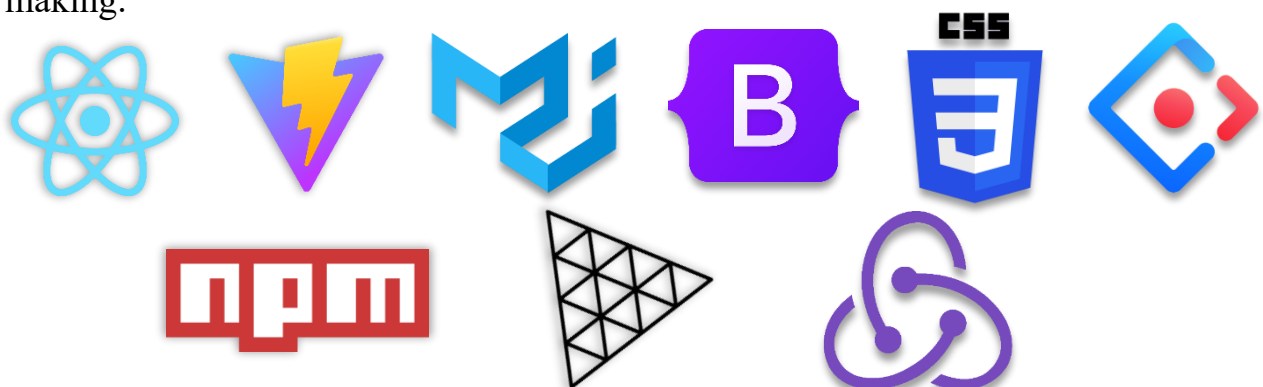
To design a clean, modern, and user-friendly interface, we used a combination of UI libraries including Material UI, Ant Design, Bootstrap, and React Icons. These libraries allowed us to create responsive layouts, stylish components, and intuitive navigation, ensuring a smooth experience for admin users across all devices.

The admin panel includes full CRUD functionality, enabling administrators to create, update, and delete product entries. In addition to product management, admins can post announcements, track user interactions, and manage platform-wide content. The panel also offers comprehensive analytics and reporting tools, displaying key metrics such as sales performance, user activity, inventory status, and trends over time. These insights are presented through dynamic charts and tables for easy interpretation.

All actions performed through the admin interface communicate with the back-end using secure and efficient RESTful APIs, ensuring real-time data updates and consistency across the system. The panel also supports filtering, searching, and category-based views, allowing administrators to manage large volumes of data with ease.

Scalability was a key consideration during development. The admin dashboard is structured to support future expansion, such as adding role-based access control, vendor-specific tools, and live notifications. Its architecture is modular, maintainable, and ready for growth as the platform evolves.

Overall, the admin panel serves as a powerful and centralized control system, designed to streamline platform management and support data-driven decision-making.



6.4 Backend Development

The backend of our application was built using PHP with the powerful Laravel framework, providing a structured, modern, and scalable approach to server-side development. We deployed the application in a local environment using XAMPP, which enabled smooth management of PHP services and MySQL, our chosen relational database.

Laravel's expressive syntax and built-in tools significantly streamlined the development process. It allowed us to efficiently handle routing, database migrations, validation, and controller logic. We followed MVC (Model-View-Controller) architecture to ensure a clean separation of concerns, making the codebase more maintainable and scalable.

The system integrates a fully functional RESTful API layer, enabling seamless communication with the front end. This API handles everything from user authentication and product management to shopping cart operations and data retrieval. We implemented authentication and authorization using Laravel's built-in features like Sanctum for token-based authentication, ensuring secure access to protected routes.

Data security was a top priority. We applied best practices such as hashed password storage, form request validation, middleware for role-based access control,. All sensitive actions are guarded by layered security policies, ensuring only authorized users can perform critical operations.

The backend supports comprehensive CRUD operations for products, users, favorites, and orders. Admins can manage the platform through dedicated endpoints with full control over product inventory, announcements, and user activity tracking. We also used Laravel's Eloquent ORM to simplify complex database interactions.

In summary, the backend serves as a robust and secure core for the entire system. With Laravel's powerful features, RESTful API structure, and strong security implementations, the server side is both reliable and ready for future growth and integration.



6.5 Tools Used

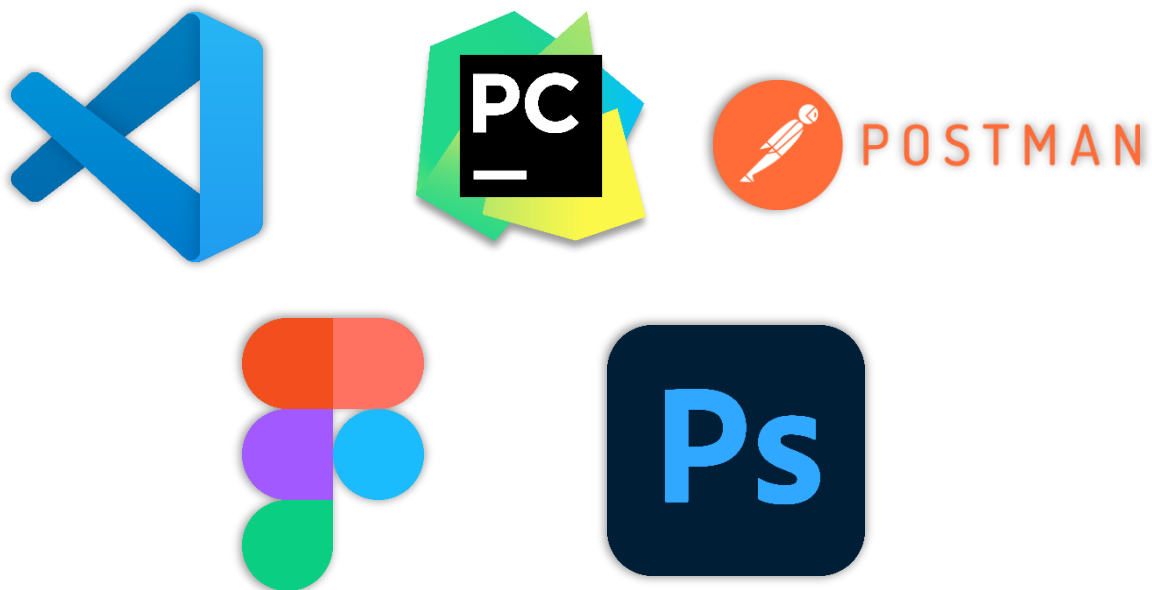
To ensure a smooth and efficient development workflow, we utilized a wide range of tools throughout the project. For code development, we primarily used Visual Studio Code and PyCharm, which provided powerful features such as intelligent code completion, debugging tools, and Git integration. Visual Studio Code was mainly used for front-end and full-stack development, while PyCharm supported backend scripting and logic, especially in earlier development phases.

For API development and testing, we relied on Postman, a robust tool that allowed us to test all RESTful API endpoints efficiently. Postman helped us verify request and response data, set authentication headers, and debug issues during client-server integration.

In terms of UI/UX design, we used Figma, a collaborative design tool that enabled us to build wireframes, high-fidelity prototypes, and ensure consistency across components before development began. This helped align the frontend implementation closely with the intended user experience.

Additionally, Adobe Photoshop was used for graphic editing and preparing visual assets such as icons, logos, and image optimizations to enhance the visual appeal of both the user and admin interfaces.

These tools collectively improved our productivity, design accuracy, and overall project quality, ensuring a professional and streamlined development process.



6.6 Deployment For The Project

The deployment strategy of our full-stack application was designed to ensure flexibility, scalability, and reliability by using two cloud platforms: Heroku for the frontend and Railway for the backend.

For the frontend, which was developed using React and Vite, we used Heroku to host and serve the application. Heroku's buildpack allowed us to deploy the static frontend files effortlessly. The integration with GitHub made the deployment process smooth every push to the main branch automatically triggered a new build and redeployment. We also configured Heroku environment variables to securely store backend API endpoints and other necessary keys. The result was a fast, responsive, and scalable frontend interface accessible from anywhere.

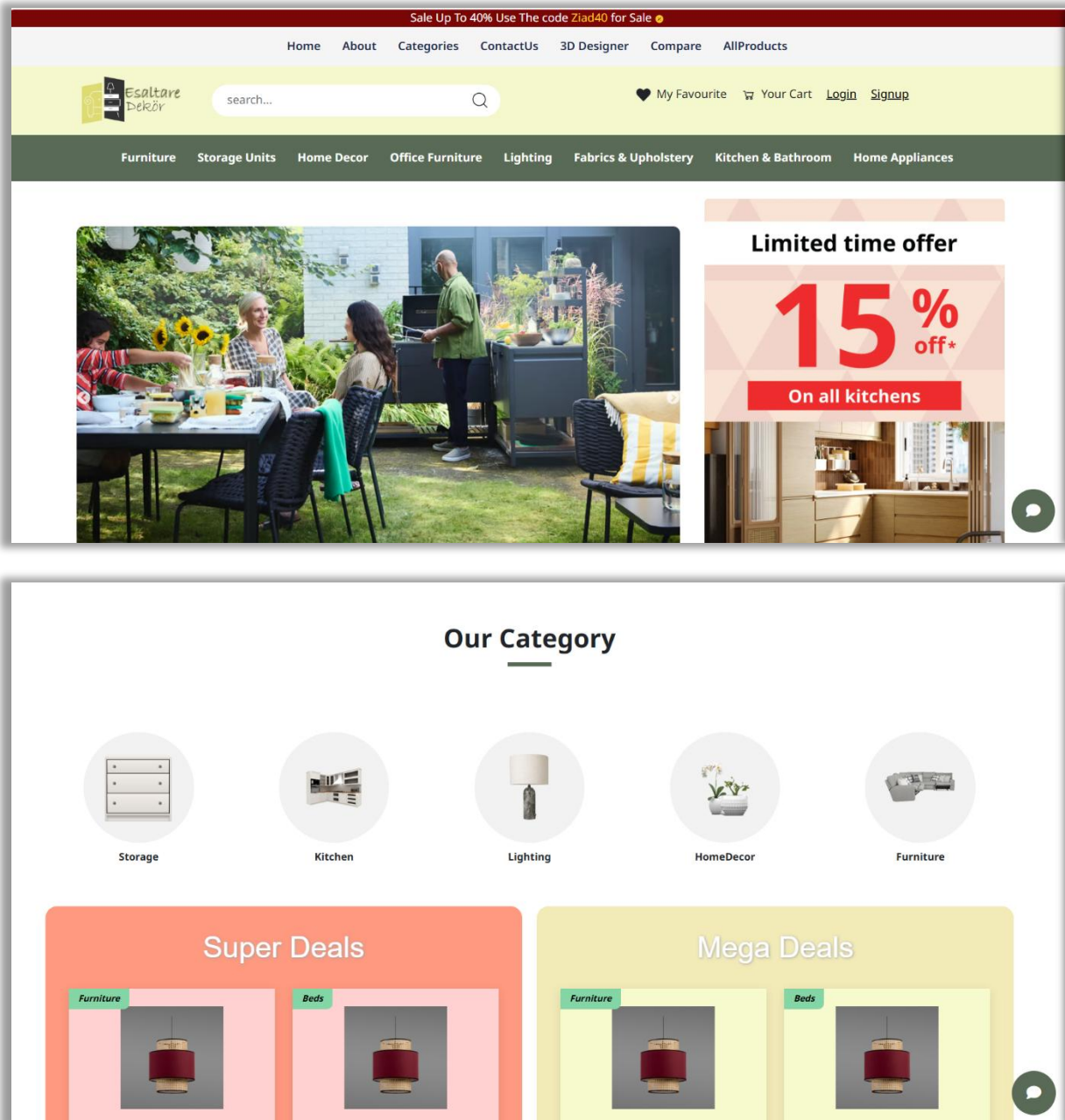
The backend, built with Laravel (PHP), was deployed using Railway, a developer-friendly cloud platform that supports modern frameworks and services. Railway allowed us to set up our Laravel application and connect it to a managed MySQL database. Environment variables, such as database credentials, API tokens, and secret keys, were securely managed through Railway's dashboard. The Laravel backend provided complete functionality, including user authentication, vendor management, RESTful API endpoints, and business logic. We also implemented security measures such as route protection and input validation.

The backend APIs hosted on Railway communicated seamlessly with the frontend on Heroku. Tools like Postman were used to test endpoints and ensure smooth integration. This combination of platforms enabled a clear separation of the application layers and made it easy to scale, monitor, and maintain each service independently.

By leveraging Heroku and Railway, we ensured a clean, efficient, and production-ready deployment environment, delivering high performance and robust functionality for both users and administrators.



6.7 Results After Implementation



Super Deals

Furniture



Ziad Cahir Chair

589645

294823

Beds



Tonato King Bed

12468

10598

Furniture



Classic Accent Chair

23550

Furniture



Premium Leather Chair

23550

Mega Deals

Furniture



Ziad Cahir Chair

589645

294823

Beds



Tonato King Bed

12468

10598

Furniture



Classic Accent Chair

23550

Furniture



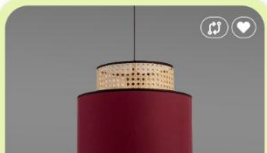
Premium Leather Chair

23550

Our Brands



New Arrivals








Ziad Cahir Chair

~~fairly 589645.00~~

294823.00

discount 50%

Add to cart



Tonato King Bed

~~fairly 12468.00~~

10598.00

discount 15%

Add to cart



Classic Accent Chair

~~fairly 23550.00~~

18369.00

discount 22%

Add to cart



Premium Leather Chair

~~fairly 23550.00~~

18840.00

discount 20%

Add to cart



Ergo Comfort Chair

~~fairly 18550.00~~

15397.00

discount 17%

Add to cart



The company

- about us
- Contact us
- Jobs

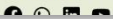
Important Links

- Privacy policy
- Terms and Conditions
- Join as a partner
- Return Policy

Need Help ?

Esaltare@help.org

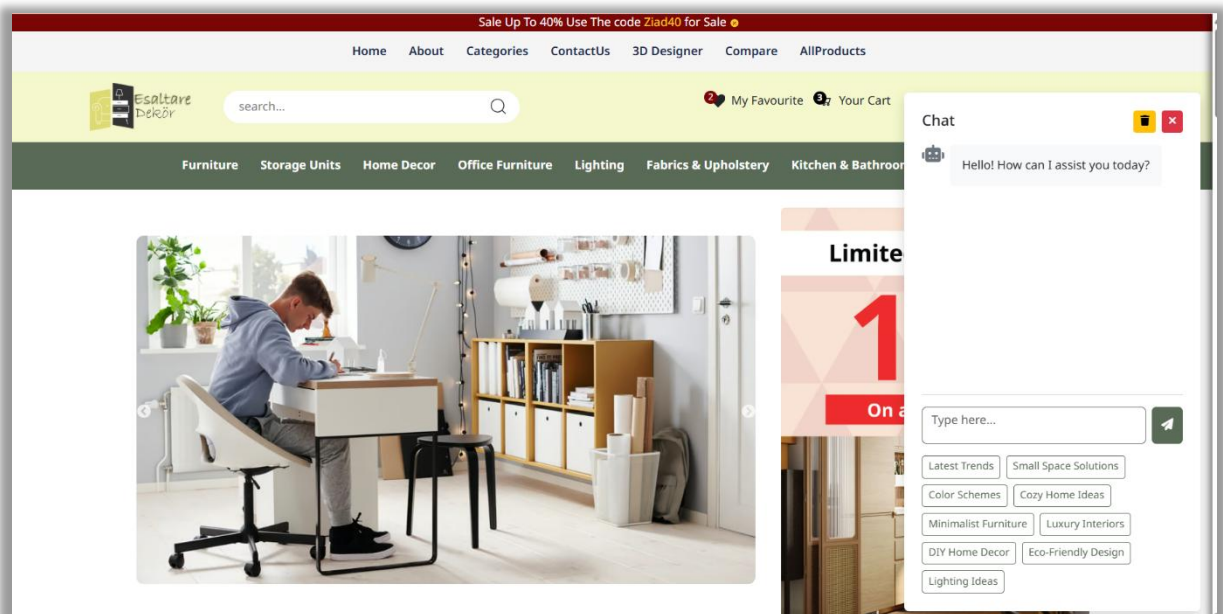
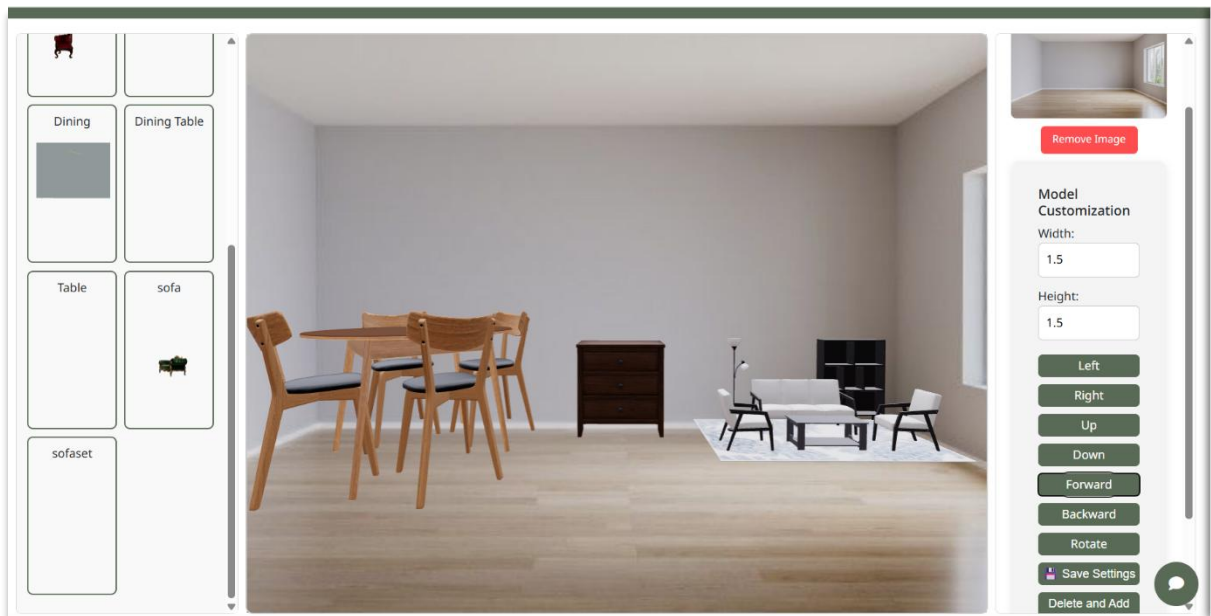
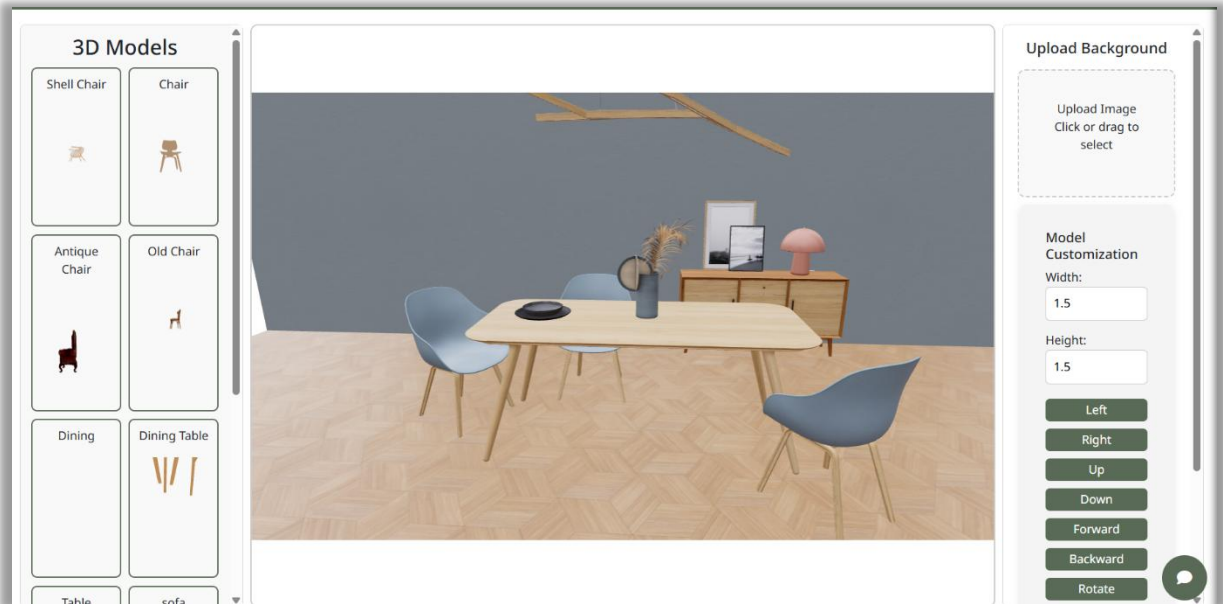
Follow Us



Payment Methods



186




My Account

[MyPackage](#)
[Profile](#)
[Add Product](#)



Welcome to Esaltare Dekor Console Packages Please Select A Package To Subscribe

basic package

BASIC

\$10,99

Compact Size

Easy Assembly

Neutral Colors

BUY NOW

standard package

STANDARD

\$29,99

Modern Look

Comfort Padding

Modern Look

BUY NOW

premium package

PREMIUM

\$49,99

Smart Features

Designer Touch

consectetur

BUY NOW


My Account

[MyPackage](#)
[Profile](#)
[Add Product](#)

Vendor Decor Plus

Full Name

Decor

Phone Number

015469887452

Address

Cairo Egypt

Full Name

Decor@organ.com

ZipCode

45698214542

MyPackage ▶

Product Posted

Product ID	Status	Item Qty	Total
7bF987	Processing	1	17,800
A857523	Processing	1	5,800


My Account

[MyPackage](#)
[Profile](#)
[Add Product](#)

Add New Product

Upload Images

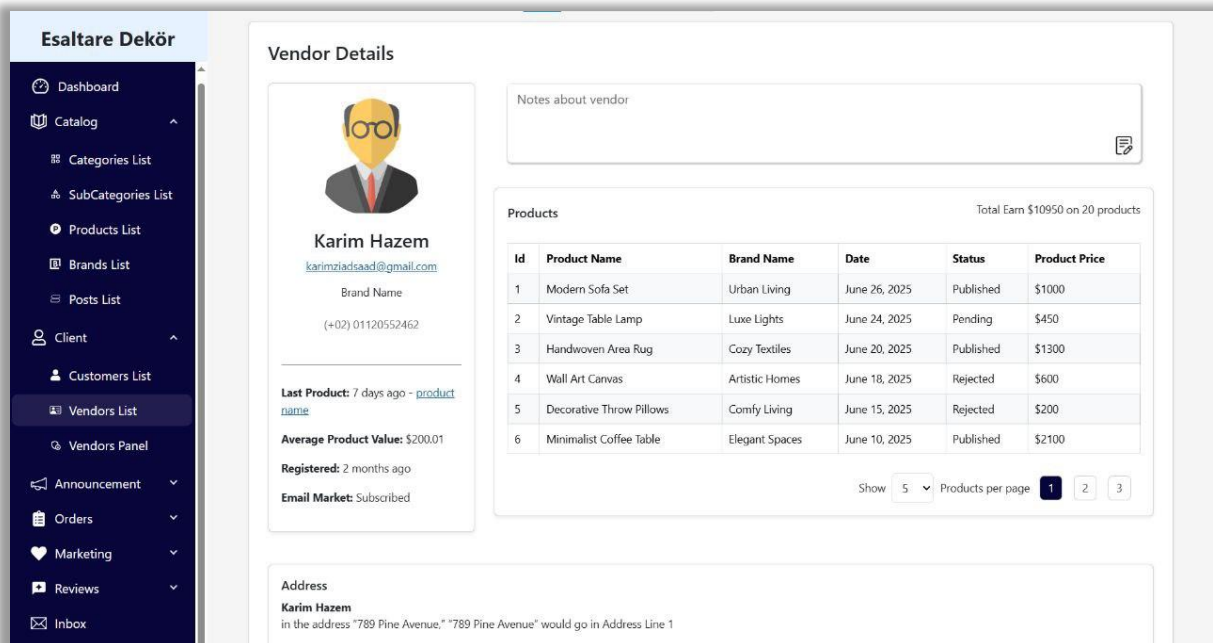




Product Name

Enter product name

Category



Esaltare Dekor

Dashboard

Catalog

Client

Customers List

Vendors List

Vendors Panel

Announcement

Orders

Orders List

Marketing

Reviews

Inbox

Setting

Search For Items

English

Admin Master Admin

Order List

Search by order number

Export

ID	ORDER NUMBER	DATE	CUSTOMER	PAID	ORDER STATUS	ITEMS	TOTAL	STATUS	ACTION
1	#3201	June 26, 2025	Karim	Yes	New	3 items	\$1000	Active	 
2	#3209	June 26, 2025	ziad	Yes	New	3 items	\$1000	Inactive	 
3	#3208	June 26, 2025	ziad	Yes	New	3 items	\$1000	Active	 
4	#3202	June 26, 2025	saad	Yes	New	3 items	\$1000	Inactive	 
5	#3203	June 26, 2025	saad	No	Pending	3 items	\$1000	Active	 

Show 5 reviews per page12

Esaltare Dekor HC. Copyright ZSK@2025

Business SetupProfileHome

Esaltare Dekor

Dashboard

Catalog

Categories List

SubCategories List

Products List

Brands List

Posts List

Client

Announcement

Orders

Marketing

Reviews

Inbox

Setting

Search For Items

English

Admin Master Admin

Add Category

Basic Information

Category Name

Enter category name

Priority

Enter priority (e.g., 1, 2, 3...)

Description

Normal B I U   

Status

☒ Active☐ InActive

Images



Add Category

Dekor HC

Dashboard

Catalog

Client

Customers List

Vendors List

Vendors Panel

Announcement

Orders

Marketing

Reviews

Inbox

Setting

Search For Items

English

Admin Master Admin

Vendor Panel List

Search by Brand, Vendor, P

Export

GE	VENDOR NAME	BRAND NAME	PRODUCT NAME	LENGTH	WIDTH	HEIGHT	PRODUCT DESCRIPTION	PRODUCT CATEGORY	PRODUCT SUBCATEGORY	PRODUCT PRICE	REGISTERED	APPROVAL
	Karim	Bonus brand Arflex	Sofa	100cm	100cm	100cm	how do you... Read More	Sofa	Living Room	\$100	May 15, 2025	Rejected
	Alice	Luxury Living	Dining Table	100cm	100cm	100cm	how do you... Read More	Table	Dining Room	\$300	June 20, 2025	Published
	John	Comfort Zone	Armchair	100cm	100cm	100cm	how do you... Read More	Chair	Living Room	\$150	July 10, 2025	Pending
	Sophia	Modern Home	Coffee Table	100cm	100cm	100cm	how do you... Read More	Table	Living Room	\$200	August 5, 2025	Published
	Michael	Classic Designs	Bookshelf	100cm	100cm	100cm	how do you... Read More	Storage	Living Room	\$250	September 15, 2025	Rejected

Show 5 Vendor Panel per page12

Chapter 7: Conclusion

As this project reaches its final stage, we reflect on the journey, the challenges we encountered, the growth we experienced, and the results we have achieved. This was more than just an academic requirement; it was a complete learning experience that combined creativity, collaboration, problem-solving, and shared vision. It gave us the opportunity to explore the real-world needs of users in the home décor industry and transform those insights into a fully functioning and thoughtfully designed e-commerce platform.

What began as a simple concept—an idea to build a digital space for home décor shopping—gradually evolved into a comprehensive and interactive platform. Through continuous research, experimentation, planning, and improvement, we built a system aimed at enhancing the experience of browsing, comparing, and purchasing décor items. Our goal was to deliver a modern and intuitive shopping journey for users, while also supporting vendors and administrators with practical tools to manage their operations.

Throughout the process, we kept user experience at the heart of our design. We quickly realized that a successful platform is not defined by the number of features it has, but by how effectively it helps users achieve their goals with ease. From viewing high-quality images to saving favorites, tracking orders in real time, and receiving product suggestions through a built-in chatbot, each feature was intentionally crafted to deliver value, convenience, and satisfaction.

Of course, the journey came with its share of challenges. Some were technical, while others were related to planning, time constraints, and decision-making. Finding the balance between functionality and simplicity was a continuous task. Including too many features risked overwhelming users, while offering too little could make the platform feel limited. Navigating this balance taught us the importance of thoughtful decisions and structured planning.

We also had to manage expectations and ambitions. As we explored new ideas, we often found ourselves eager to implement cutting-edge technologies and features. However, we had to stay grounded and practical, always keeping in mind our deadlines, tools, and the core goals of the project. This helped us prioritize what truly mattered and taught us the value of strategic focus and teamwork.

From a development perspective, this project became a space for skill-building and growth. We strengthened our abilities in organization, collaboration, time management, and user-centered design. We learned how to predict potential issues, adapt to unexpected changes, and stay motivated in the face of obstacles.

It also helped us grow more confident in our capacity to transform ideas into usable, impactful solutions.

Alongside the technical growth, we deepened our understanding of the human and commercial elements that define successful digital platforms. E-commerce is not just about listing products, it's about trust, communication, and reliability. Offering secure payment options, giving users control over their shopping experience, and providing clear and transparent updates are all essential to building that trust.

Vendors also played an essential role in shaping the platform. We recognized the need to give sellers flexible tools to manage their listings, adjust pricing, and track performance. By supporting vendors with intuitive systems, we not only empowered them to succeed but also enriched the platform with more diverse and dynamic content, ultimately benefiting the end users.

One of the most exciting aspects of the project was implementing an AI-powered chatbot. Although still in its early form, this feature opened our eyes to the potential of artificial intelligence in enhancing user interaction. It served as a gateway to future possibilities like personalization, automation, and intelligent recommendations, all of which could elevate the platform's capabilities and user satisfaction.

In the end, what makes this project most meaningful is not just the code or the interface, but the journey we shared and the knowledge we gained. It stands as a reflection of what a team can achieve through commitment, vision, and collaboration. Together, we have created something that not only works but also has the potential to make a real difference for users, vendors, and administrators alike.

To conclude, this was not just a technical endeavor, but a creative and collaborative journey. It challenged us to think like developers, designers, users, and problem-solvers all at once. While the project may be ending, the experiences we've gained and the ideas we've sparked will carry forward into future work. This is not just the end of a project, it is the beginning of what we hope will be many more innovations to come.

References

<https://threejs.org/>

<https://r3f.docs.pmnd.rs/getting-started/introduction>

<https://ant.design/docs/react/introduce>

<https://react-icons.github.io/react-icons/>

<https://getbootstrap.com/>

<https://www.php.net/>

<https://redux.js.org/>

<https://www.npmjs.com/>

<https://getcomposer.org/>

<https://mui.com/material-ui/>